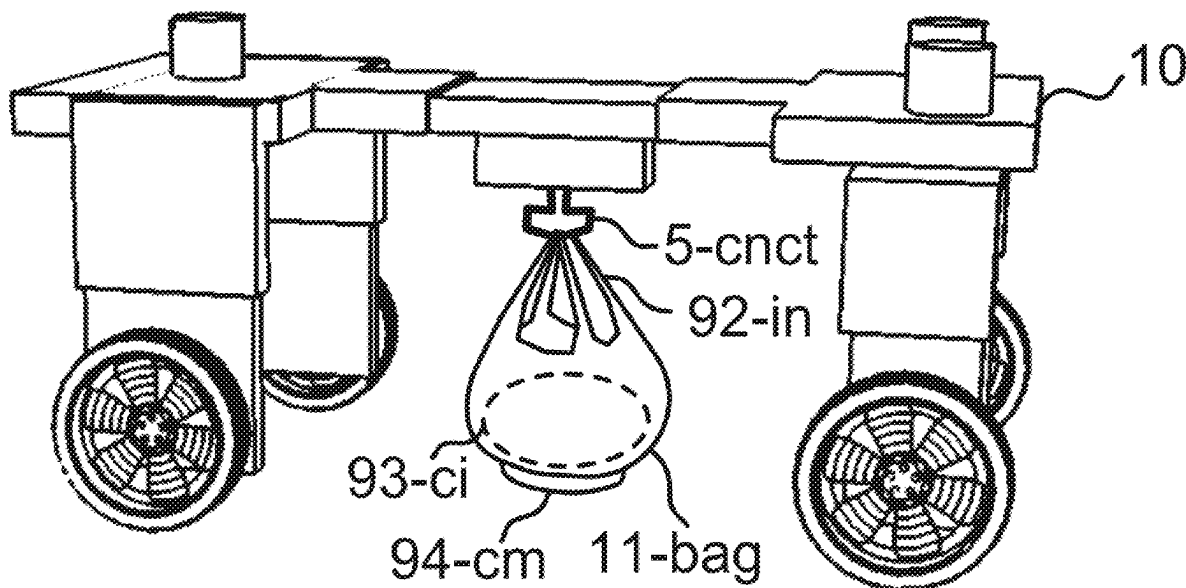




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(19) **United States**(12) **Patent Application Publication**
Zuckerman et al.(10) **Pub. No.: US 2025/0282273 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **AUTONOMOUS VEHICLE WITH ACTIVE
WHEEL POSITIONING FOR SUSPENSION
AND LOAD MANIPULATION**(52) **U.S. Cl.**
CPC *B60P 1/6445* (2013.01); *B60G 17/0157*
(2013.01); *B60G 17/016* (2013.01); *B60G*
2400/25 (2013.01)(71) Applicants: **Gal Zuckerman**, Holon (IL); **Moshe
Salhov**, Herzeliya (IL)(72) Inventors: **Gal Zuckerman**, Holon (IL); **Moshe
Salhov**, Herzeliya (IL)(21) Appl. No.: **19/218,737**(22) Filed: **May 27, 2025****Related U.S. Application Data**(63) Continuation-in-part of application No. 18/105,889,
filed on Feb. 6, 2023, now Pat. No. 12,337,875, which
is a continuation-in-part of application No. 17/246,
745, filed on May 3, 2021, now Pat. No. 11,599,845,
which is a continuation-in-part of application No.
16/502,834, filed on Jul. 3, 2019, now Pat. No.
11,046,230.**Publication Classification**(51) **Int. Cl.**
B60P 1/64 (2006.01)
B60G 17/015 (2006.01)
B60G 17/016 (2006.01)(57) **ABSTRACT**

An autonomous vehicle comprises a main body, wheels, and powered actuation systems modifying the relative positioning between each wheel and the main body. A grabbing mechanism holds a load beneath the main body. The actuation systems serve dual purposes: providing active suspension during forward motion and enabling coordinated movements to manipulate the load relative to the ground (e.g., lifting or lowering). The vehicle may be sized for sidewalk traversal and deliver consumer items, potentially in disposable packages held by the grabbing mechanism. Manipulation includes lowering and releasing the load, or positioning over, grabbing, and lifting the load. Active suspension maintains load-to-ground clearance during transport, potentially functioning as springs/dampers. Sensors provide environmental data for generating 3D representations used in controlling actuation for load manipulation. Methods include transporting the load using wheels while providing active suspension, and upon arrival, using the actuation systems for coordinated load manipulation and release.



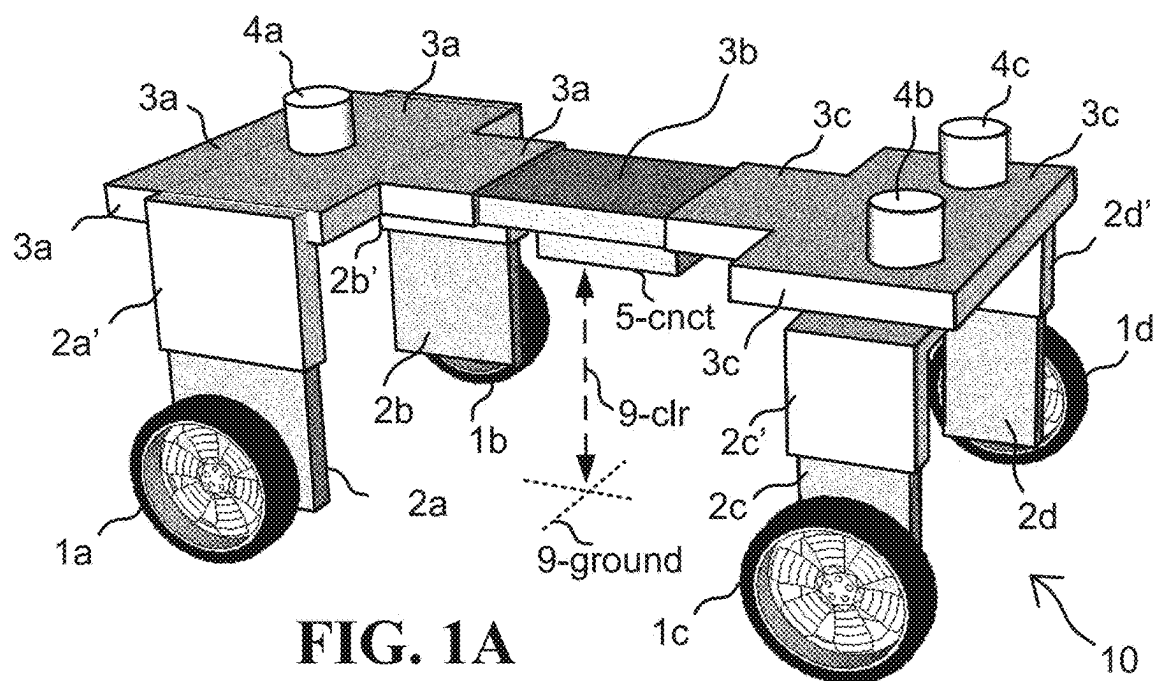


FIG. 1A

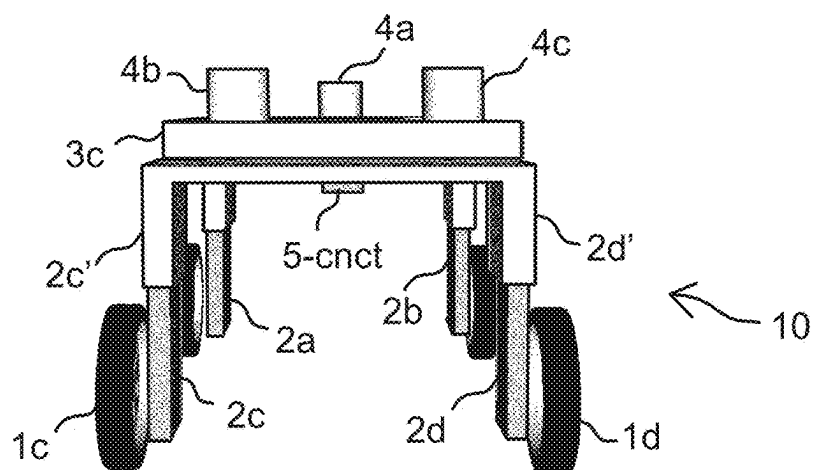


FIG. 1B

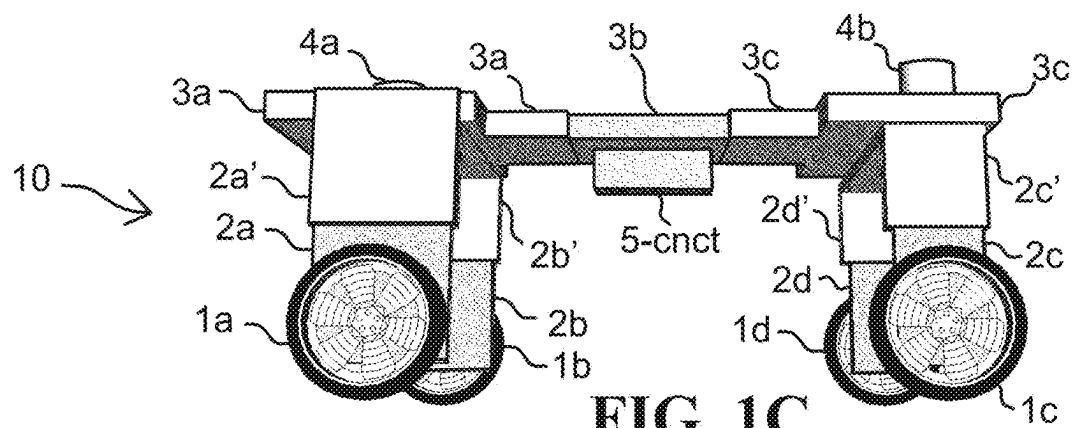


FIG. 1C

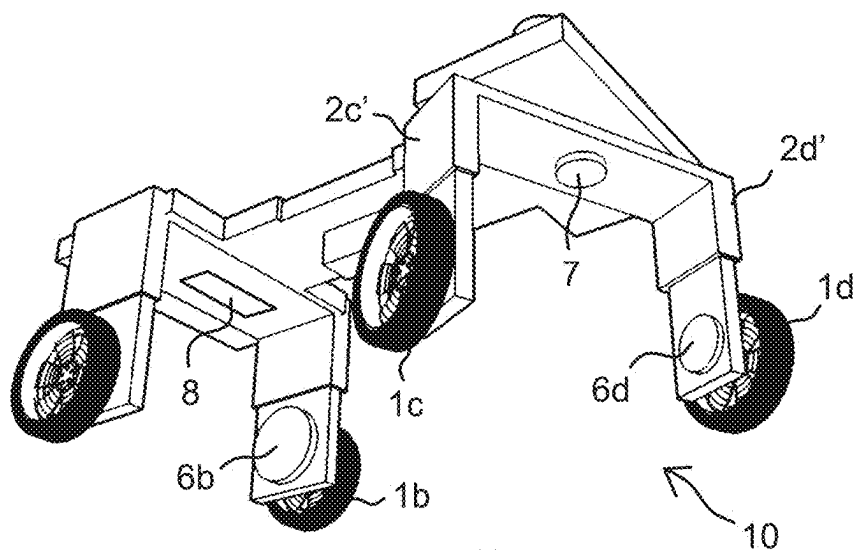


FIG. 1D

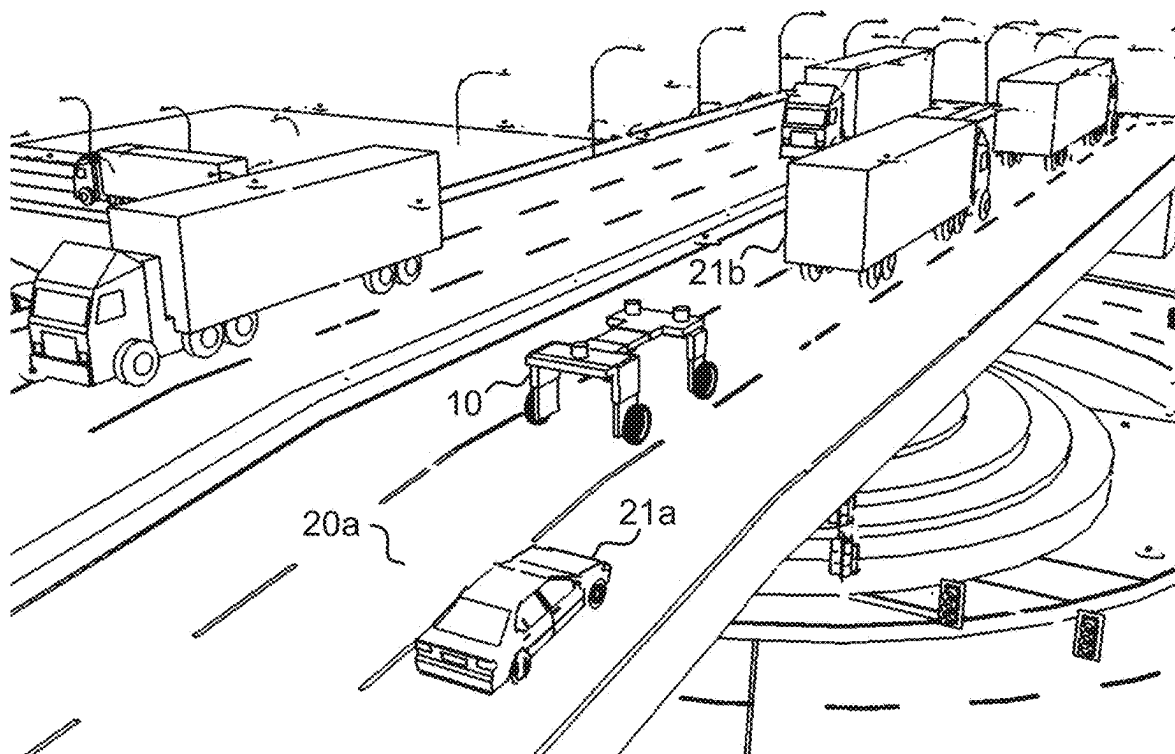


FIG. 2A

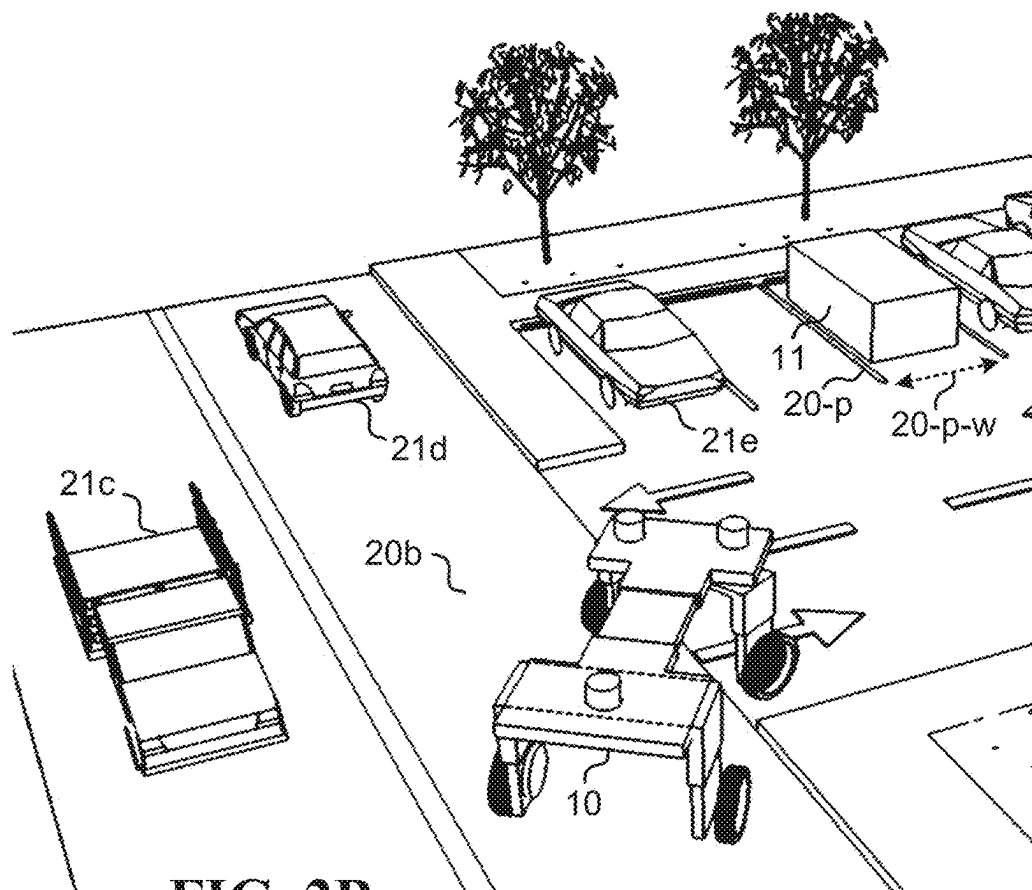


FIG. 2B

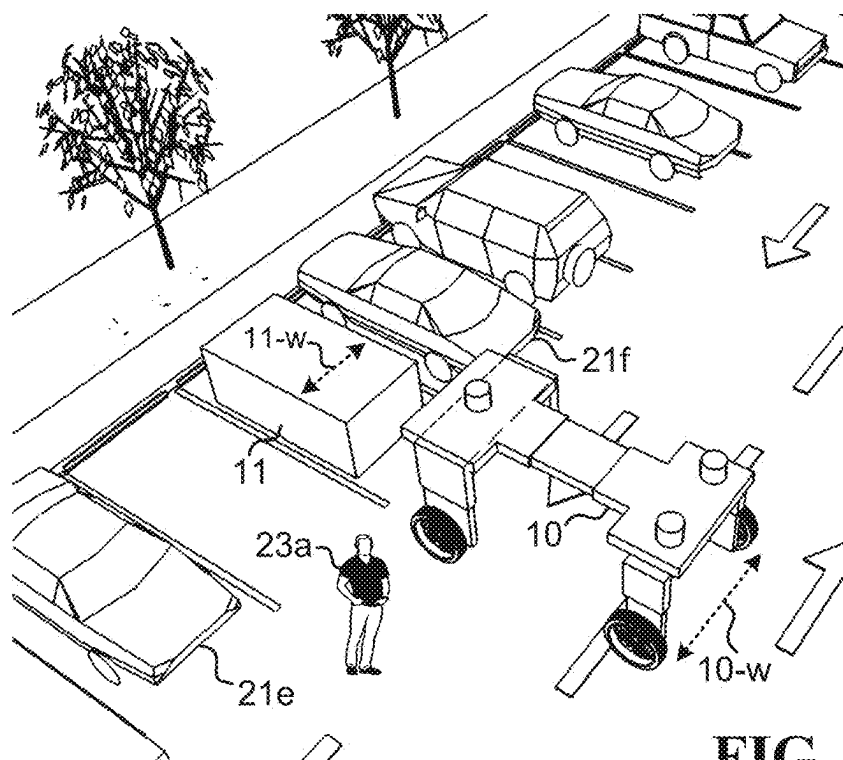


FIG. 2C

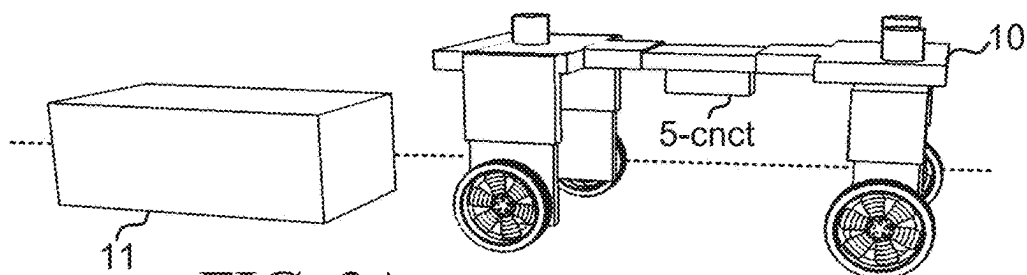


FIG. 3A

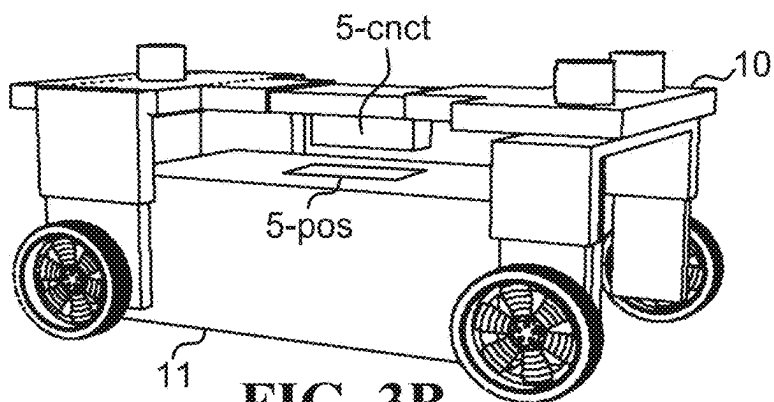


FIG. 3B

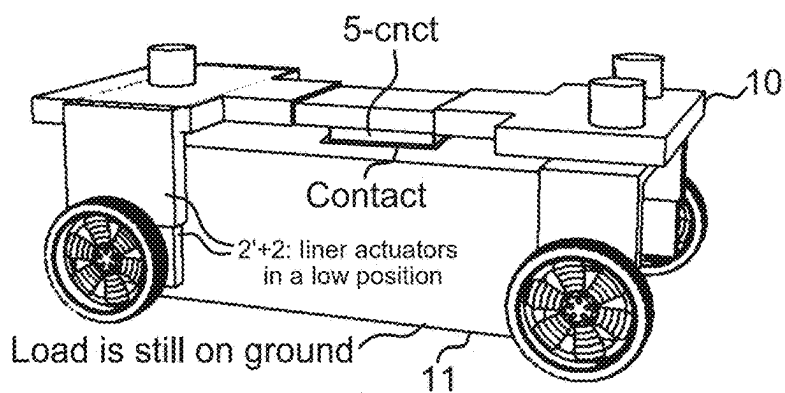


FIG. 3C

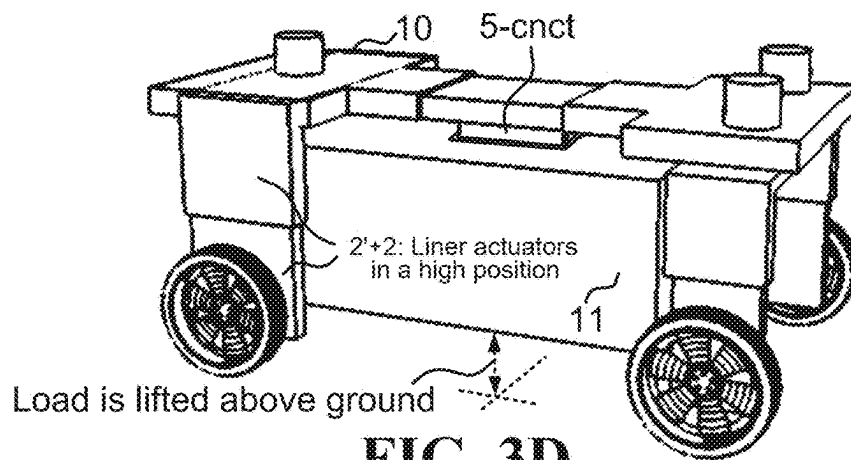


FIG. 3D

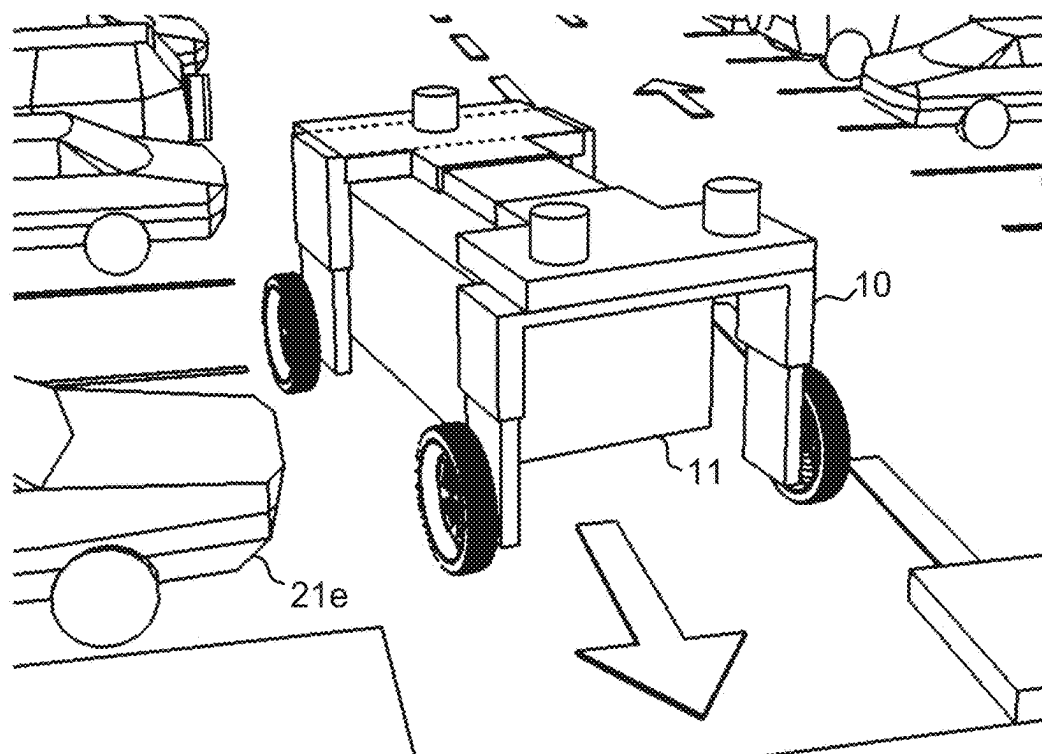


FIG. 4A

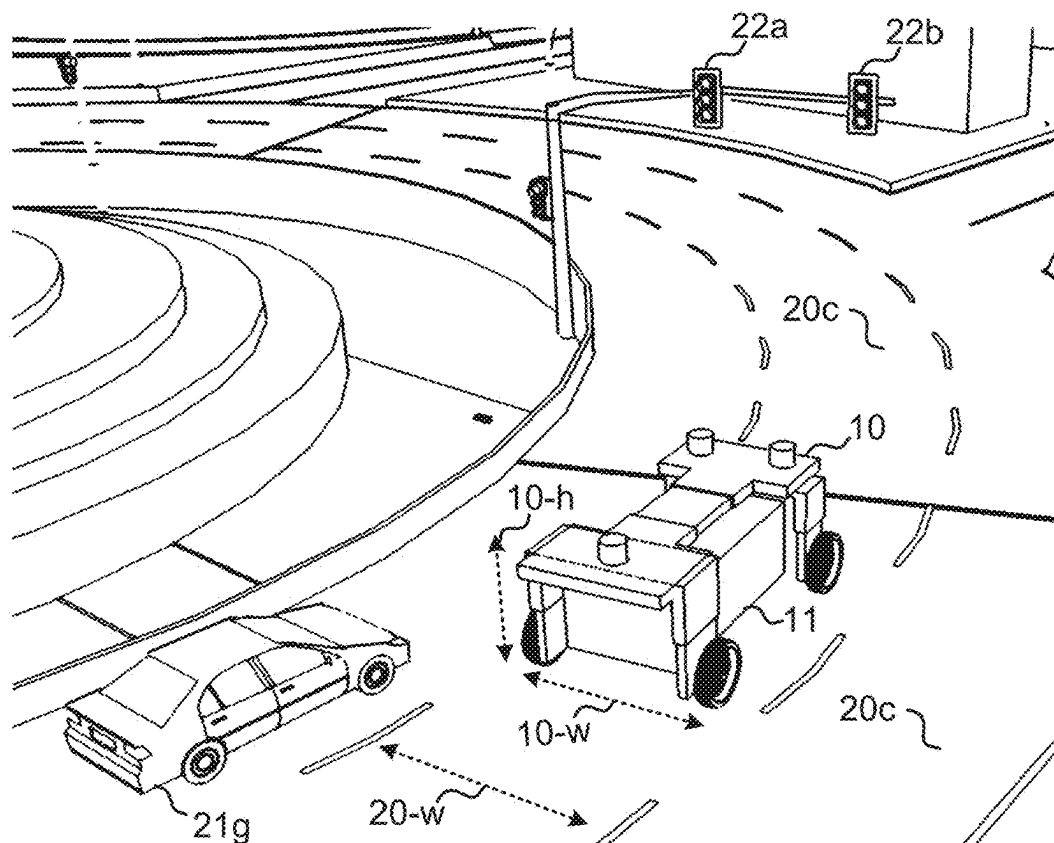


FIG. 4B

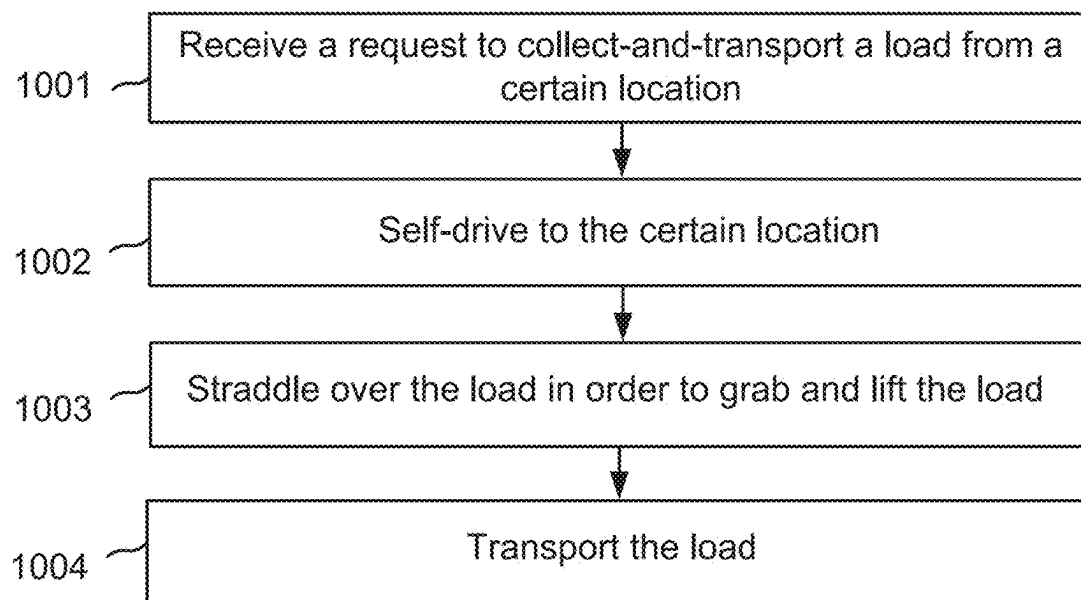


FIG. 4C

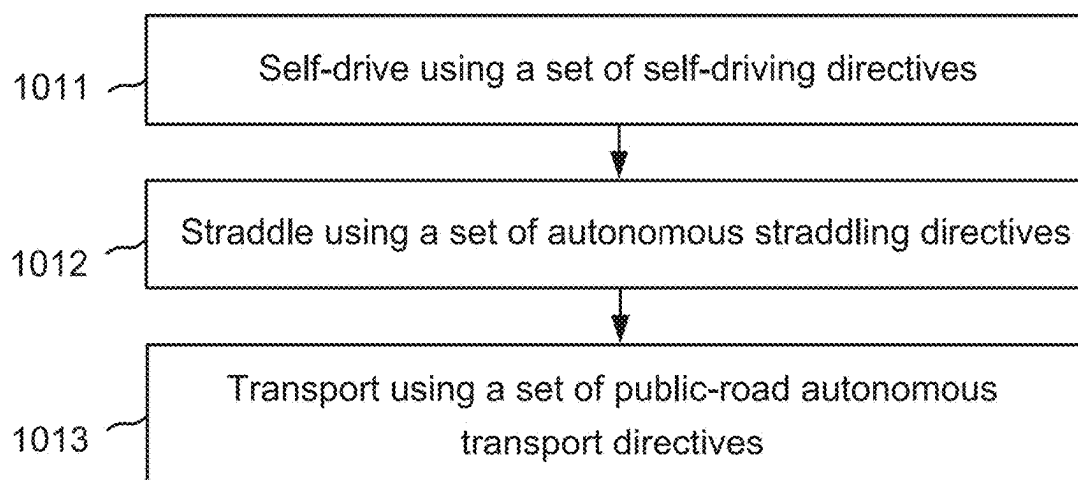


FIG. 4D

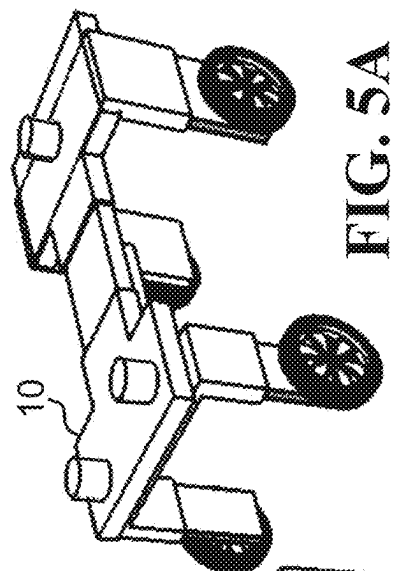


FIG. 5A

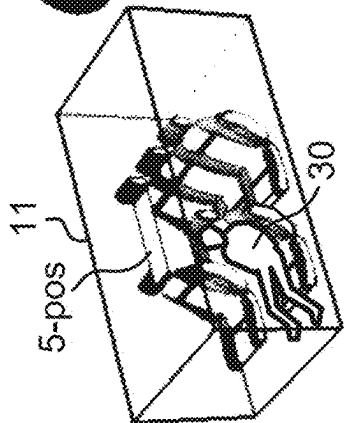


FIG. 5B

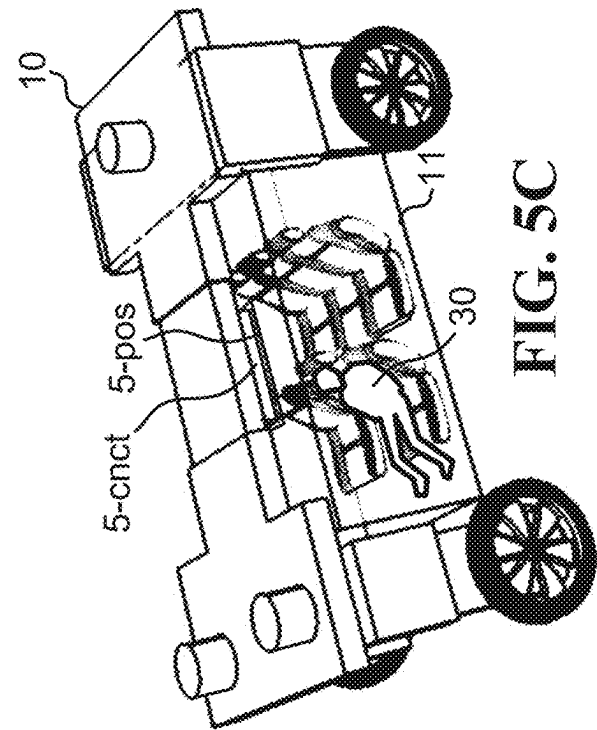


FIG. 5C

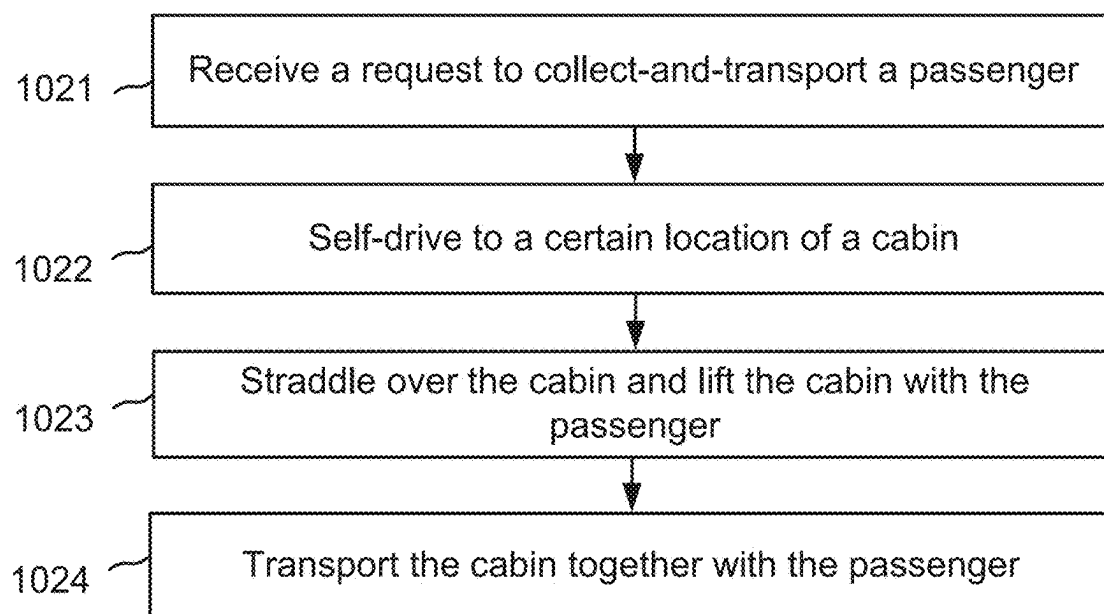


FIG. 5D

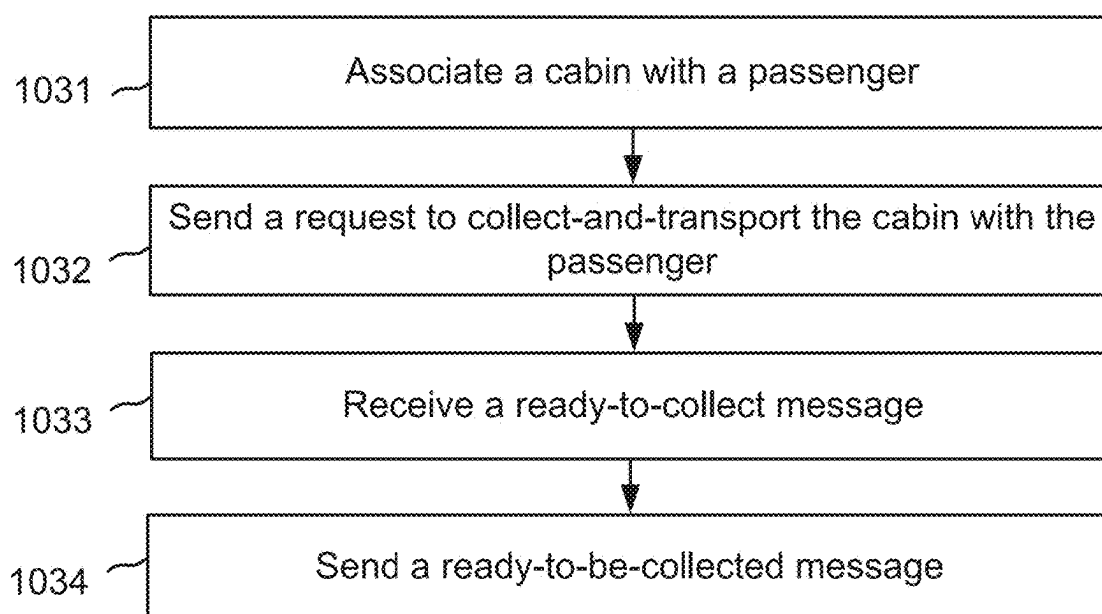
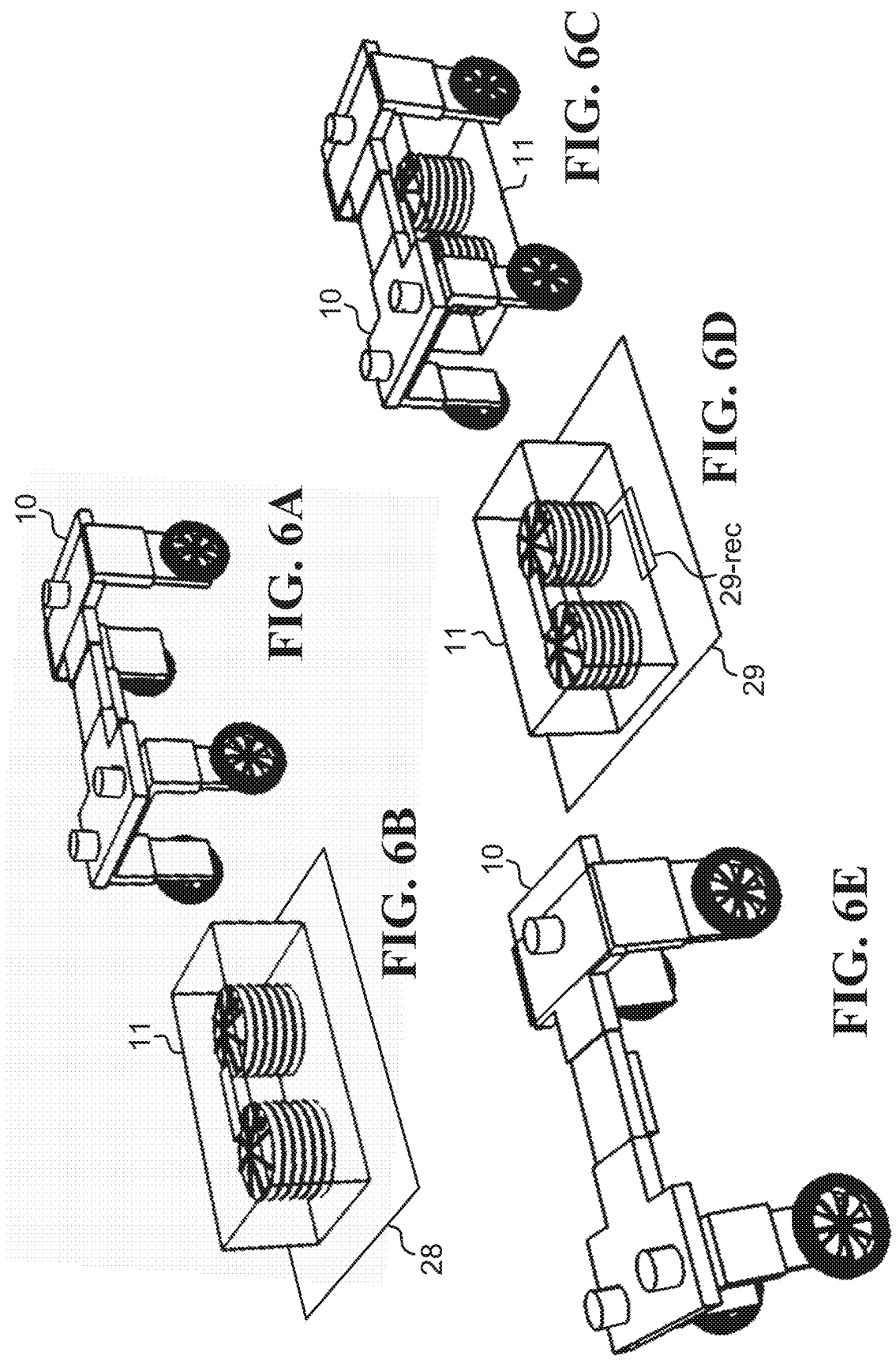


FIG. 5E



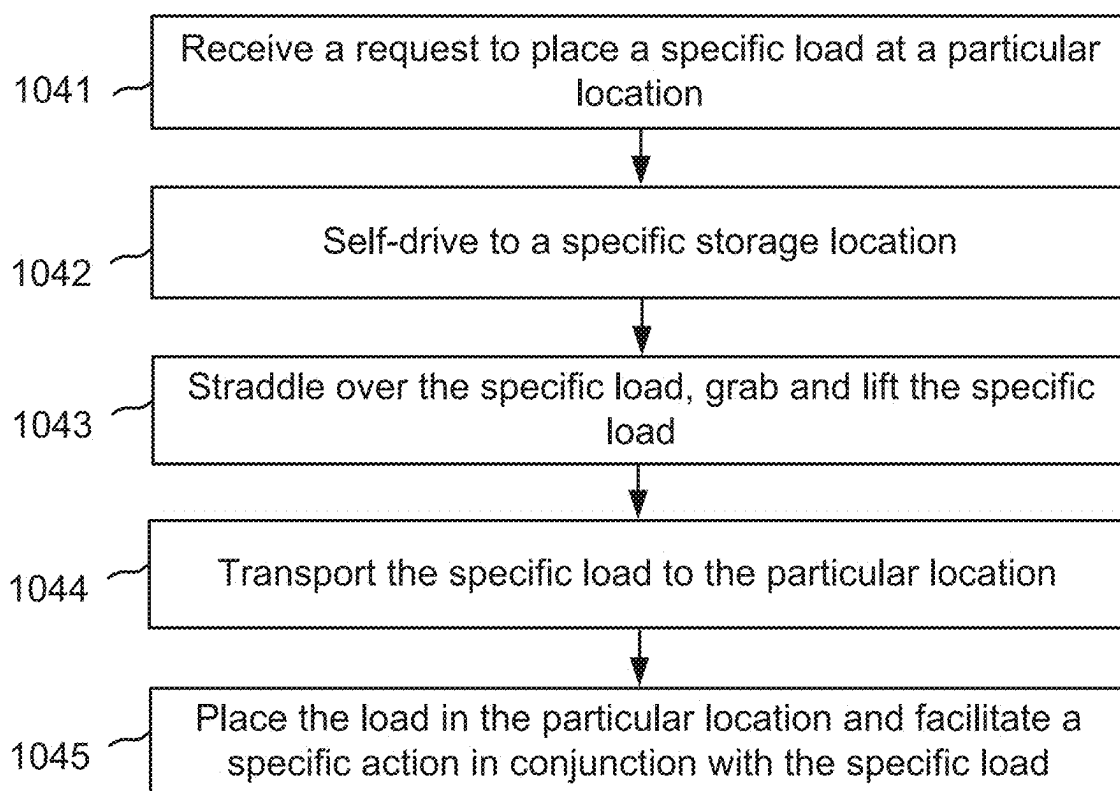
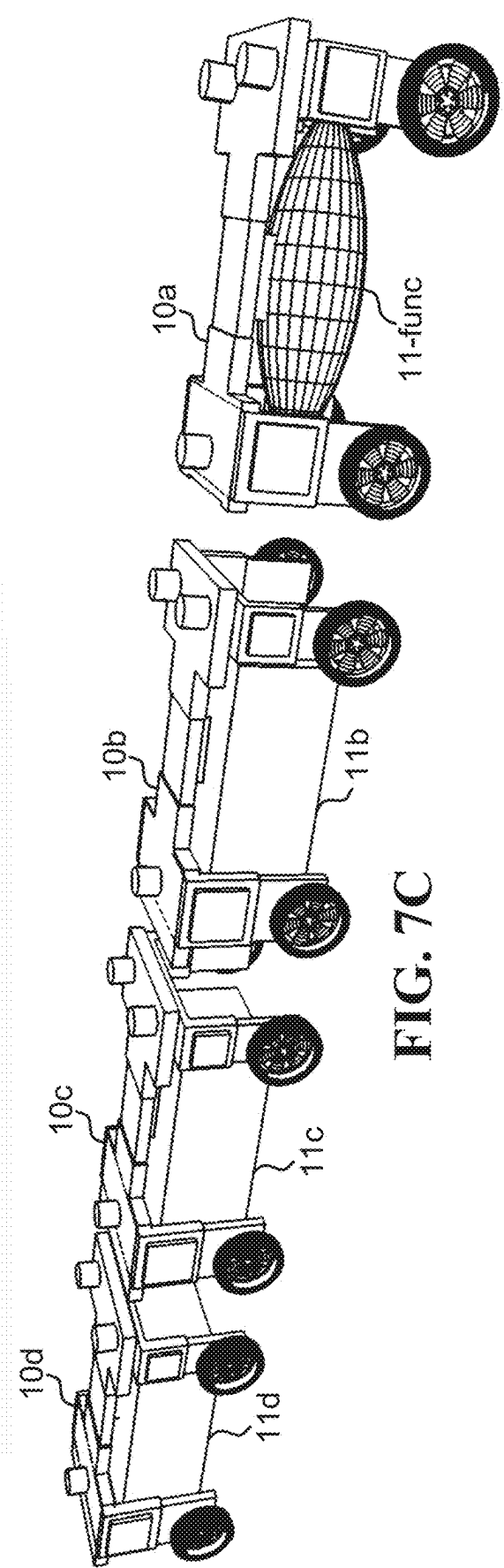
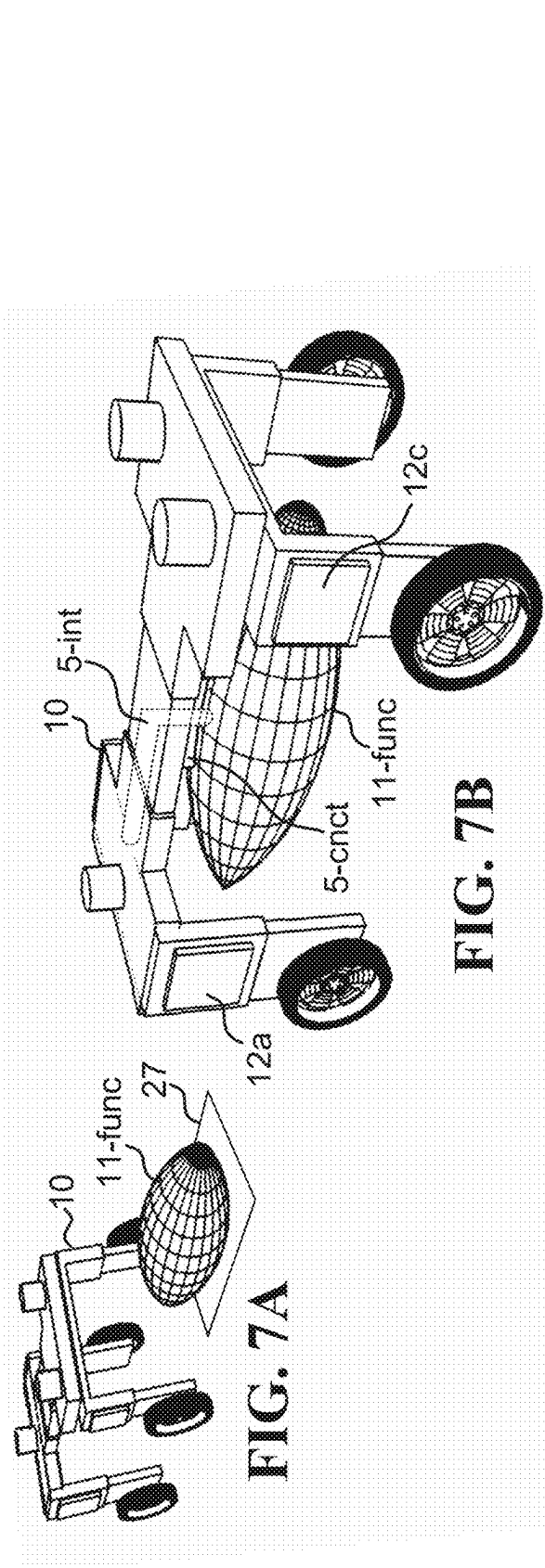


FIG. 6F



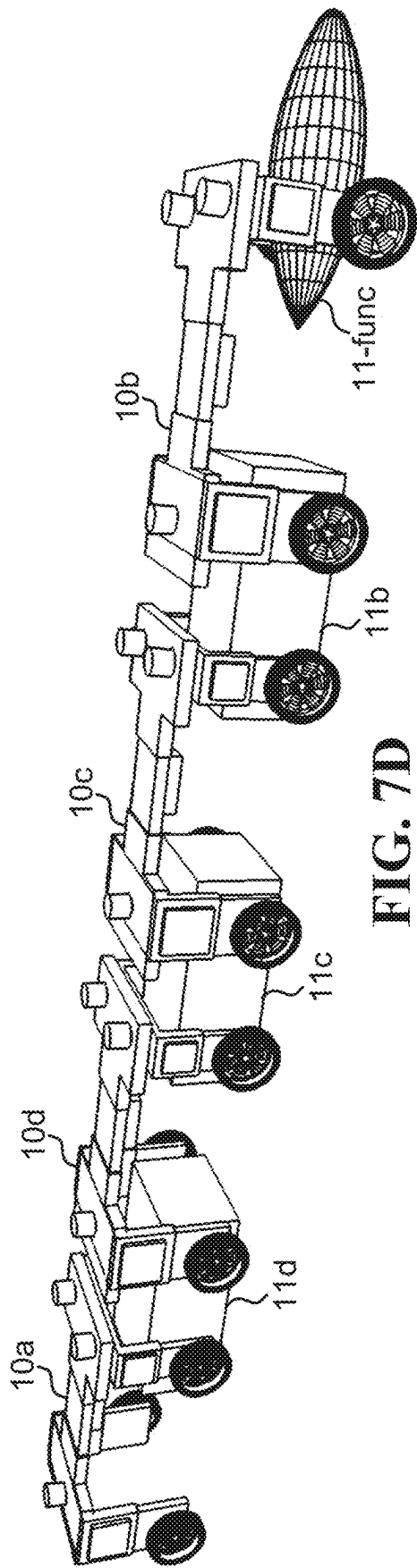


FIG. 7D

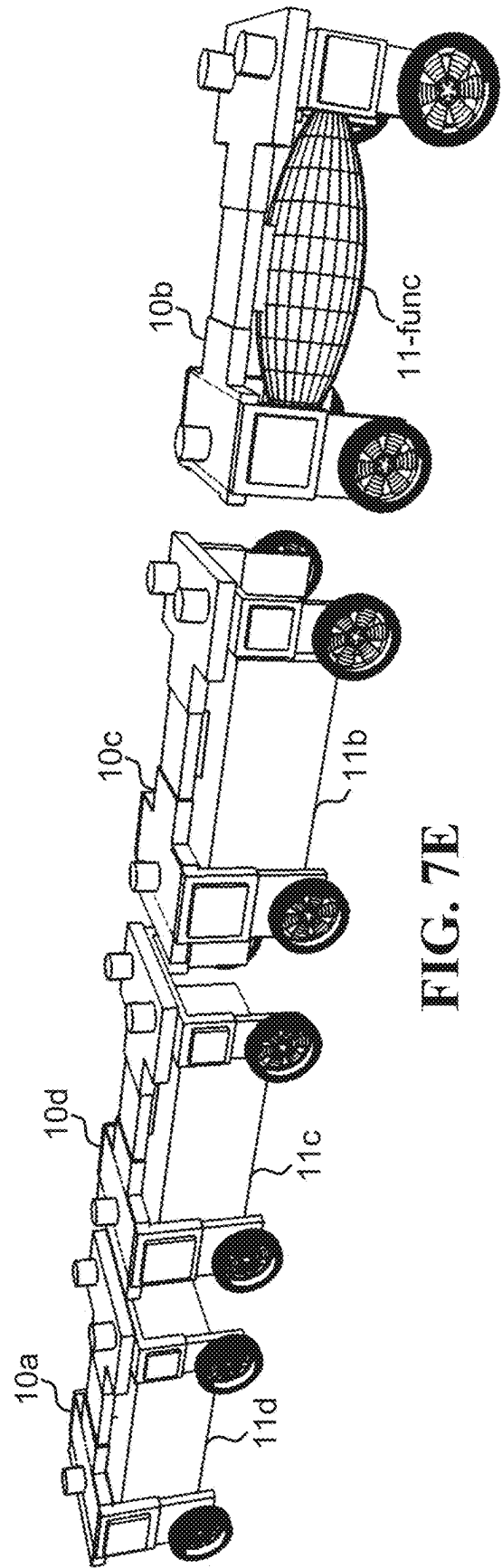


FIG. 7E

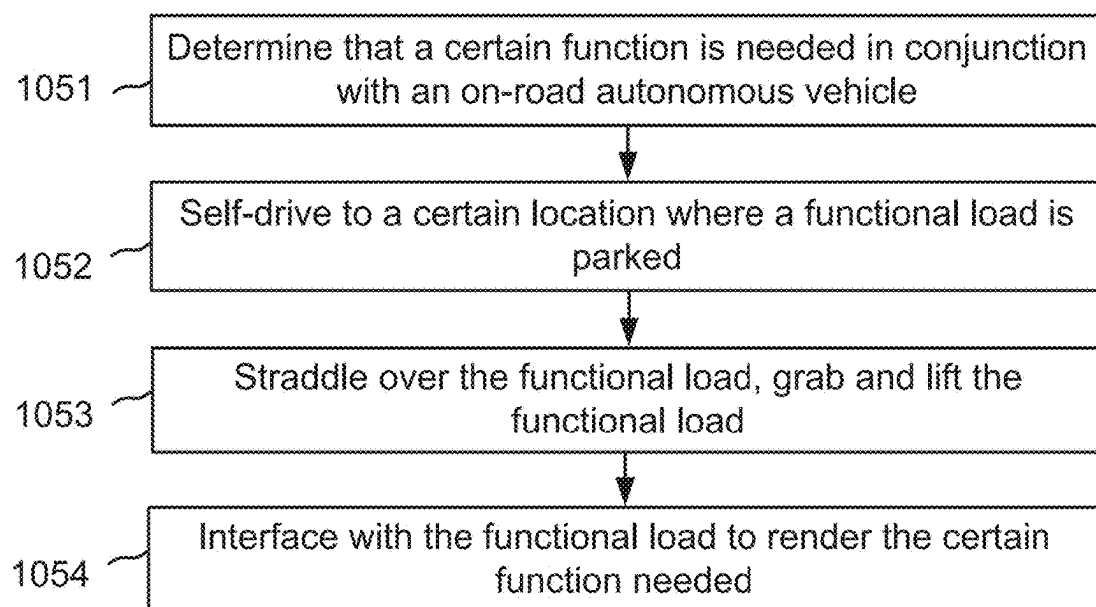


FIG. 7F

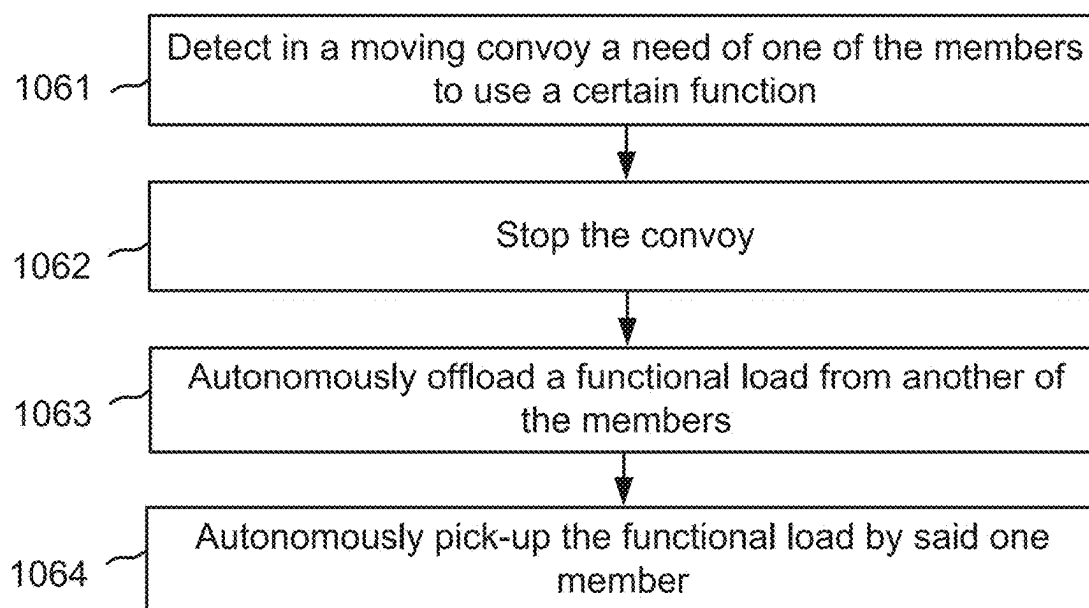
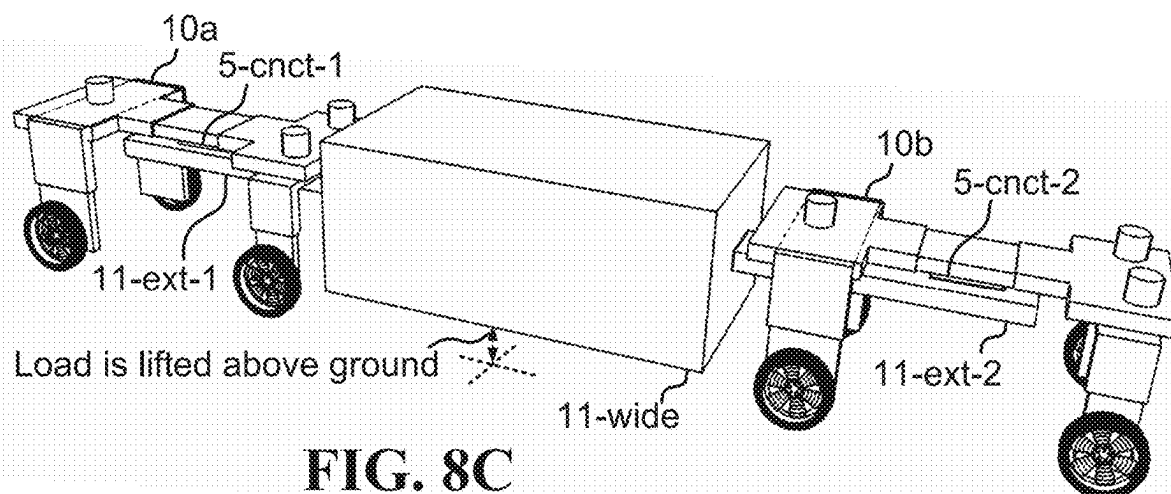
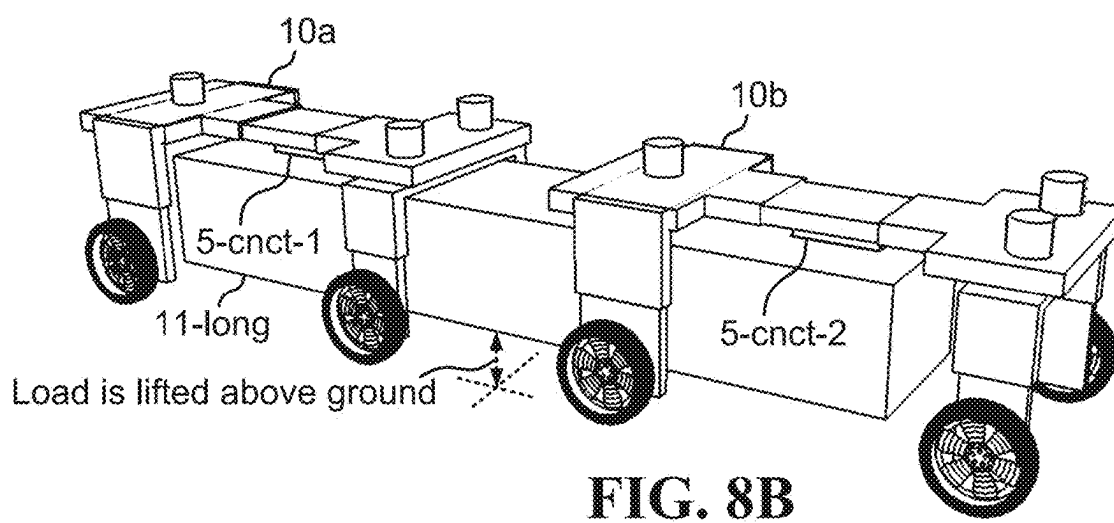
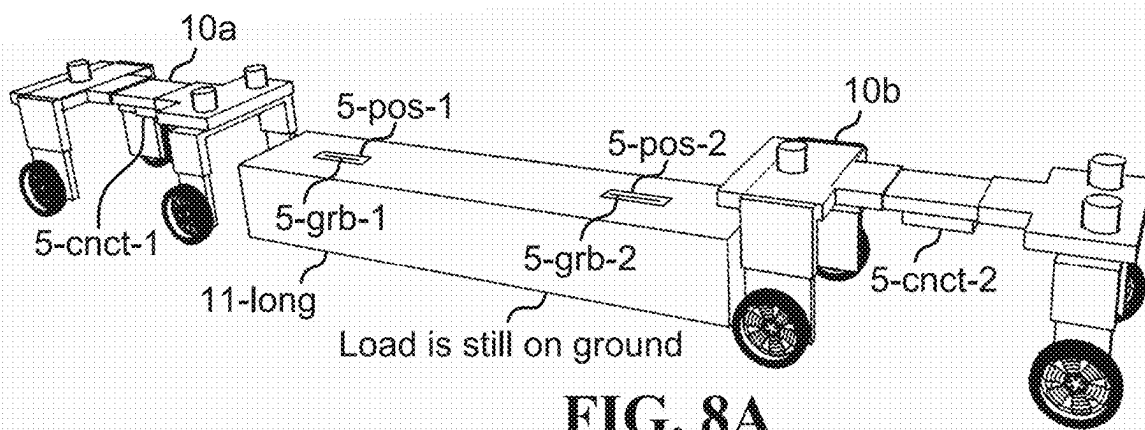


FIG. 7G



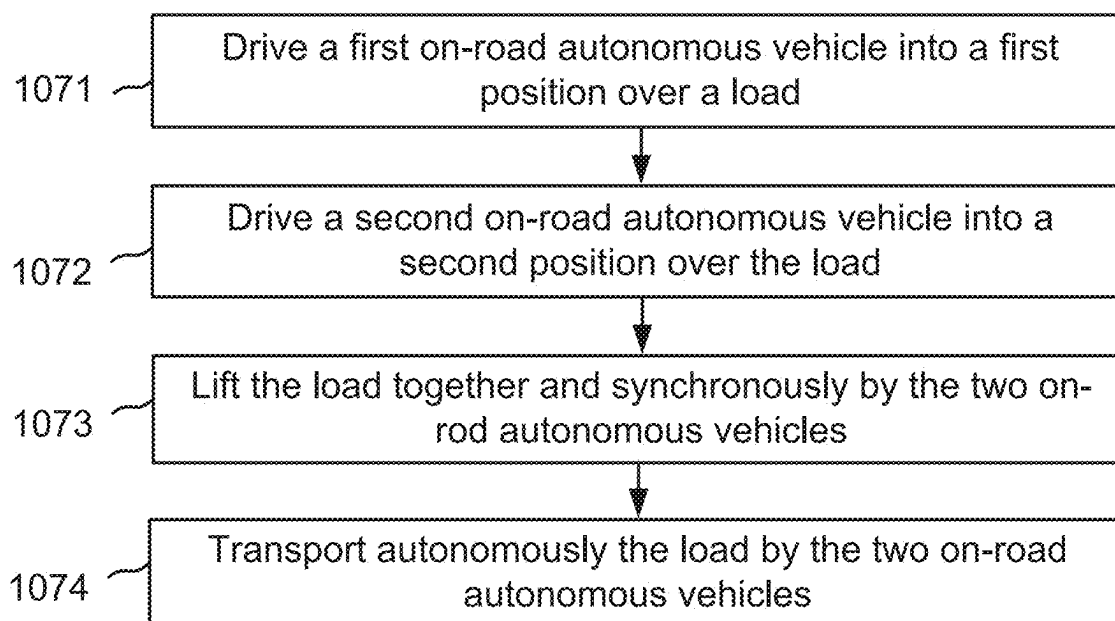
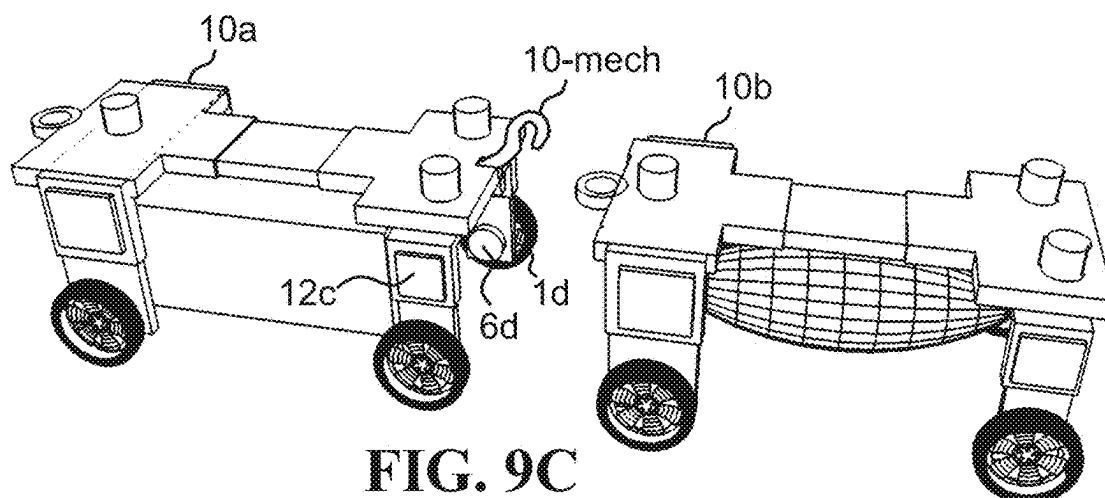
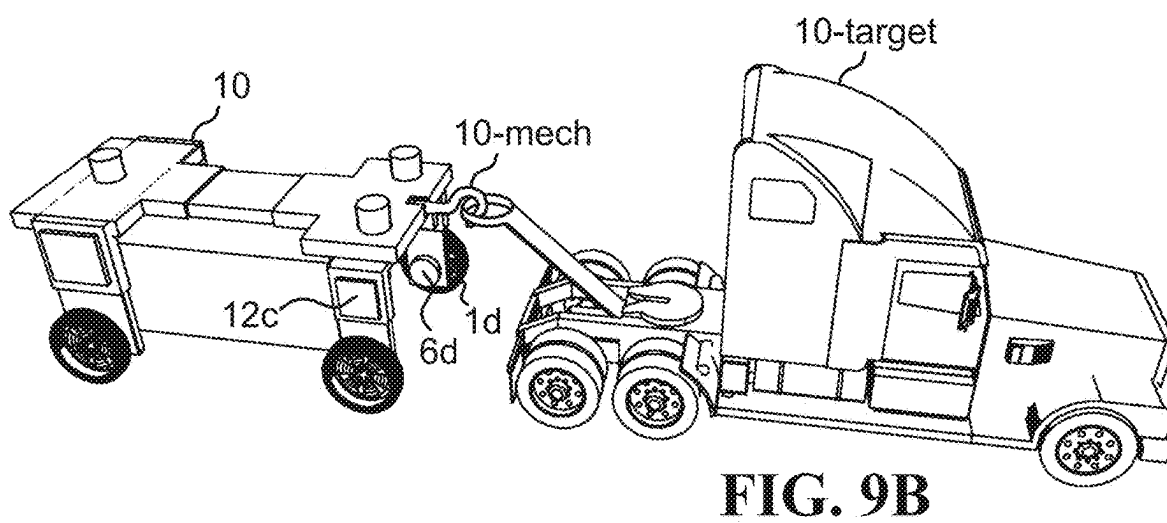
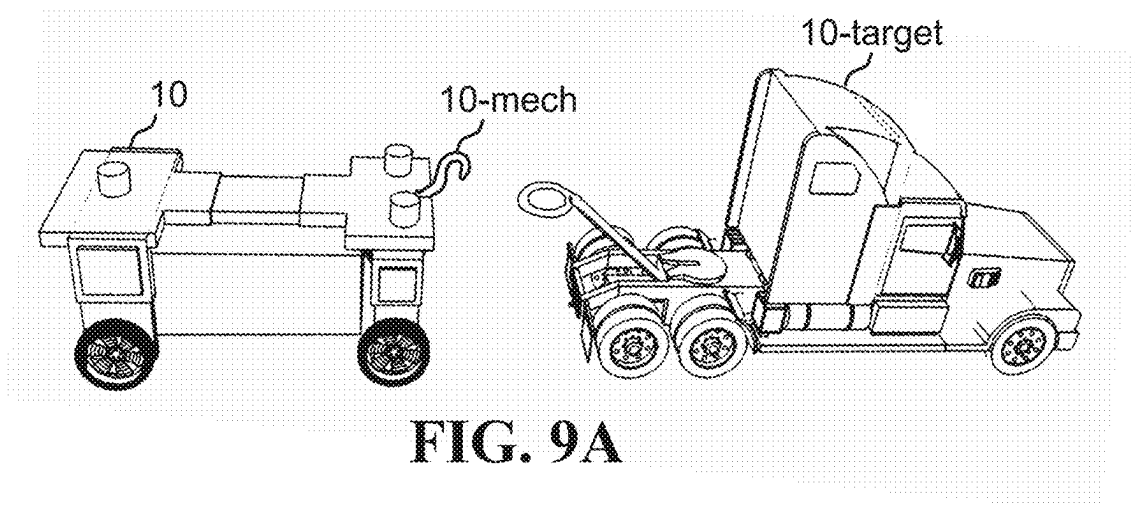


FIG. 8D



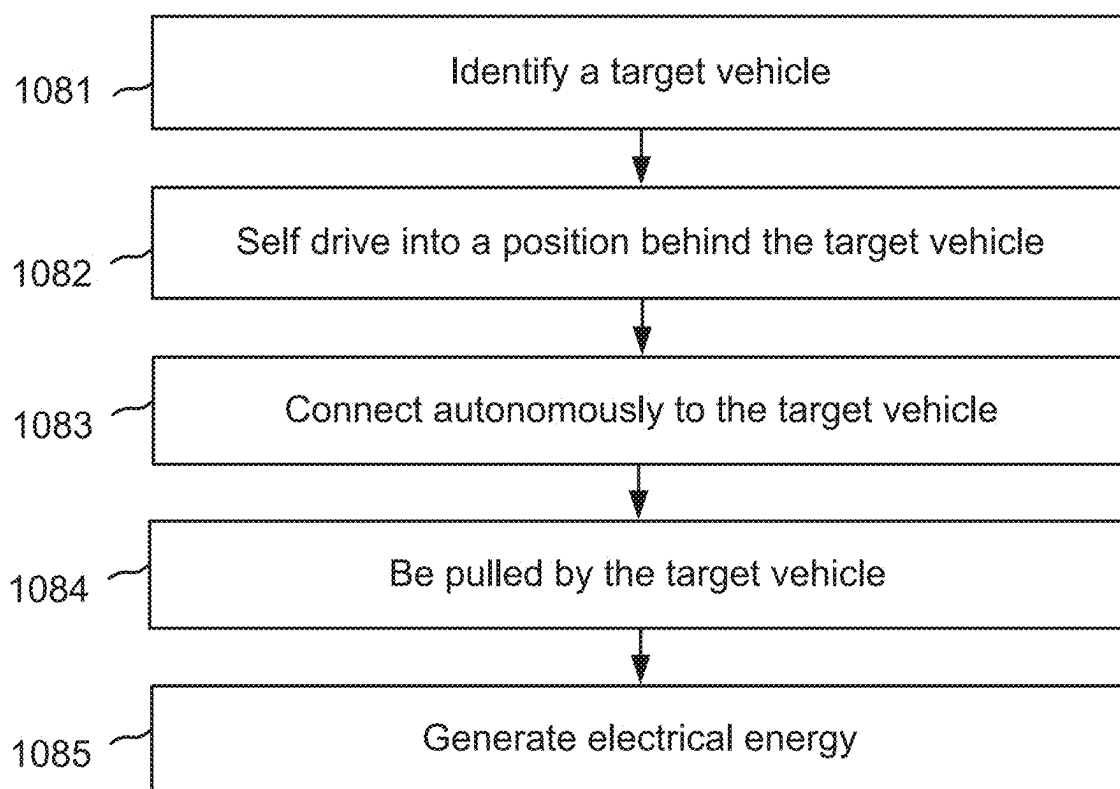


FIG. 9D

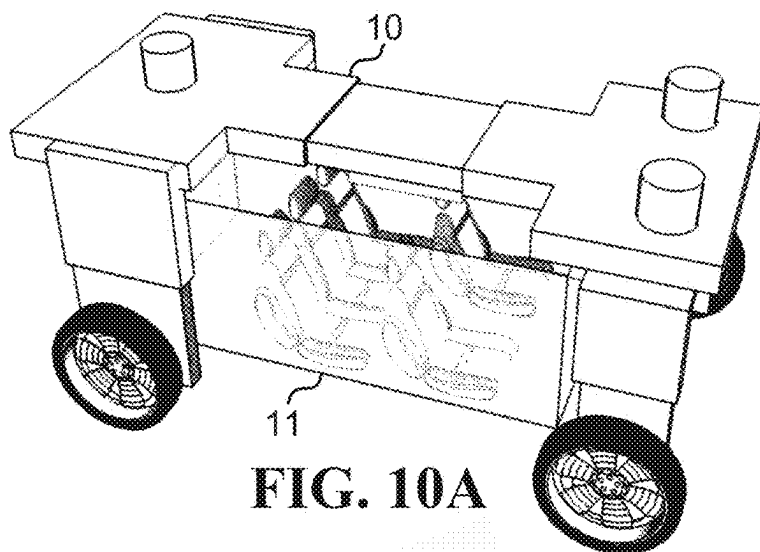


FIG. 10A

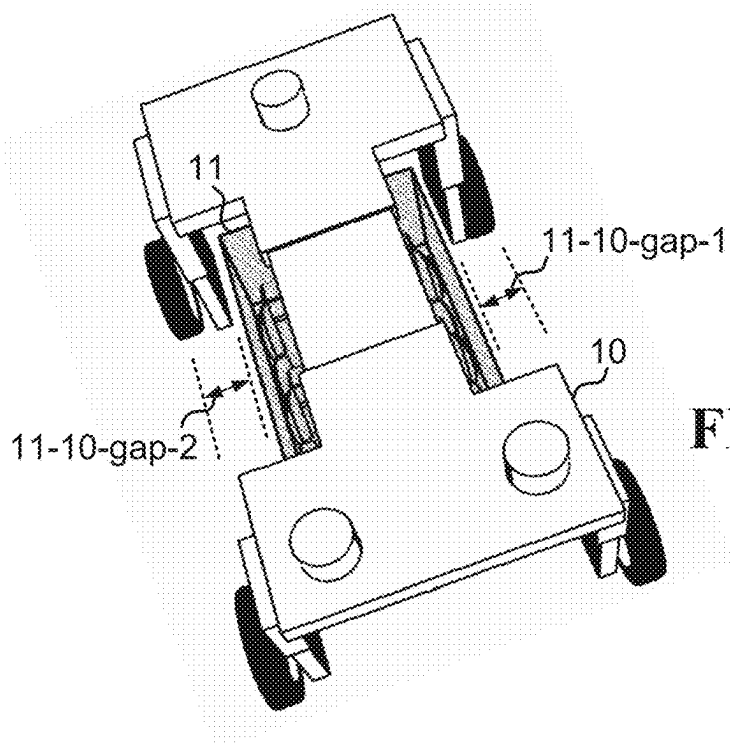


FIG. 10B

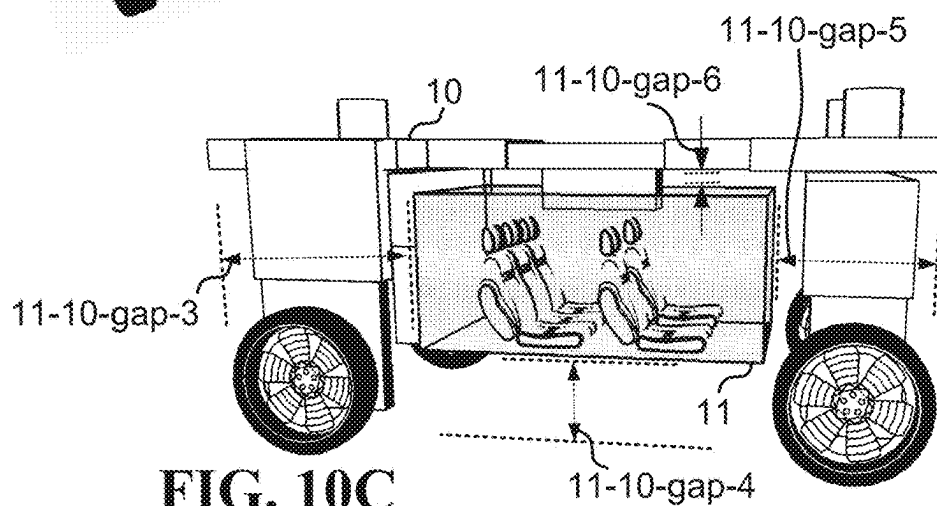
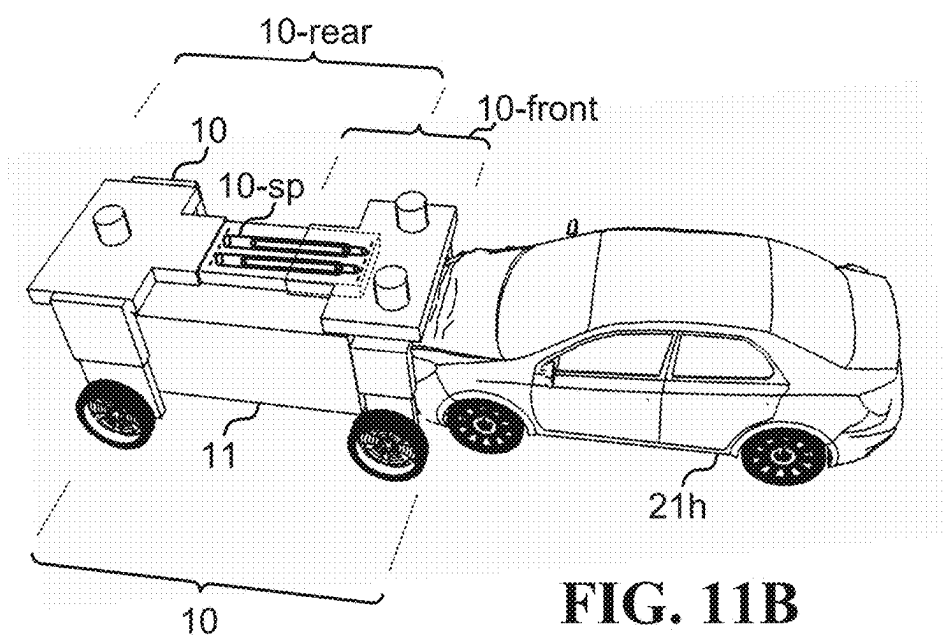
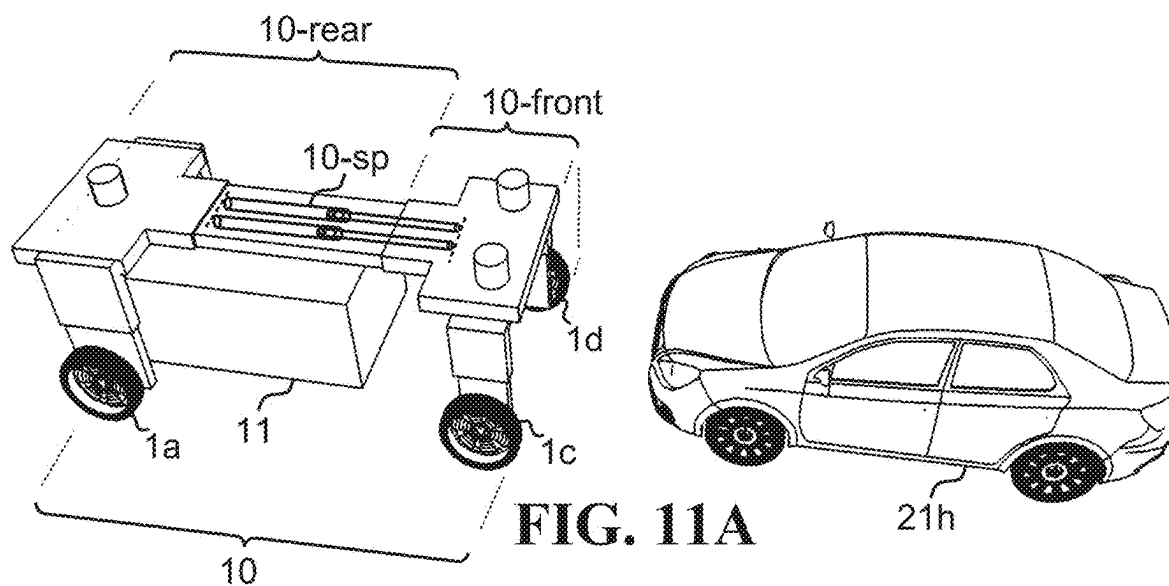


FIG. 10C



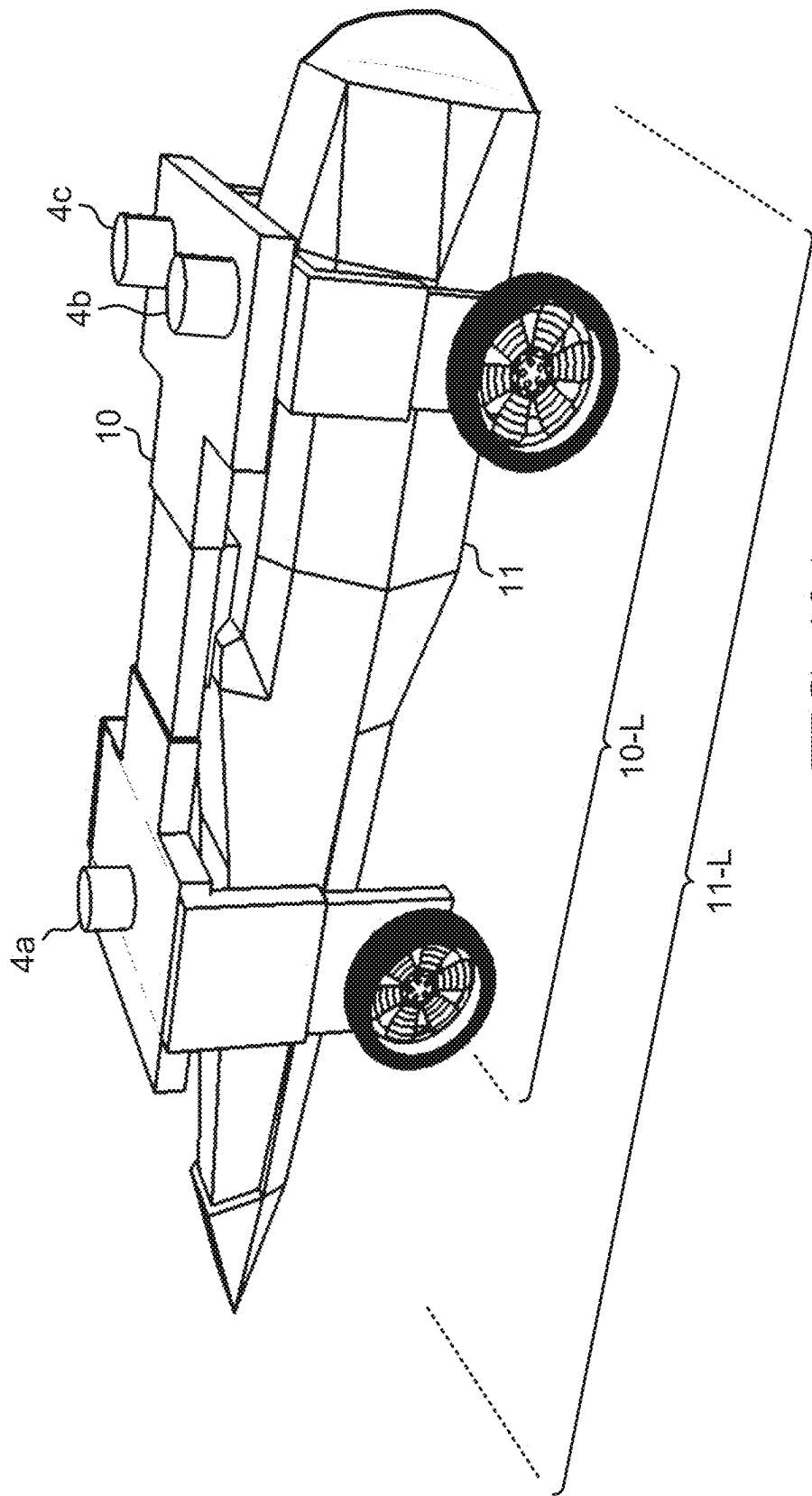


FIG. 12A

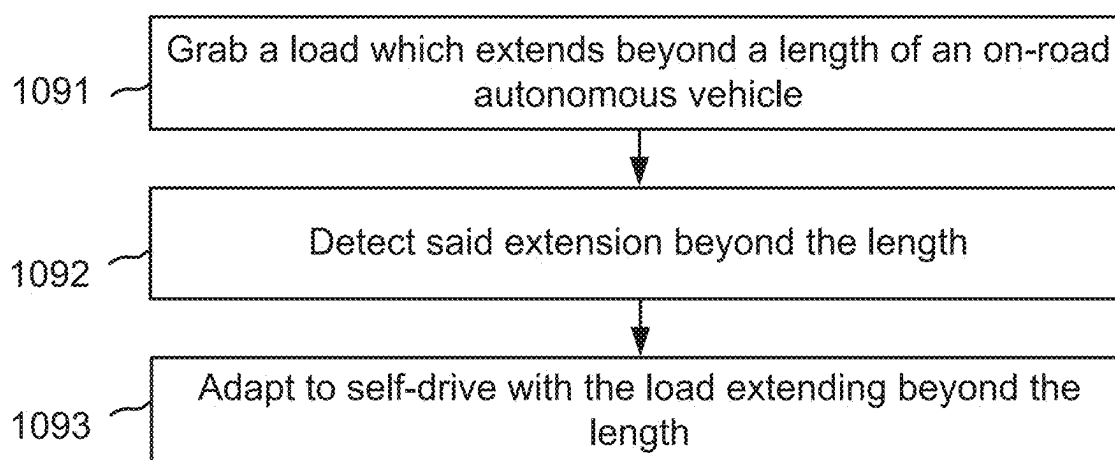
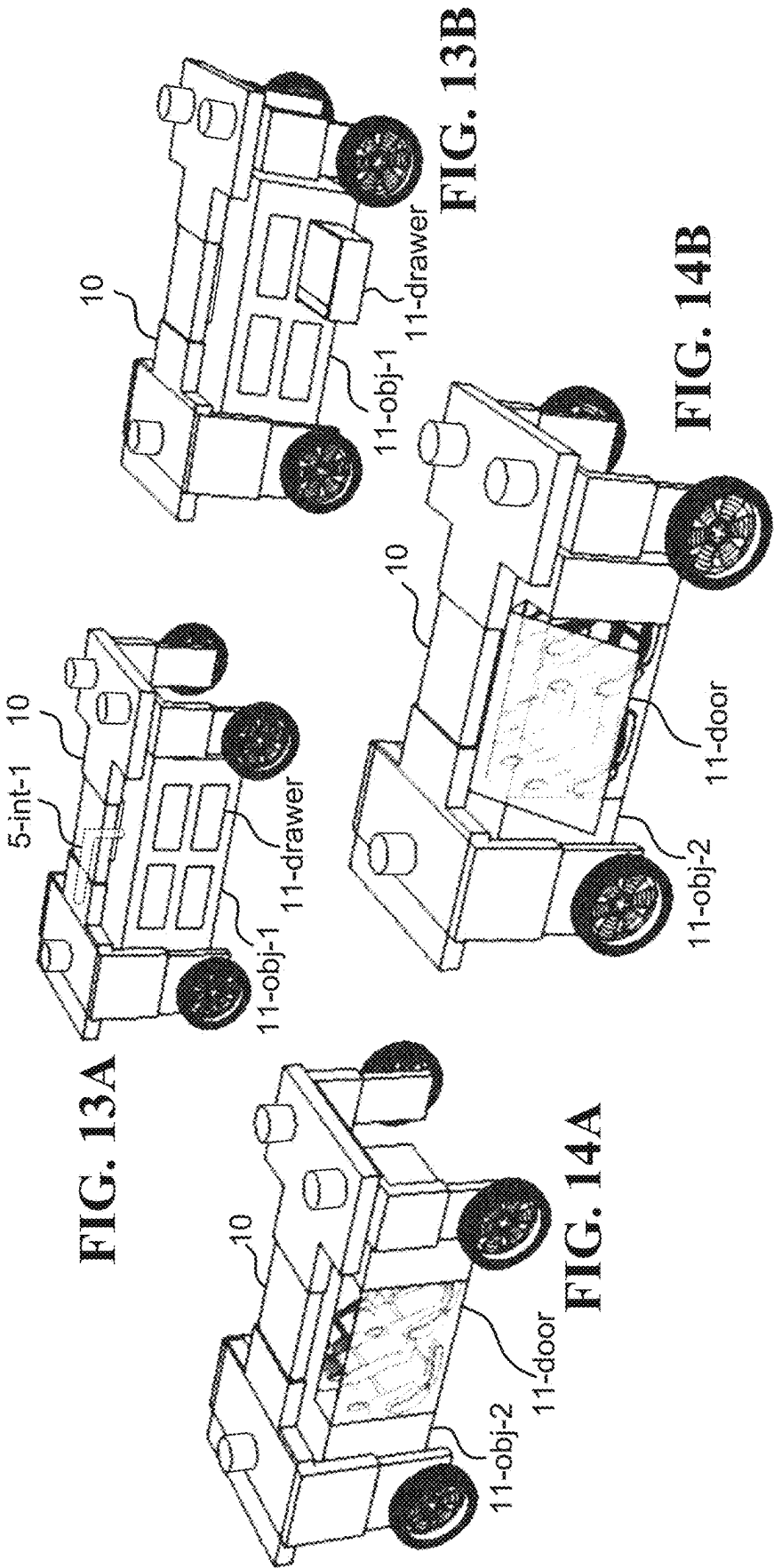


FIG. 12B



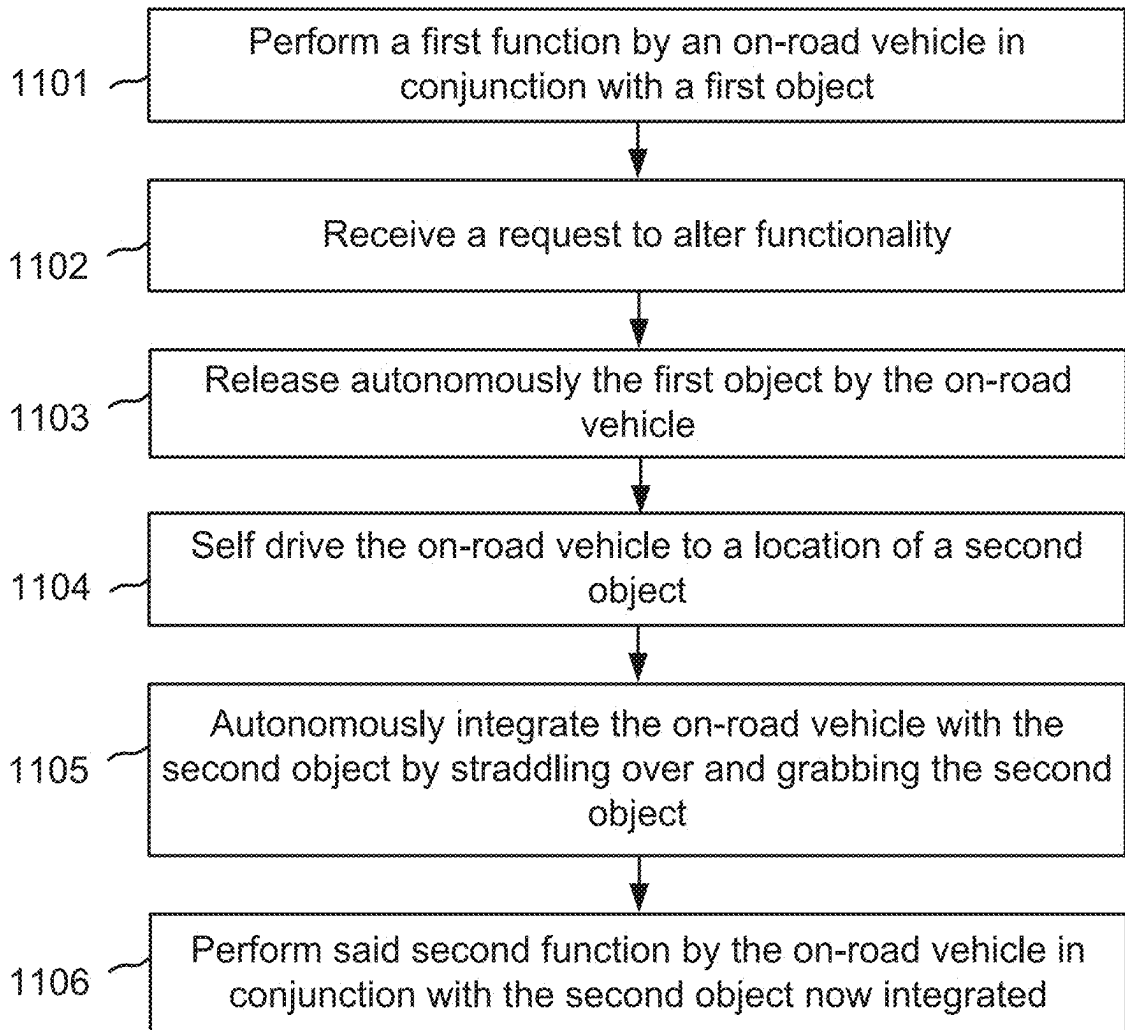


FIG. 15

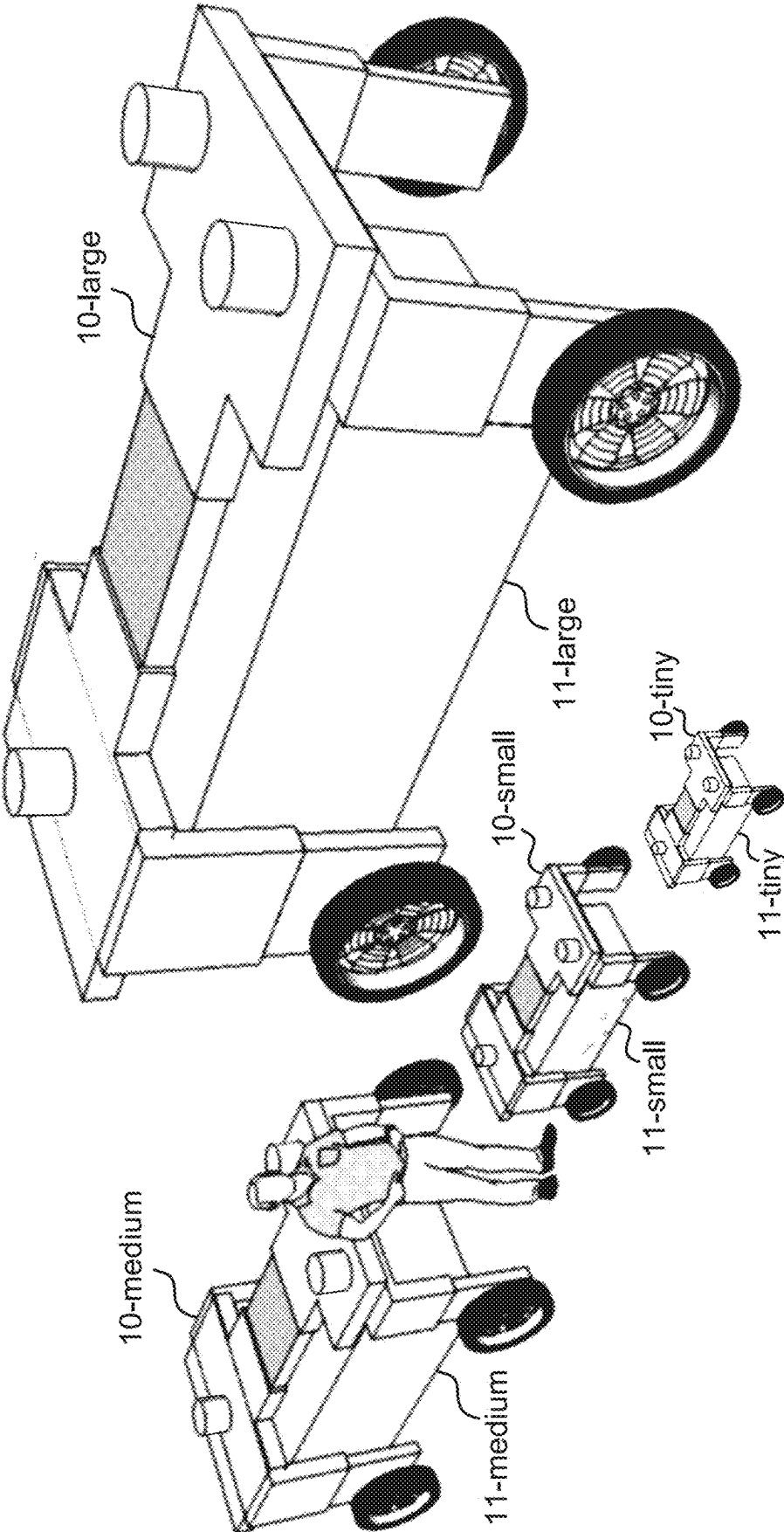


FIG. 16

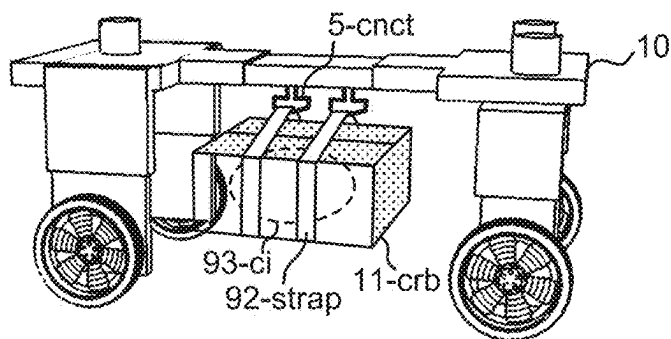


FIG. 17A

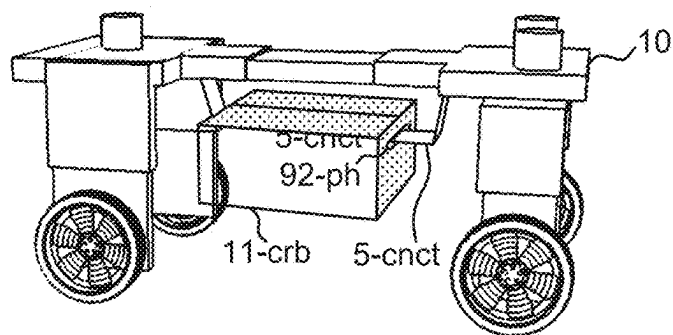


FIG. 17B

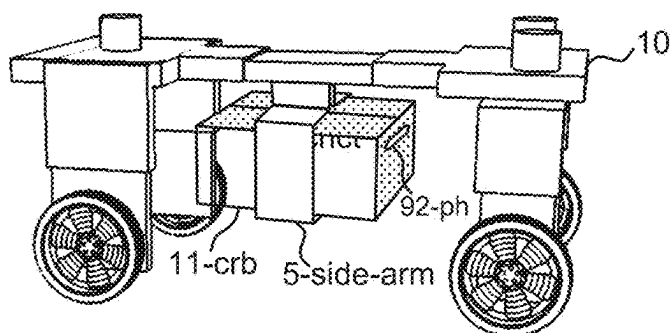


FIG. 17C

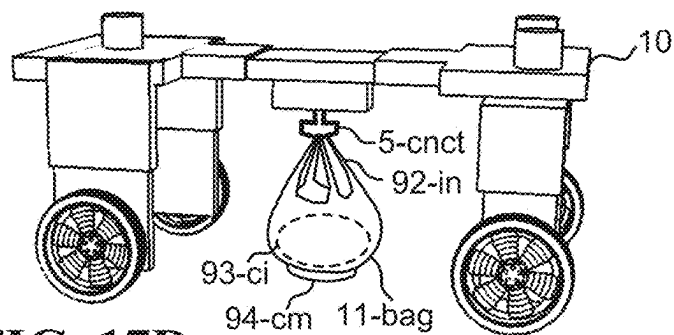


FIG. 17D

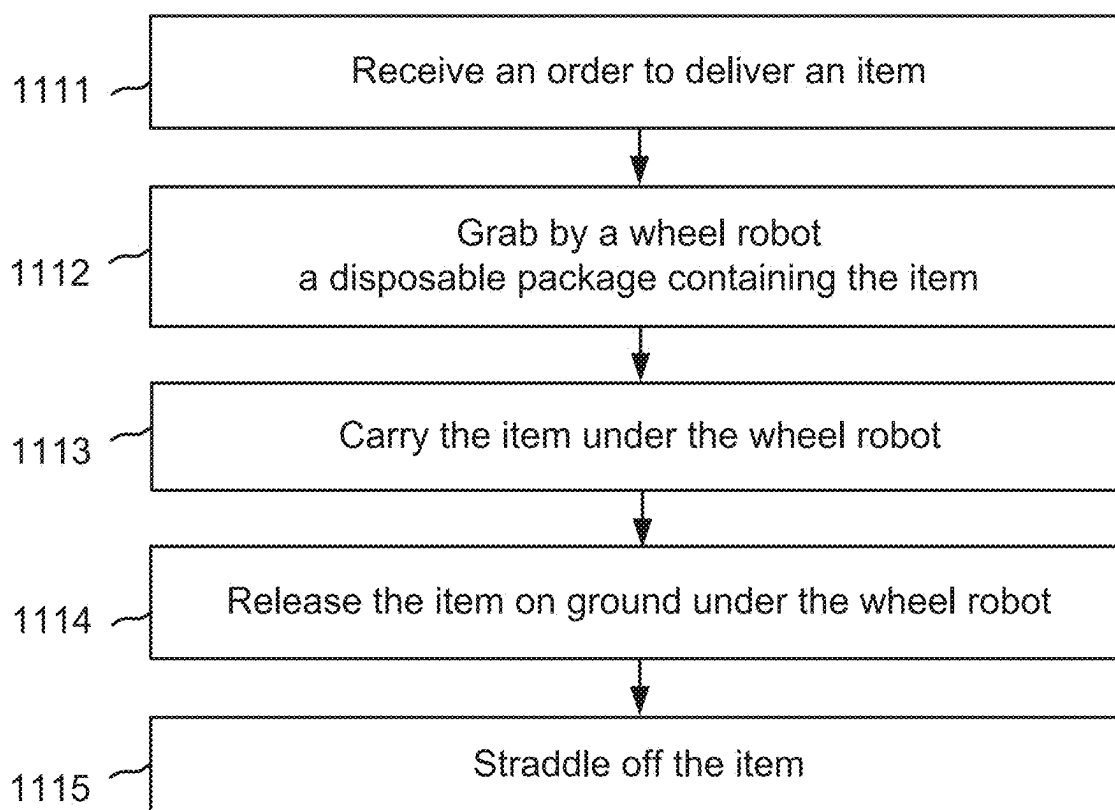


FIG. 18

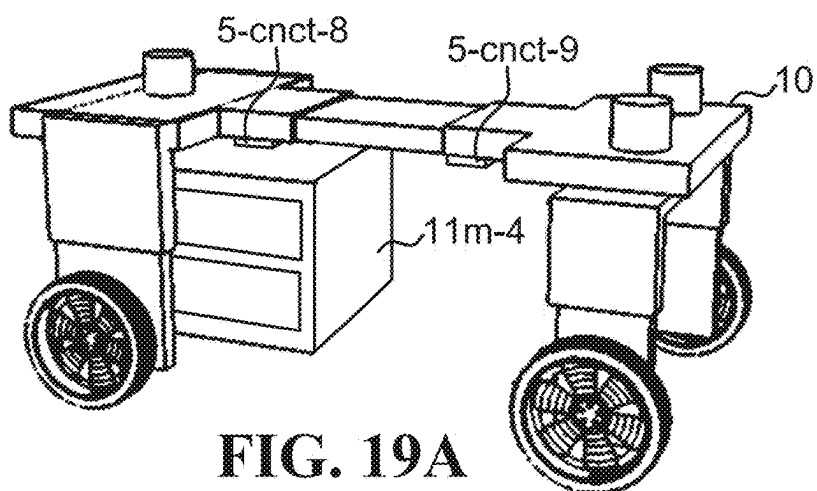


FIG. 19A

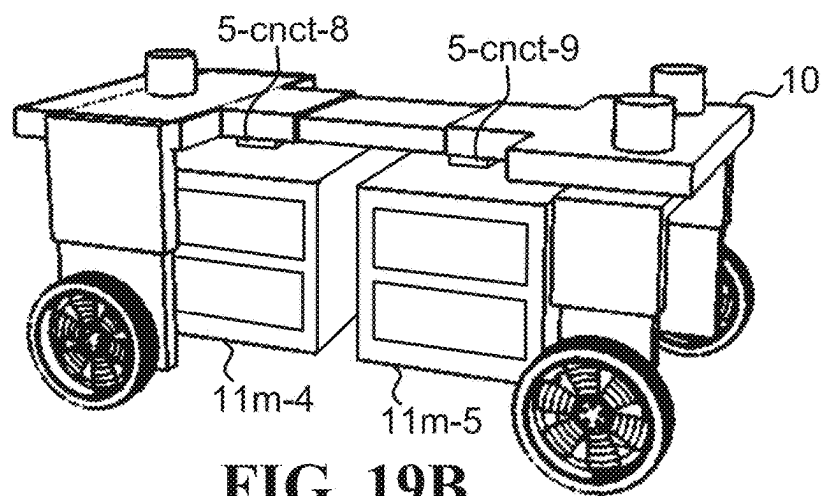


FIG. 19B

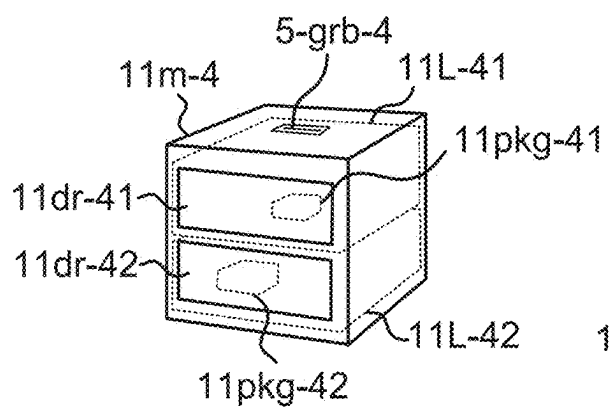


FIG. 19C

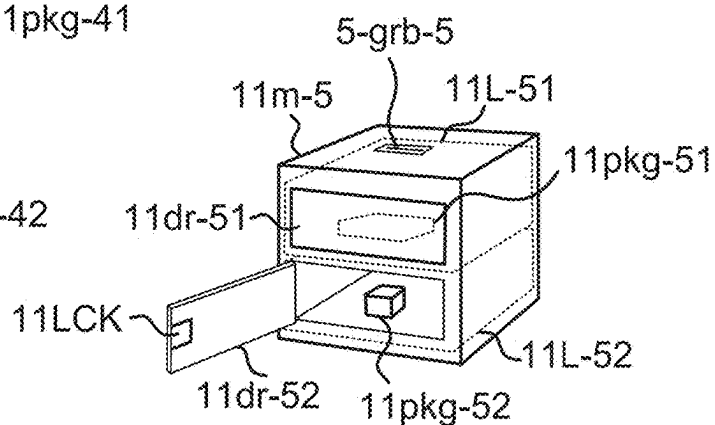
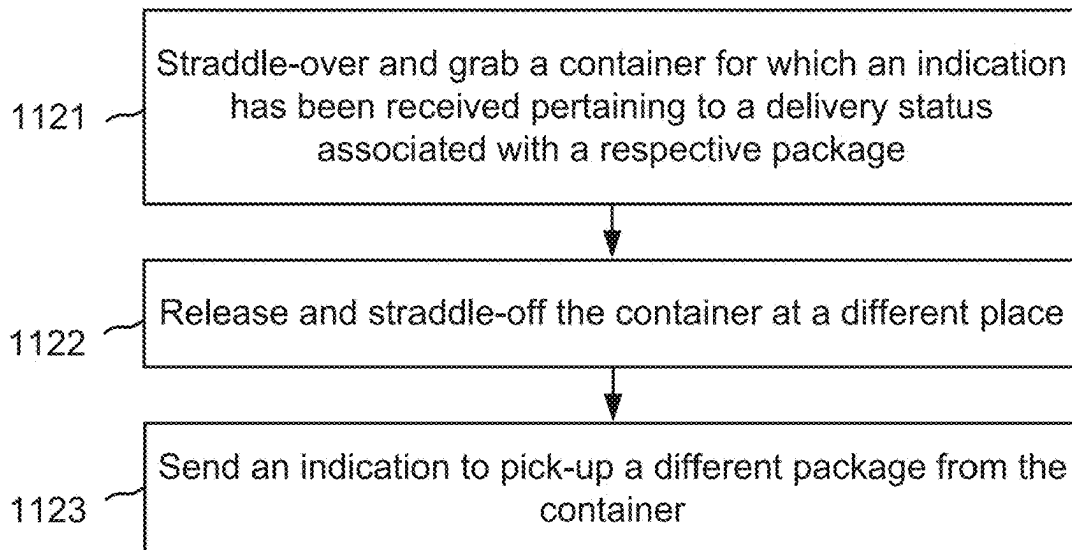
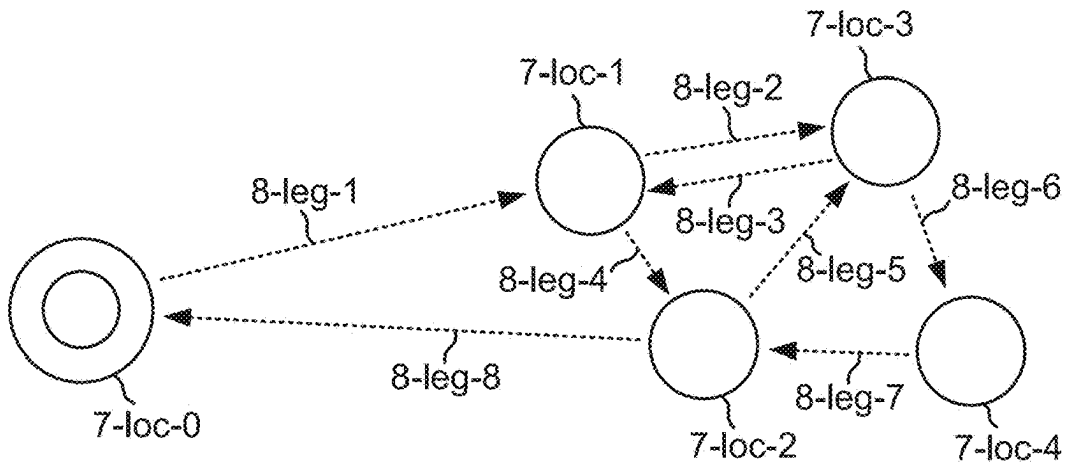
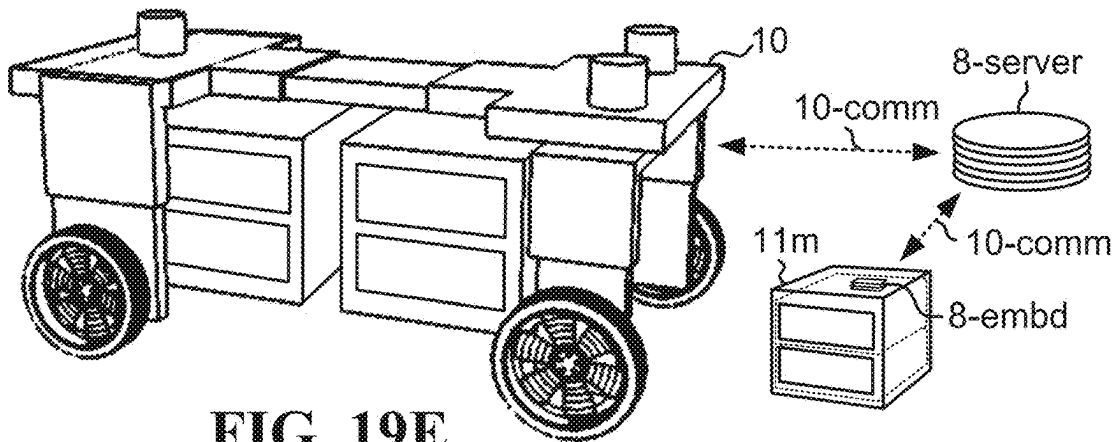


FIG. 19D



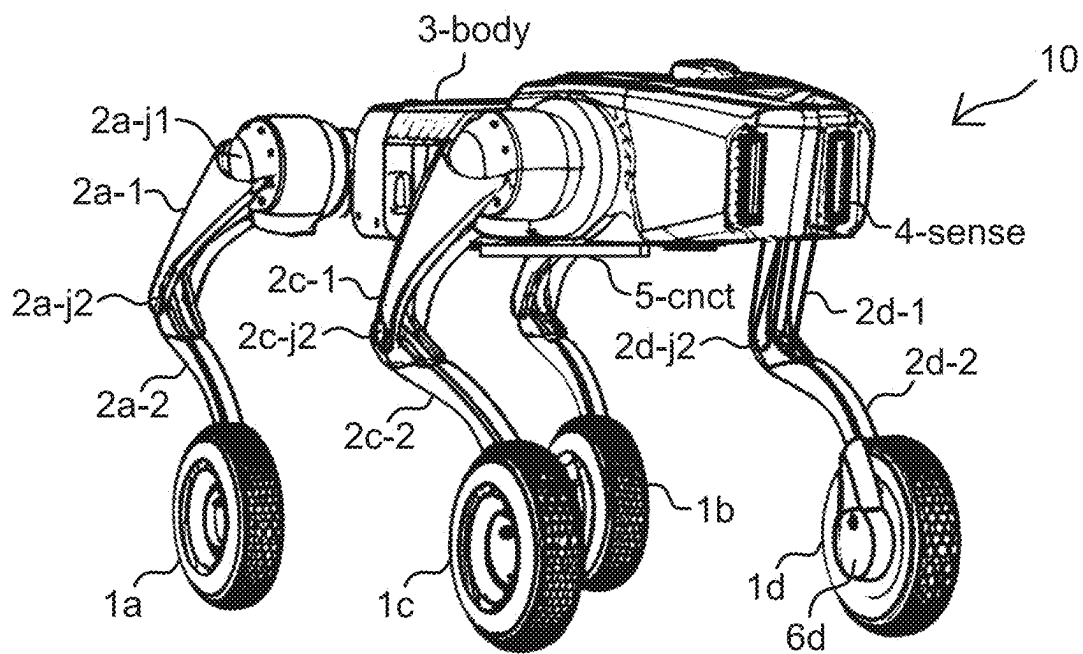


FIG. 21A

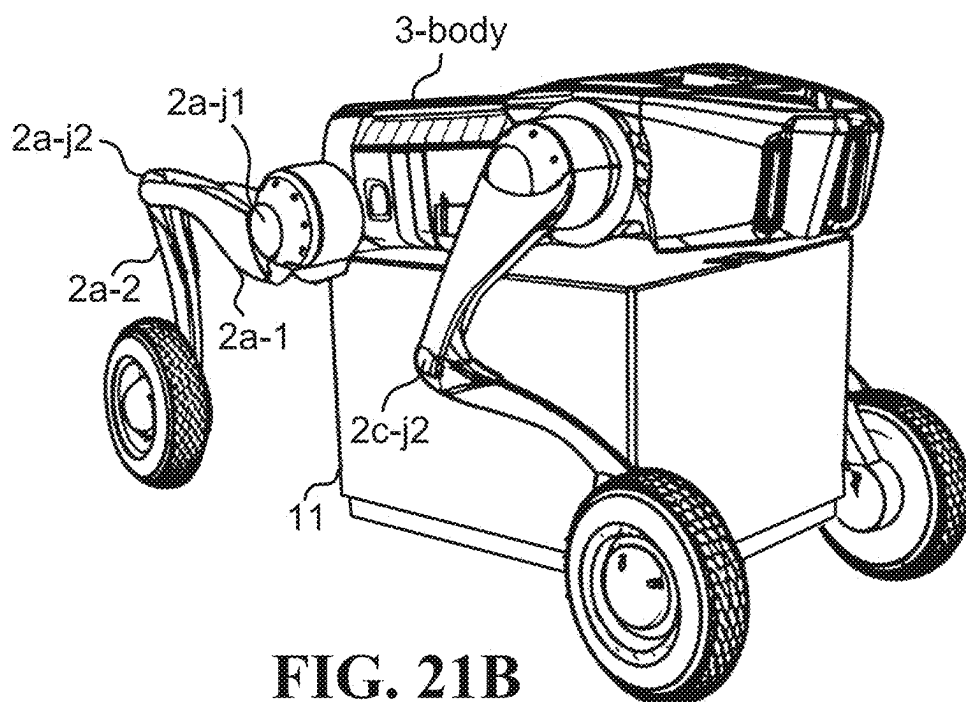


FIG. 21B

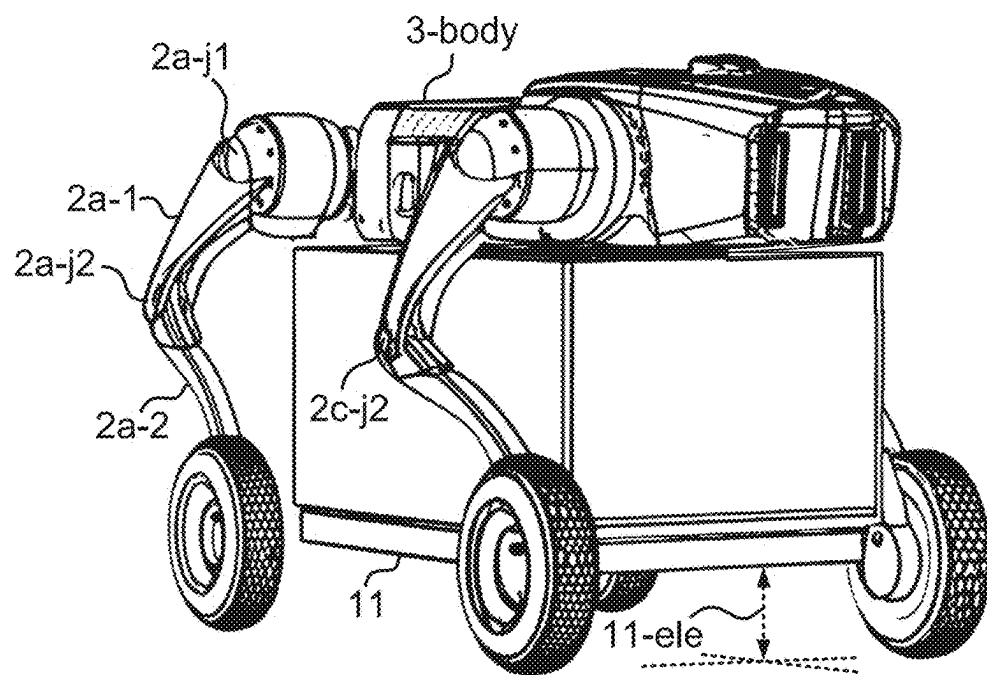


FIG. 21C

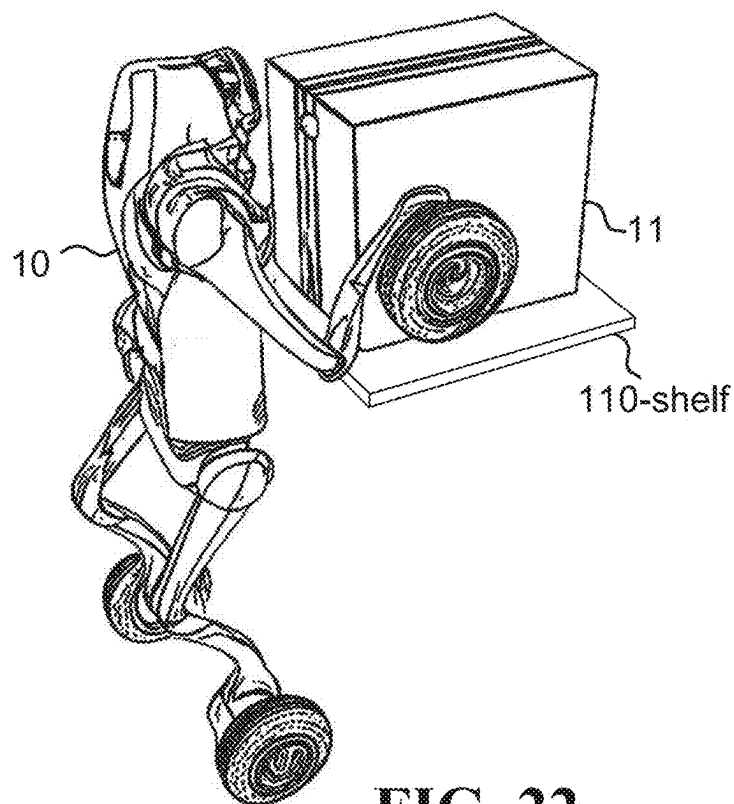


FIG. 22

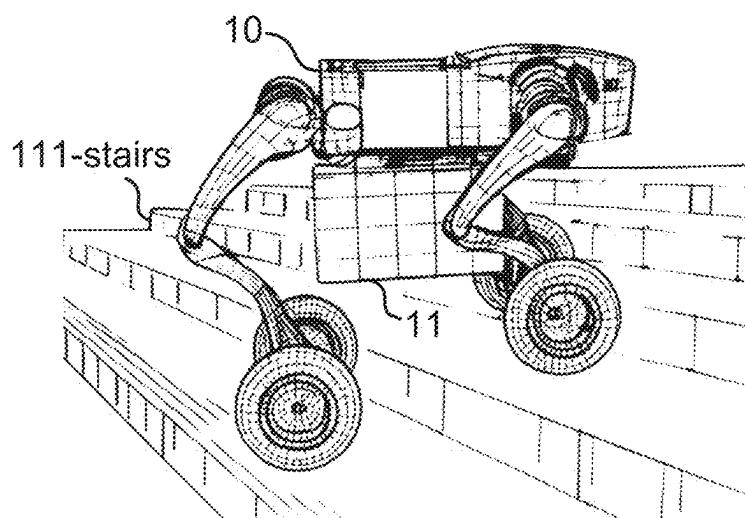


FIG. 23

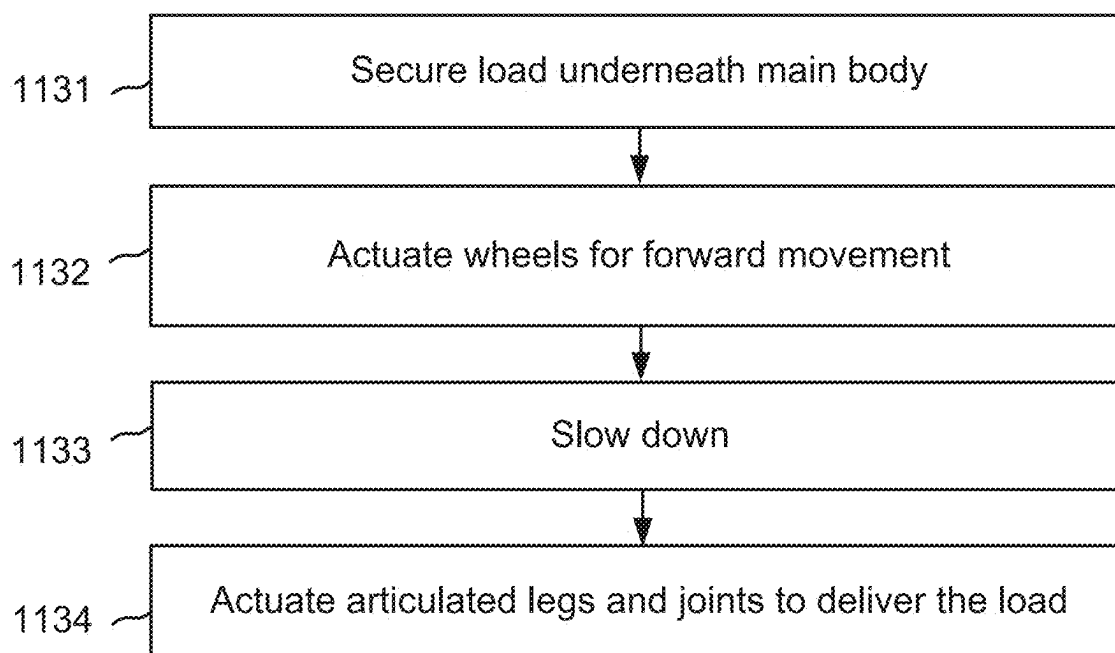


FIG. 24

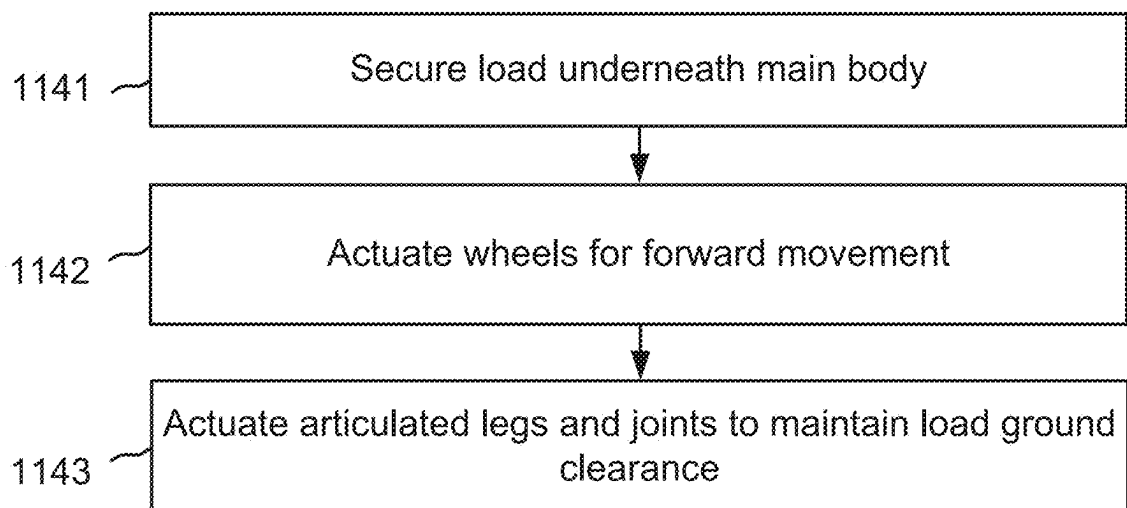


FIG. 25

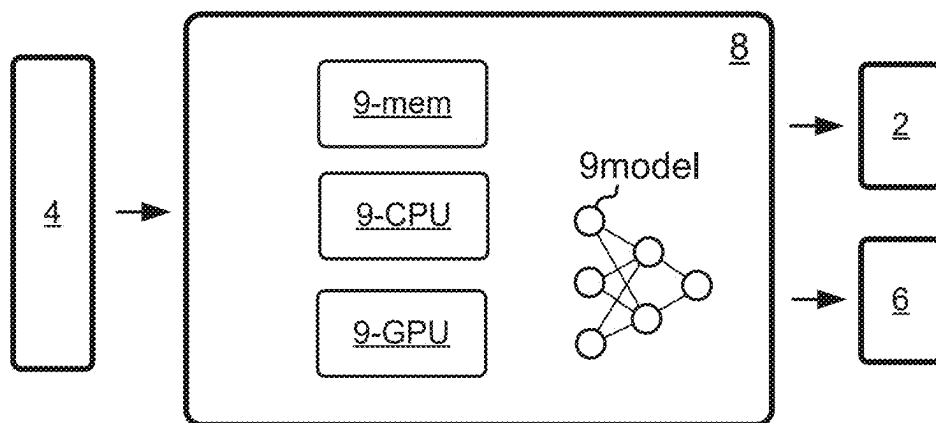


FIG. 26

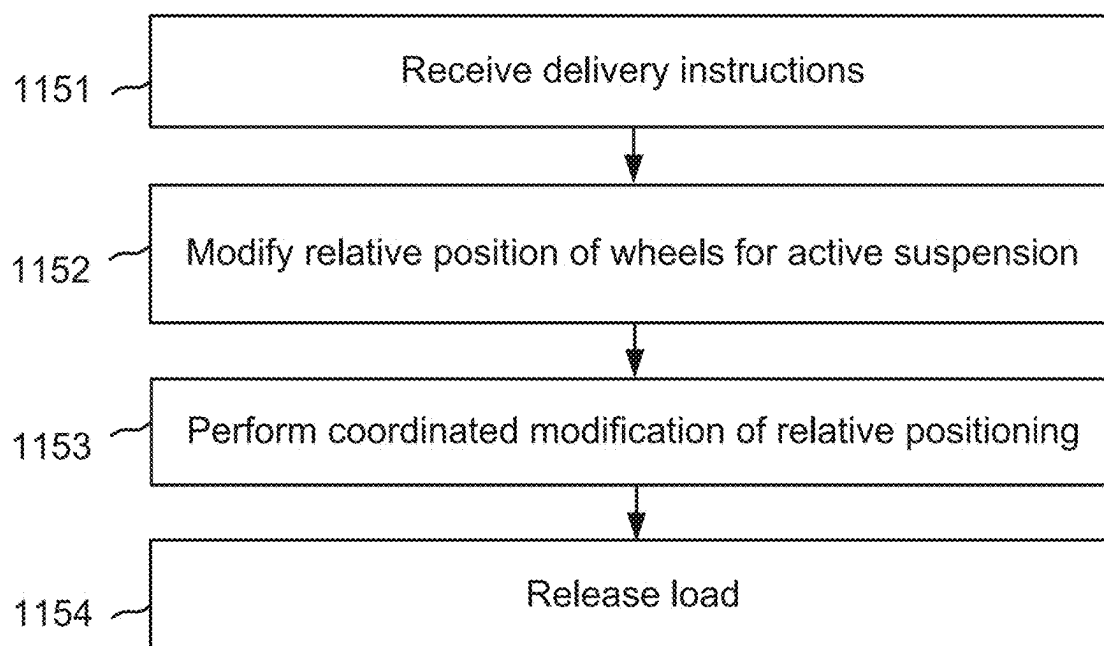


FIG. 27

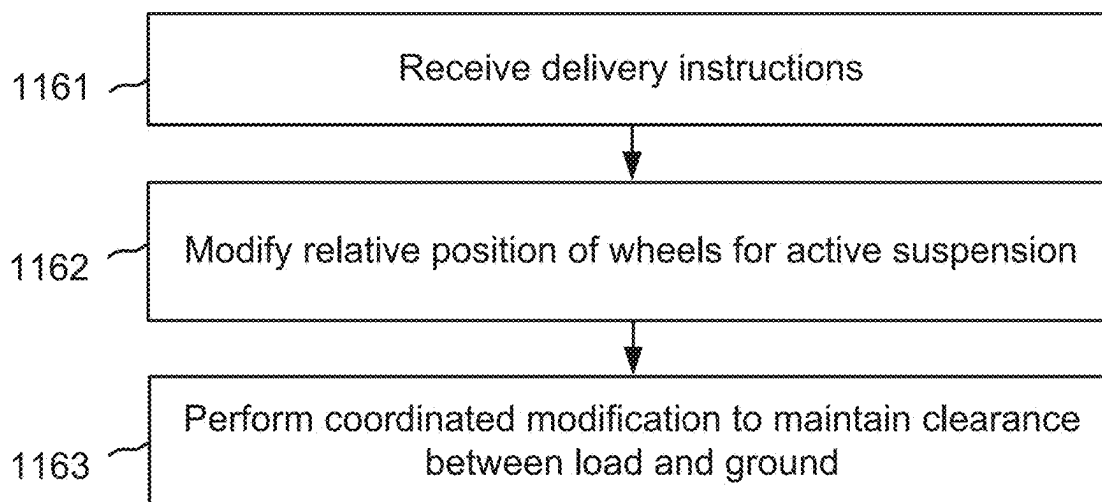


FIG. 28

AUTONOMOUS VEHICLE WITH ACTIVE WHEEL POSITIONING FOR SUSPENSION AND LOAD MANIPULATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This Application is a continuation-in-part of patent application Ser. No. 18/105,889, filed on Feb. 6, 2023, which is a continuation-in-part of patent application Ser. No. 17/246,745, filed on May 3, 2021, now U.S. Pat. No. 11,599,845, which is a continuation-in-part of patent application Ser. No. 16/502,834, filed on Jul. 3, 2019, now U.S. Pat. No. 11,046,230, which is a continuation-in-part of patent application Ser. No. 15/898,473, filed on Feb. 17, 2018, now U.S. Pat. No. 10,384,871, which claims priority to U.S. Provisional Patent Application No. 62/473,436, filed on Mar. 19, 2017.

BACKGROUND

[0002] Autonomous delivery systems are increasingly being developed to automate logistics. However, current ground-based autonomous delivery vehicles often rely on wheels, limiting their operational range to relatively smooth and level terrain. This restricts access to destinations involving stairs, curbs, or significantly uneven ground. While legged robots offer superior mobility over complex terrain, they can be less energy-efficient and slower than wheeled vehicles on flat surfaces.

SUMMARY

[0003] One embodiment is an autonomous vehicle, comprising: a main body structure; a plurality of wheels; a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is configured to actively modify a relative positioning between said wheel and the main body structure; and a grabbing mechanism coupled to an underside of the main body structure, the grabbing mechanism configured to hold a load beneath the main body structure and above ground; wherein the autonomous vehicle is configured to control the plurality of powered actuation systems such that said active modification of the relative positioning between each of the wheels and the main body structure serves dual purposes of (i) providing active suspension for each said wheel at least when the vehicle is in forward motion and (ii) enabling coordinated modification of said relative positioning so as to manipulate the load relative to the ground.

[0004] One embodiment is a method for autonomously delivering a load, the method comprising: receiving, by a control system of an autonomous vehicle comprising a main body structure and a plurality of wheels, instructions associated with delivering the load to a drop-off location; transporting the load, held by a grabbing mechanism coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active suspension is provided by a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is actively modifying a relative positioning between said wheel and the main body structure; upon arrival at the drop-off location, performing, via control of said plurality of powered actuation systems, a coordinated

modification of said relative positioning between each of the wheels and the main body structure to manipulate the load relative to a ground surface; and operating the grabbing mechanism to release the load.

[0005] One embodiment is a method for autonomously delivering a load, the method comprising: receiving, by a control system of an autonomous vehicle comprising a main body structure and a plurality of wheels, instructions associated with delivering the load to a drop-off location; transporting the load, held by a grabbing mechanism coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active suspension is provided by a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is actively modifying a relative positioning between said wheel and the main body structure; and coordinately controlling, via the control system during at least a portion of said transporting, said plurality of powered actuation systems for maintaining a clearance between a bottom surface of the load and a ground surface.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The embodiments are herein described by way of example only, with reference to the accompanying drawings. No attempt is made to show structural details of the embodiments in more detail than is necessary for a fundamental understanding of the embodiments. In the drawings:

[0007] FIG. 1A illustrates one embodiment of an on-road autonomous vehicle operative to autonomously collect and transport a load over public roads;

[0008] FIG. 1B illustrates one embodiment of the on-road autonomous vehicle as seen from a front view;

[0009] FIG. 1C illustrates one embodiment of the on-road autonomous vehicle as seen from a side view;

[0010] FIG. 1D illustrates one embodiment of the on-road autonomous vehicle as seen from below;

[0011] FIG. 2A illustrates one embodiment of an on-road autonomous vehicle self-driving without a load over a public road and alongside regular traffic;

[0012] FIG. 2B illustrates one embodiment of the on-road autonomous vehicle arriving at a parking lot in which a load is parked;

[0013] FIG. 2C illustrates one embodiment of the on-road autonomous vehicle getting ready to pick up the load;

[0014] FIG. 3A illustrates one embodiment of an on-road autonomous vehicle getting ready to pick up a load;

[0015] FIG. 3B illustrates one embodiment of the on-road autonomous vehicle moving/straddling over the load;

[0016] FIG. 3C illustrates one embodiment of the on-road autonomous vehicle getting into a lower position and grabbing the load;

[0017] FIG. 3D illustrates one embodiment of the on-road autonomous vehicle picking up the load;

[0018] FIG. 4A illustrates one embodiment of an on-road autonomous vehicle driving away with a load that was previously parked in a parking lot;

[0019] FIG. 4B illustrates one embodiment of the on-road autonomous vehicle self-driving with the load over a public road and alongside regular traffic;

[0020] FIG. 4C illustrates one embodiment of a method for autonomously collecting and transporting a load over public roads;

[0021] FIG. 4D illustrates one embodiment of another method for autonomously collecting and transporting a load over public roads;

[0022] FIG. 5A illustrates one embodiment of an on-road autonomous vehicle self-driving to a location in which a passenger in a cabin is located;

[0023] FIG. 5B illustrates one embodiment of the passenger in the cabin awaiting arrival of the on-road autonomous vehicle;

[0024] FIG. 5C illustrates one embodiment of the on-road autonomous vehicle picking up the cabin with the passenger;

[0025] FIG. 5D illustrates one embodiment of a method for autonomously collecting and transporting a passenger in a cabin;

[0026] FIG. 5E illustrates one embodiment of a method for requesting autonomous collection and transporting of a passenger in a cabin;

[0027] FIG. 6A illustrates one embodiment of an on-road autonomous vehicle self-driving to a location in which a functional load is located;

[0028] FIG. 6B illustrates one embodiment of the functional load awaiting arrival of the on-road autonomous vehicle;

[0029] FIG. 6C illustrates one embodiment of the on-road autonomous vehicle picking up and transporting the functional load;

[0030] FIG. 6D illustrates one embodiment of the functional load after being placed by the on-road autonomous vehicle at a particular location operative to work in conjunction with or support the functional load;

[0031] FIG. 6E illustrates one embodiment of the on-road autonomous vehicle driving away after placing the functional load at the particular location;

[0032] FIG. 6F illustrates one embodiment of a method for autonomously collecting transporting and placing a functional load according to a request;

[0033] FIG. 7A illustrates one embodiment of an on-road autonomous vehicle self-driving to a location in which a functional load is located;

[0034] FIG. 7B illustrates one embodiment of the on-road autonomous vehicle picking up the functional load and using a function associated with the functional load;

[0035] FIG. 7C illustrates one embodiment of a convoy of several on-road autonomous vehicles in which one of the on-road autonomous vehicles is carrying a functional load;

[0036] FIG. 7D illustrates one embodiment of the convoy of several on-road autonomous vehicles in which the on-road autonomous vehicles switch at least some of the loads between themselves so as to pass the functional load from one of the on-road autonomous vehicles to another of the on-road autonomous vehicles;

[0037] FIG. 7E illustrates one embodiment of the convoy of several on-road autonomous vehicles in which another of the on-road autonomous vehicles is now carrying the functional load;

[0038] FIG. 7F illustrates one embodiment of a method for autonomously collecting and using a functional load;

[0039] FIG. 7G illustrates one embodiment of a method for exchanging a functional load between at least two on-road autonomous vehicles in a convoy;

[0040] FIG. 8A illustrates one embodiment of two on-road autonomous vehicles getting into positions in conjunction with a load that is too big to be carried by only one on-road autonomous vehicle;

[0041] FIG. 8B illustrates one embodiment of the two on-road autonomous vehicles cooperatively lifting the load;

[0042] FIG. 8C illustrates one embodiment of the two on-road autonomous vehicles cooperatively lifting another load;

[0043] FIG. 8D illustrates one embodiment of a method for cooperatively lifting and transporting a load by at least two on-road autonomous vehicles;

[0044] FIG. 9A illustrates one embodiment of an on-road autonomous vehicle getting into position behind a target vehicle;

[0045] FIG. 9B illustrates one embodiment of the on-road autonomous vehicle now mechanically connected to the target vehicle that pulls the on-road autonomous vehicle thereby allowing for self generation of electrical energy in the on-road autonomous vehicle;

[0046] FIG. 9C illustrates one embodiment of an on-road autonomous vehicle getting into position behind another on-road autonomous vehicle in order to be pulled thereby;

[0047] FIG. 9D illustrates one embodiment of a method for charging batteries of an on-road autonomous vehicle on the move;

[0048] FIG. 10A illustrates one embodiment of an on-road autonomous vehicle carrying a passenger cabin;

[0049] FIG. 10B illustrates one embodiment of the on-road autonomous vehicle providing air gap protection to the passenger cabin;

[0050] FIG. 10C illustrates one embodiment of the on-road autonomous vehicle providing further air gap protection to the passenger cabin;

[0051] FIG. 11A illustrates one embodiment of an on-road autonomous vehicle about to be hit by a foreign object;

[0052] FIG. 11B illustrates one embodiment of the on-road autonomous vehicle being hit by the foreign object;

[0053] FIG. 12A illustrates one embodiment of an on-road autonomous vehicle carrying a load having a certain aerodynamic design and extending beyond a length of the on-road autonomous vehicle;

[0054] FIG. 12B illustrates one embodiment of a method for adjusting an on-road autonomous vehicle to carry a long load;

[0055] FIG. 13A illustrates one embodiment of an on-road autonomous vehicle currently integrated with a first object having drawers that are presently closed;

[0056] FIG. 13B illustrates one embodiment of the on-road autonomous vehicle still integrated with the first object and getting one of the drawers opened;

[0057] FIG. 14A illustrates one embodiment of an on-road autonomous vehicle currently integrated with a second object having a door that is presently closed;

[0058] FIG. 14B illustrates one embodiment of the on-road autonomous vehicle still integrated with the second object and getting the door opened;

[0059] FIG. 15 illustrates one embodiment of a method for autonomously altering functionality of an on-road vehicle;

[0060] FIG. 16 illustrates several embodiments of on-road autonomous vehicles having several different sizes respectively;

[0061] FIG. 17A illustrates one embodiment of a wheel robot carrying a disposable package using straps;

[0062] FIG. 17B illustrates one embodiment of a wheel robot carrying a disposable package by inserting grabbing elements into handle holes embedded in the package;

[0063] FIG. 17C illustrates one embodiment of a wheel robot carrying a disposable package using side arms operative to move sideways to release the package;

[0064] FIG. 17D illustrates one embodiment of a wheel robot carrying a disposable package comprising a bag;

[0065] FIG. 18 illustrates one embodiment of a method for autonomously delivering consumer items in disposable packages;

[0066] FIG. 19A illustrates one embodiment of an on-road autonomous vehicle carrying a first package container;

[0067] FIG. 19B illustrates one embodiment of the on-road autonomous vehicle carrying a first package container and a second package container at the same time and in tandem;

[0068] FIG. 19C illustrates one embodiment of the first package container comprising a plurality of lockers containing a respective plurality of packages;

[0069] FIG. 19D illustrates one embodiment of the second package container comprising a plurality of lockers containing a respective plurality of packages;

[0070] FIG. 19E illustrates one embodiment of the on-road autonomous vehicle and package containers being in communicative contact with a server;

[0071] FIG. 19F illustrates one embodiment of a route taken by the on-road autonomous vehicle in conjunction with moving and placing the package containers in different delivery locations;

[0072] FIG. 20 illustrates one embodiment of a method for autonomously delivering packages using a plurality of package containers;

[0073] FIG. 21A illustrates one embodiment of an autonomous vehicle with a legged-wheeled hybrid locomotion system;

[0074] FIG. 21B illustrates one embodiment of the autonomous vehicle adjusting its articulated legs to straddle over and lower its main body towards a load;

[0075] FIG. 21C illustrates one embodiment of the autonomous vehicle transporting the load underneath its main body at a certain elevation above the ground;

[0076] FIG. 22 illustrates one embodiment of the autonomous vehicle manipulating a load onto a shelf structure;

[0077] FIG. 23 illustrates one embodiment of the autonomous vehicle navigating stairs while carrying a load;

[0078] FIG. 24 illustrates a one embodiment of a method for delivering a load autonomously using the legged-wheeled vehicle;

[0079] FIG. 25 illustrates one embodiment of a method for transporting a load autonomously while actively maintaining ground clearance using the legged-wheeled vehicle;

[0080] FIG. 26 illustrates one embodiment of a control sub-system for the autonomous vehicle;

[0081] FIG. 27 illustrates one embodiment of a method for autonomously delivering a load; and

[0082] FIG. 28 illustrates one embodiment of another method for autonomously delivering a load.

DETAILED DESCRIPTION

[0083] FIG. 1A illustrates one embodiment of an on-road autonomous vehicle 10 operative to autonomously collect and transport a load over public roads. The on-road autonomous vehicle 10 comprises wheels 1 (1a, 1b, 1c, 1d) touching ground 9-ground, vertical structures 2 (2a, 2b, 2c), 2' (2a', 2b', 2c') mounted on the wheels, an upper horizontal

structure 3 (3a, 3b, 3c), a plurality of sensors 4 (4a, 4b, 4c), and a first connector 5-cnct having a certain clearance 9-clr above ground 9-ground.

[0084] FIG. 1B illustrates one embodiment of the on-road autonomous vehicle 10 as seen from a front view.

[0085] FIG. 1C illustrates one embodiment of the on-road autonomous vehicle 10 as seen from a side view.

[0086] FIG. 1D illustrates one embodiment of the on-road autonomous vehicle 10 as seen from below. The on-road autonomous vehicle 10 further comprises a control sub-system 4 (FIG. 1A), 6, 7, 8 comprising a processing unit 8 and a plurality of sensors 4 (4a, 4b, 4c, FIG. 1A) and actuators 6, 7.

[0087] FIG. 2A illustrates one embodiment of the on-road autonomous vehicle 10 self-driving without the load over a public road 20a and alongside regular traffic 21a, 21b.

[0088] FIG. 2B illustrates one embodiment of the on-road autonomous vehicle 10 arriving at a parking lot 20-p in which the load 11 is parked. The parking lot 20-p has a certain width 20-p-w. 21c, 21d, 21e are other vehicles, and 20b is a public road.

[0089] FIG. 2C illustrates one embodiment of the on-road autonomous vehicle 10 getting ready to pick up the load 11. The load has a certain width 11-w, and the on-road autonomous vehicle 10 has a certain width 10-w. 23a is a pedestrian, and 21f is a vehicle parking alongside the load 11.

[0090] FIG. 3A illustrates one embodiment of the on-road autonomous vehicle 10 getting ready to pick up the load 11.

[0091] FIG. 3B illustrates one embodiment of the on-road autonomous vehicle 10 moving/straddling over the load 11 and getting the connector 5-cnct over a certain position 5-pos associated with the load 11.

[0092] FIG. 3C illustrates one embodiment of the on-road autonomous vehicle 10 getting into a lower position and grabbing the load 11.

[0093] FIG. 3D illustrates one embodiment of the on-road autonomous vehicle 10 picking up the load 11.

[0094] FIG. 4A illustrates one embodiment of the on-road autonomous vehicle 10 driving away with the load 11 that was previously parked in the parking lot.

[0095] FIG. 4B illustrates one embodiment of the on-road autonomous vehicle 10 self-driving with the load 11 over a public road 20c and alongside regular traffic 21g. 22a, 22b are traffic lights or other traffic signs. The public road 20c has a certain lane width 20-w. The on-road autonomous vehicle 10 has a certain width 10-w and a certain height 10-h.

[0096] In one embodiment, the on-road autonomous vehicle 10 is operative to autonomously collect and transport loads over public roads.

[0097] In one embodiment, the on-road autonomous vehicle comprises an upper horizontal structure 3 (3a, 3b, 3c) elevated above ground 9-ground (FIG. 1A) by vertical structures 2 (2a, 2b, 2c), 2' (2a', 2b', 2c') mounted on at least four wheels 1 (1a, 1b, 1c, 1d) touching ground 9-ground, so as to create a certain clearance 9-clr above ground for at least a first connector 5-cnct associated with the upper horizontal structure 3 and attached thereunder.

[0098] In one embodiment, the on-road autonomous vehicle comprises a control sub-system 4, 6, 7, 8 comprising a processing unit 8 (FIG. 1D) and a plurality of sensors 4 (4a, 4b, 4c) and actuators 6, 7 (FIG. 1D), in which the control sub-system is configured to generate, in real-time, a

three-dimensional representation of surrounding environment using data collected by the plurality of sensors 4.

[0099] In one embodiment, the on-road autonomous vehicle comprises at least a first linear actuator 2'+2 (2' moving up and down relative to 2, i.e. 2a' moving relative to 2a, 2b' moving relative to 2b, 2c' moving relative to 2c, and 2d' moving relative to 2d) configured to control and set said certain clearance 9-clr of the first connector 5-cnct, by causing the first connector 5-cnct, or the entire upper horizontal structure 3, to move up or down relative to ground 9-ground.

[0100] In one embodiment, the control sub-system 4, 6, 7, 8 is further configured to use said three-dimensional representation, said actuators 6, 7, and said processing unit 8 in conjunction with a set of public-road self-driving directives, to: (i) self-drive (FIG. 2A, FIG. 2B) the on-road autonomous vehicle 10, over public roads 20a (FIG. 2A), 20b (FIG. 2B) and alongside regular car traffic 21a, 21b, 21c, 21d, 21e, 21f (FIG. 2A, FIG. 2B, FIG. 2C), to a location in which a first load 11 (FIG. 2B) is parked, (ii) position the on-road autonomous vehicle 10 in front of the load 11 (FIG. 2C, FIG. 3A), and (iii) straddle the on-road autonomous vehicle 10 over the first load 11 (FIG. 3B), such that said first connector 5-cnct is brought to a predetermined position 5-pos over the first load 11 (FIG. 3B).

[0101] In one embodiment, the control sub-system 4, 6, 7, 8 is further configured to use the first linear actuator 2'+2 to: (i) lower (FIG. 3C) the first connector 5-cnct into mechanical contact with the first load 11 thereby allowing the connector to connect to or grab the first load, and (ii) lift (FIG. 3D) the first load 11 above ground into a position operative to self-transport (FIG. 4A, FIG. 4B) the first load 11 over public roads 20c (FIG. 4B) and alongside regular car traffic 21e (FIG. 4A), 21g (FIG. 4B).

[0102] In one embodiment, the plurality of sensors 4 comprises a plurality of digital cameras together covering front, sides, and back of the on-road autonomous vehicle 10, thereby facilitating said generation of three-dimensional representation of surrounding environment.

[0103] In one embodiment, the plurality of sensors 4 comprises at least one radar device, thereby farther facilitating said generation of three-dimensional representation of surrounding environment.

[0104] In one embodiment, the plurality of sensors 4 comprises at least one light-detection-and-ranging (LIDAR) laser device, thereby farther facilitating said generation of three-dimensional representation of surrounding environment.

[0105] In one embodiment, the plurality of sensors 4 comprises at least two of: (i) a plurality of digital cameras, (ii) a LIDAR laser device, and (iii) a radar device, in which said generation of three-dimensional representation of surrounding environment is achieved using data fusion techniques acting on a combination of signals generated in real time by the plurality of sensors.

[0106] In one embodiment, said surrounding environment comprises at least the public roads 20a (FIG. 2A), 20b (FIG. 2B), 20c (FIG. 4B) surrounding the on-road autonomous vehicle 10, regular car traffic 21a, 21b, 21c, 21d, 21e, 21f, 21g surrounding the on-road autonomous vehicle 10, traffic lights 22a, 22b (FIG. 4B) and other traffic signs surrounding the on-road autonomous vehicle 10, and pedestrians 23a (FIG. 2C) surrounding the on-road autonomous vehicle 10, in which said three-dimensional representation in conjunc-

tion with said set of public-road self-driving directives are operative to facilitate said self-driving.

[0107] In one embodiment, said plurality of sensors 4 in conjunction with said data fusion and three-dimensional representation of surrounding environment are operative to detect said load 11 and to facilitate said positioning (FIG. 2C, FIG. 3A) of the on-road autonomous vehicle 10 in front of the load and said straddling (FIG. 3B) of the on-road autonomous vehicle 10 over the load 11.

[0108] In one embodiment, said on-road autonomous vehicle 10 is narrow enough to facilitate said self driving over public roads 20 and in conjunction with a width 20-w (FIG. 4B) associated with a standard lane in public roads 20c, in which said standard lane has the width 20-w of between 2.5 (two point five) meters and 3.7 (three point seven) meters. In one embodiment, the on-road autonomous vehicle 10 has a width 10-w (FIG. 4B) of below 2.5 (two point five) meters. In one embodiment, said load 11 has a width 11-w (FIG. 2C) of below 2 meters, and said load 11 is narrower than said on-road autonomous vehicle 10 (i.e., 11-w is narrower than 10-w in FIG. 2C), in order to facilitate said straddling (FIG. 3B) of the on-road autonomous vehicle 10 over the load 11.

[0109] In one embodiment, said location in which the load 11 is parked is a standard parking lot 20-p (FIG. 2B) in a parking area, in which said standard parking lot has a width 20-p-w of between 2.3 (two point three) meters and 2.7 (two point seven) meters, and said on-road autonomous vehicle 10 is narrow enough to facilitate said straddling of the on-road autonomous vehicle over the load 11 while the load is parked in said standard parking lot 20-p and without the on-road autonomous vehicle exceeding the width 20-p-w of the standard parking lot (i.e., 11-w is narrower than 10-w, and 10-w is narrower than 20-p-w). In one embodiment, the on-road autonomous vehicle 10 has a width 10-w of below 2.3 (two point three) meters. In one embodiment, said load 11 has a width 11-w of below 1.8 (one point eight) meters, and said load 11 is narrower than said on-road autonomous vehicle 10, in order to facilitate said straddling of the on-road autonomous vehicle 10 over the load 11 while the load is parked in said standard parking lot 20-p.

[0110] In one embodiment, a height 10-h (FIG. 4B) of said on-road autonomous vehicle 10 does not exceed a width 10-w of the on-road autonomous vehicle 10, thereby facilitating a center of gravity which is low enough to facilitate regular car traffic maneuvers in conjunction with said public roads 20a (FIG. 2A), 20b (FIG. 2B), 20c (FIG. 4B) and alongside regular car traffic 21a, 21b, 21c, 21d, 21e, 21f, 21g. In one embodiment, said maneuvers comprise traveling at a velocity exceeding 80 (eighty) kilometers-per-hour. In one embodiment, said maneuvers comprise a centripetal acceleration of above 3 (three) meters-per-second-square (m/s*s). In one embodiment, said width 10-w is between 2 (two) meters and 2.5 (two point five) meters, and said height 10-h is below 2 (two) meters.

[0111] In one embodiment, said actuators 6, 7 comprise electrical motors 6b, 6d capable of linearly accelerating and de-accelerating the on-road autonomous vehicle 10 at a rate of at least 3 (three) meters-per-second-square (m/s*s), thereby avoiding car-accidents in conjunction with said public roads 20a (FIG. 2A), 20b (FIG. 2B), 20c (FIG. 4B) and regular car traffic 21a, 21b, 21c, 21d, 21e, 21f, 21g.

[0112] In one embodiment, said lifting (FIG. 3D) is done so as to lift the load 11 to a position which is not more than

50 cm (fifty centimeters) above ground, thereby facilitating a center of gravity which is low enough to facilitate regular car traffic maneuvers in conjunction with said public roads **20a** (FIG. 2A), **20b** (FIG. 2B), **20c** (FIG. 4B) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, **21g**.

[0113] In one embodiment, said positioning and straddling comprises: using said three-dimensional representation of surrounding environment to accurately represent the load **11** and position the on-road autonomous vehicle **10** in front to the load, and using slow and controlled acceleration to slowly move forward while maintaining constant and regular spacing between the load **11** and the inner surfaces of the on-road autonomous vehicle **10**, until predicting accurate positioning of the on-road autonomous vehicle **10** above the load **11**.

[0114] In one embodiment, said first liner actuator **2'+2** is a distributed linear actuator comprising several sub-actuators (e.g., four sub-actuators: **2a'** moving relative to **2a**, **2b'** moving relative to **2b**, **2c'** moving relative to **2c**, and **2d'** moving relative to **2d**), in which each sub-actuator is associated with one of the wheels, such that the entire upper horizontal structure **3** is operative to move up and down relative to the wheels and ground.

[0115] In one embodiment, the sub-actuators are also operative to act as springs or mechanical dampers for the wheels relative to the upper horizontal structure.

[0116] In one embodiment, said first liner actuator is embedded in the first connector **5-cnct**, thereby causing only the connector to move up and down relative to ground, and such that the upper horizontal structure **3** remains in place.

[0117] FIG. 4C illustrates one embodiment of a method for autonomously collecting and transporting a load over public roads. In step **1001**, receiving, in conjunction with an on-road autonomous vehicle **10**, a request to collect-and-transport a load **11** which is currently parked in a certain location **20-p**. In step **1002**, self-driving (FIG. 2A, FIG. 2B, FIG. 2C), by the on-road autonomous vehicle, over public roads **20a** (FIG. 2A), **20b** (FIG. 2B) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, from a current location of the on-road autonomous vehicle to said certain location **20-p** of the load **11**. In step **1003**, upon arrival to said certain location **20-p**, straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), by the on-road autonomous vehicle **10**, over the load **11**, thereby allowing the on-road autonomous vehicle to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the load **11** above ground in a linear upward movement that creates a full clearance of the load **11** above ground. In step **1004**, transporting autonomously (FIG. 4A, FIG. 4B) the load **11**, by the on-road autonomous vehicle **10**, over public roads **20c** (FIG. 4B) and alongside regular traffic **21e** (FIG. 4A), **21g** (FIG. 4B), while the load **11** is hanging underneath the on-road autonomous vehicle **10** and such that the entire load **11** maintains said full clearance above ground during transport. In one embodiment, autonomously navigating, by the on-road autonomous vehicle, to a destination location, and lowering the load **11** at the destination location in a linear downward movement that places the load on the ground (a reverse transition from FIG. 3D to FIG. 3C).

[0118] FIG. 4D illustrates one embodiment of a method for autonomously collecting and transporting a load over public roads. In step **1011**, self-driving using a set of public-road self-driving directives, by an on-road autonomous vehicle **10**, over public roads **20a** (FIG. 2A), **20b** (FIG.

2B) and alongside regular car traffic **21a**, **21b**, **21c**, **21d**, **21e**, **21f**, from a current location of the on-road autonomous vehicle to a certain location **20-p** of the load **11**. In step **1012**, upon arrival to said certain location **20-p**, straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), using a set of autonomous straddling directives, by the on-road autonomous vehicle **10**, over the load **11**, thereby allowing the on-road autonomous vehicle to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the load **11** above ground in a linear upward movement that creates a full clearance of the load **11** above ground. In step **1013**, transporting autonomously (FIG. 4A, FIG. 4B) the load **11** using a set of public-road autonomous transport directives, by the on-road autonomous vehicle **10**, over public roads **20c** (FIG. 4B) and alongside regular traffic **21e** (FIG. 4A), **21g** (FIG. 4B), while the load **11** is hanging underneath the on-road autonomous vehicle **10** and such that the entire load **11** maintains said full clearance above ground during transport. In one embodiment, said public-road self-driving directives are fine-tuned to facilitate said self-driving during a period that the on-road autonomous vehicle **10** does not transport the load **11**, and is therefore (i) lighter and (ii) has a higher center of gravity; said public-road autonomous transport directives are fine-tuned to facilitate said transporting during a period that the on-road autonomous vehicle **10** transports the load **11**, and is therefore (i) heavier and (ii) has a lower center of gravity; and said autonomous straddling directives are fine-tuned to facilitate said straddling by directing a slow approach and slow straddling of the on-road autonomous vehicle **10** over the load **11** and into an accurate final position above the load.

[0119] FIG. 5A illustrates one embodiment of an on-road autonomous vehicle **10** self-driving to a location in which a passenger in a cabin is located.

[0120] FIG. 5B illustrates one embodiment of the passenger **30** in the cabin **11** awaiting arrival of the on-road autonomous vehicle **10**. **5-pos** is a position in the cabin **11** that facilitates grabbing or connecting to the cabin **11**.

[0121] FIG. 5C illustrates one embodiment of the on-road autonomous vehicle **10** picking up the cabin **11** with the passenger **30**. The cabin **11** is now connected to the on-road autonomous vehicle **10** via connector **5-cnct** and in conjunction with position **5-pos**.

[0122] FIG. 5D illustrates one embodiment of a method for autonomously collecting and transporting a passenger in a cabin. In step **1021**, receiving, in conjunction with an on-road autonomous vehicle **10** (FIG. 5A), from a passenger **30** (FIG. 5B) associated with a cabin **11** (FIG. 5B) located in a certain location, a request to collect-and-transport the passenger together with the cabin. In step **1022**, self-driving (FIG. 2A), by the on-road autonomous vehicle **10**, from a current location of the on-road autonomous vehicle to said certain location of the cabin **11** (FIG. 5B). In step **1023**, upon arrival to said certain location: (i) confirming that the passenger **30** is indeed in the cabin **11** (FIG. 5B), and (ii) straddling autonomously, by the on-road autonomous vehicle **10**, over the cabin **11** (FIG. 3B), thereby allowing the on-road autonomous vehicle **10** to grab and lift autonomously (FIG. 3C, FIG. 3D) the cabin **11**, together with the passenger **30**, above ground (FIG. 5C). In step **1024**, transporting autonomously the cabin **11** together with the passenger **30**, by the on-road autonomous vehicle **10**, while the cabin **11** is hanging underneath the on-road autonomous vehicle (FIG. 4B). In one embodiment, the method further

includes: receiving, in conjunction with the on-road autonomous vehicle 10, a second request to collect-and-transport a cargo 11 which is currently located in a second location 20-*p* (FIG. 2B); self-driving (FIG. 2A, FIG. 2B, FIG. 2C), by the on-road autonomous vehicle 10, over public roads 20*a* (FIG. 2A), 20*b* (FIG. 2B) and alongside regular car traffic 21*a*, 21*b*, 21*c*, 21*d*, 21*e*, 21*f*, from a current location of the on-road autonomous vehicle to said second location 20-*p*; upon arrival to said second location 20-*p*, straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), by the on-road autonomous vehicle 10, over the cargo 11, thereby allowing the on-road autonomous vehicle 10 to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the cargo 11 above ground in a linear upward movement that creates a full clearance of the cargo 11 above ground; and transporting autonomously (FIG. 4A, FIG. 4B) the cargo 11, by the on-road autonomous vehicle 10, over public roads 20*c* (FIG. 4B) and alongside regular traffic 21*e* (FIG. 4A), 21*g* (FIG. 4B), while the cargo 11 is hanging underneath the on-road autonomous vehicle 10 and such that the entire cargo 11 maintains said full clearance above ground during transport, thereby facilitating dual use of the on-road autonomous vehicle 10 for both said transporting of the passenger 30 during a certain period of time and said transporting of the cargo 11 during another period of time. In one embodiment, said transporting of the passenger 30 or other passengers is done during the mornings or the evenings and in conjunction with work rush hours, while said transporting of the cargo 11 or other cargo is done during mid-day hours, while most people are working.

[0123] FIG. 5E illustrates one embodiment of a method for requesting autonomous collection and transporting of a passenger in a cabin. In step 1031, associating a cabin 11 with a passenger 30 (FIG. 5B). In step 1032, sending, in conjunction with the cabin 11, a request to collect-and-transport the cabin together with the passenger 30. In step 1033, receiving, in conjunction with the cabin 11, a ready-to-collect message from an on-road autonomous vehicle 10 (FIG. 5A). In step 1034, sending, in conjunction with the cabin 11, a ready-to-be-collected message to the on-road autonomous vehicle 10, thereby facilitating collection and transporting of the passenger 30 in the cabin 11 (FIG. 5C).

[0124] FIG. 6A illustrates one embodiment of an on-road autonomous vehicle 10 self-driving to a location 28 (FIG. 6B) in which a functional load 11 is located.

[0125] FIG. 6B illustrates one embodiment of the functional load 11 awaiting arrival of the on-road autonomous vehicle 10.

[0126] FIG. 6C illustrates one embodiment of the on-road autonomous vehicle 10 picking up and transporting the functional load 11.

[0127] FIG. 6D illustrates one embodiment of the functional load 11 after being placed by the on-road autonomous vehicle 10 at a particular location 29 operative to work in conjunction with or support the functional load 11 using an interface 29-rec.

[0128] FIG. 6E illustrates one embodiment of the on-road autonomous vehicle 10 driving away after placing the functional load 11 at the particular location.

[0129] FIG. 6F illustrates one embodiment of a method for autonomously collecting transporting and placing a functional load according to a request. In step 1041, receiving, in conjunction with an on-road autonomous vehicle 10 (FIG.

6A), a request to place a specific load 11 (FIG. 6B) at a particular location 29 (FIG. 6D), in which the specific load 11 is operative to perform a specific function. In step 1042, self-driving, by the on-road autonomous vehicle 10, to a specific storage location 28 (FIG. 6B) of the specific load 11. In step 1043, upon arrival to said specific storage location 28, straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), by the on-road autonomous vehicle 10, over the specific load 11, thereby allowing the on-road autonomous vehicle to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the specific load 11 above ground in a linear upward movement that creates a full clearance of the specific load 11 above ground. In step 1044, transporting autonomously (FIG. 4B) the specific load 11, by the on-road autonomous vehicle 10, over public roads 20*c* (FIG. 4B) and alongside regular traffic 21*g* (FIG. 4B), to the particular location 29 (FIG. 6D), while the specific load 11 is hanging (FIG. 6C) underneath the on-road autonomous vehicle 10 and such that the entire specific load 11 maintains said full clearance above ground during transport. In step 1045, upon arrival to the particular location 29, lowering the specific load 11 at the particular location 29 (FIG. 6D) in a linear downward movement that places the load down (a reverse transition from FIG. 3D to FIG. 3C), thereby facilitating performance of the specific action in conjunction with the particular location 29.

[0130] In one embodiment, said linear downward movement creates a physical contact between the specific load 11 and a reception element or interface 29-rec (FIG. 6D) at the particular location 29, thereby enabling said specific function in conjunction with the physical contact.

[0131] In one embodiment, the specific function is fluid or gas transfer, and the physical contact with the interface 29-rec, which comprises ducts, enables said fluid or gas transfer.

[0132] In one embodiment, the specific function is electrical charge transfer, and the physical contact with the interface 29-rec, which comprises electrical contact, enables said electrical charge transfer.

[0133] In one embodiment, said specific function comprises at least one of: (i) fluid transfer such as water or gasoline transfer, (ii) gas transfer such as methane transfer, (iii) electrical charge transfer in which the specific load 11 is a rechargeable battery, (iv) automatic vending in which the specific load 11 is an automatic vending machine, (v) communication relaying in which the specific load 11 is a communication relay or a cellular base station, (vi) waste collection in which the specific load 11 is a waste container, (vii) monitoring, surveillance, or intelligence gathering, and (viii) distribution of objects such as drones, other autonomous vehicles, and munitions.

[0134] One embodiment further comprises straddling away (FIG. 6E) from the specific load 11, by the on-road autonomous vehicle 10, thereby leaving the specific load 11 (FIG. 6D) at the particular location 29 to perform said specific function.

[0135] FIG. 7A illustrates one embodiment of an on-road autonomous vehicle 10 self-driving to a location 27 in which a functional load 11-func is located.

[0136] FIG. 7B illustrates one embodiment of the on-road autonomous vehicle 10 picking up the functional load

11-func, interfacing 5-int with the functional load 11-func, and using a function associated with the functional load 11-func.

[0137] FIG. 7C illustrates one embodiment of a convoy of several on-road autonomous vehicles 10a, 10b, 10c, 10d in which one of the on-road autonomous vehicles 10a is carrying a functional load 10-func. 11b, 11c, 11d are general loads.

[0138] FIG. 7D illustrates one embodiment of the convoy of several on-road autonomous vehicles 10a, 10b, 10c, 10d in which the on-road autonomous vehicles switch at least some of the loads between themselves so as to pass the functional load 11-func from one of the on-road autonomous vehicles 10a to another 10b of the on-road autonomous vehicles.

[0139] FIG. 7E illustrates one embodiment of the convoy of several on-road autonomous vehicles 10a, 10b, 10c, 10d in which another of the on-road autonomous vehicles 10b is now carrying the functional load 11-func.

[0140] FIG. 7F illustrates one embodiment of a method for autonomously collecting and using a functional load. In step 1051, determining, by an on-road autonomous vehicle 10 (FIG. 7A), that a certain function is needed in conjunction with operating said on-road autonomous vehicle. In step 1052, self-driving (FIG. 2A), by the on-road autonomous vehicle 10, from a current location of the on-road autonomous vehicle to a certain location 27 (FIG. 7A) where a functional load 11-func is parked, in which said functional load is operative to render said certain function needed. In step 1053, upon arrival to said certain location (FIG. 7A), straddling autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), by the on-road autonomous vehicle 10, over the functional load 10-func, thereby allowing the on-road autonomous vehicle to grab (FIG. 3C) and lift autonomously (FIG. 3D, or the transition from FIG. 3C to FIG. 3D) the functional load 10-func above ground in a linear upward movement that creates a full clearance of the load above ground (the result is illustrated in FIG. 7B). In step 1054, interfacing 5-int (FIG. 7B), in a physical manner, between the on-road autonomous vehicle 10 and the functional load 11-func, thereby facilitating said rendering of the certain function needed from the functional load to the on-road autonomous vehicle.

[0141] In one embodiment, said certain function needed is a need to charge an electrical battery 12a, 12c (FIG. 7B) belonging to the on-road autonomous vehicle 10, said functional load 11-func is an energy source, and said interfacing 5-int is operative to transfer energy from the functional load 11-func to the on-road autonomous vehicle 10. In one embodiment, said energy source 11-func is a portable battery, and said interfacing 5-int is an electrical interface operative to transport electricity from the portable battery 11-func to the battery 12a, 12c of the on-road autonomous vehicle 10. In one embodiment, said energy source 11-func is a portable fuel cell, and said interfacing 5-int is an electrical interface operative to transport electricity from the fuel cell 11-func to the battery 12a, 12c of the on-road autonomous vehicle 10. In one embodiment, said energy source 11-func is a portable generator with on-board fuel, and said interfacing 5-int is an electrical interface operative to transport electricity from the portable generator 11-func to the battery 12a, 12c of the on-road autonomous vehicle 10.

[0142] In one embodiment, said certain function needed is a need for fuel, said functional load 11-func is a fuel tank,

and said interfacing 5-int is operative to transfer fuel from the fuel tank 11-func to the on-road autonomous vehicle 10. In one embodiment, said fuel is gasoline or diesel fuel. In one embodiment, said fuel is a fuel operative to drive a fuel cell, such as hydrogen fuel, methanol fuel, ethanol fuel, or methane fuel.

[0143] FIG. 7G illustrates one embodiment of a method for exchanging a functional load between at least two on-road autonomous vehicles in a convoy. In step 1061, detecting, in a moving convoy (FIG. 7C) comprising on-road autonomous vehicles 10a, 10b, 10c, 10d, a need of one of the on-road autonomous vehicles 10b in the convoy to use a certain function, and further detecting another on-road autonomous vehicle 10a in the moving convoy that currently makes use of said certain function in conjunction with a functional load 11-func carried therewith (i.e., 10a carries 11-func for a certain use, and 10b currently needs 11-func for that use). In step 1062, stopping the convoy (FIG. 7D) as a response to said detections. In step 1063, autonomously offloading, by said another on-road autonomous vehicle 10a, the functional load 11-func, thereby placing the functional load 11-func on ground (e.g., in FIG. 7D, 11-func has now been placed on ground, and 10a which previously carried 11-func has now moved to the back of the convoy). In step 1064, autonomously picking-up the functional load 11-func off ground and interfacing to said functional load by said one of the on-road autonomous vehicles 10b (e.g., FIG. 7E, 10b has now collected 11-func, and load 11b which was previously carried by 10b is now carried by yet another vehicle 10c), thereby facilitating said certain function in conjunction with said one of the on-road autonomous vehicles 10b.

[0144] In one embodiment, said picking-up of the functional load 11-func, by said one of the on-road autonomous vehicles 10b (FIG. 7E), is facilitated by straddling autonomously, by said one of the on-road autonomous vehicle 10b, over the functional load 11-func now on ground (FIG. 7D, 10b is driving/straddling over 11-func), thereby allowing said one of the on-road autonomous vehicle 10b to grab and lift autonomously the functional load 11-func above ground in a linear upward movement that creates a full clearance of the functional load above ground (in FIG. 7E, 10b has lifted 11-func above ground), thereby facilitating said interfacing.

[0145] In one embodiment, said certain function is an electrical charging of batteries 12a, 12c belonging to said one of the on-road autonomous vehicles 10b, in which said functional load 11-func is an energy source.

[0146] One embodiment further comprising resuming movement by the convoy.

[0147] FIG. 8A illustrates one embodiment of two on-road autonomous vehicles 10a, 10b getting into positions in conjunction with a load 11-long that is too big to be carried by only one on-road autonomous vehicle. 5-cnct-1 is a connector of on-road autonomous vehicles 10a and is operative to grab a grabbing point 5-grb-1 of load 11-long at position 5-pos-1. 5-cnct-2 is a connector of on-road autonomous vehicles 10b and is operative to grab a grabbing point 5-grb-2 of load 11-long at position 5-pos-2.

[0148] FIG. 8B illustrates one embodiment of the two on-road autonomous vehicles 10a, 10b cooperatively lifting the load 11-long.

[0149] FIG. 8C illustrates one embodiment of the two on-road autonomous vehicles 10a, 10b cooperatively lifting another load 11-wide. 11-ext-1 and 11-ext-2 are extensions of load 11-wide.

[0150] FIG. 8D illustrates one embodiment of a method for cooperatively lifting and transporting a load by at least two on-road autonomous vehicles. In step 1071, driving autonomously, by a first on-road autonomous vehicle 10a (FIG. 8A), into a first predetermined position 5-pos-1 over a load 11-long (FIG. 8A), in which the first predetermined position 5-pos-1 is associated with a first grabbing point 5-grb-1 in the load 11-long. In step 1072, driving autonomously, by a second on-road autonomous vehicle 10b (FIG. 8A), into a second predetermined position 5-pos-2 over the load 11-long, in which the second predetermined position 5-pos-2 is associated with a second grabbing point 5-grb-2 in the load 11-long. In step 1073, lifting the load 11-long together and synchronously (FIG. 8B), by the first and the second on-road autonomous vehicles 10a, 10b in conjunction respectively with the first and second grabbing points 5-grb-1, 5-grb-2, such that: (i) the load 11-long is lifted above ground, (ii) the load 11-long achieves full clearance above ground, and (iii) a weight of the load 11-long is spread between the first 10a and second 10b on-road autonomous vehicles via the first 5-grb-1 and second 5-grb-2 grabbing points respectively. In step 1074, transporting autonomously the load 11-long together and synchronously by the first and second on-road autonomous vehicles 10a, 10b, such that the load 11-long maintains said full clearance above ground during transport.

[0151] In one embodiment, the first on-road autonomous vehicle 10a constantly communicates with the second 10b on-road autonomous vehicle during said lifting and transport in order to achieve said synchronicity. In one embodiment, the first on-road autonomous vehicle 10a controls the second on-road autonomous vehicle 10b during said lifting and transport, thereby facilitating the lifting and transport autonomously. In one embodiment, the first on-road autonomous vehicle 10a receives sensory input from the second on-road autonomous vehicle 10b during said lifting and transport, thereby facilitating the lifting and transport autonomously.

[0152] In one embodiment, said driving autonomously comprises straddling autonomously over the load, by said first and second on-road autonomous vehicles 10a, 10b.

[0153] In one embodiment, said lifting comprises: lowering a first connector 5-cnct-1 of the first on-road autonomous vehicle 10a into mechanical contact with the first grabbing point 5-grb-1, lowering a second connector 5-cnct-2 of the second on-road autonomous vehicle 10b into mechanical contact with the second grabbing point 5-grb-2, grabbing the first grabbing point 5-grb-1 by the first connector 5-cnct-1, grabbing the second grabbing point 5-grb-2 by the second connector 5-cnct-2, and raising synchronously the first and second connectors 5-cnct-1, 5-cnct-2 respectively by the first and second on-road autonomous vehicles 10a, 10b.

[0154] In one embodiment, the load 11-long (FIG. 8A, FIG. 8B) is narrower than the on-road autonomous vehicles 10a, 10b, thereby enabling the entire load to be carried under the on-road autonomous vehicles.

[0155] In one embodiment, the load 11-wide (FIG. 8C) is wider than the on-road autonomous vehicles 10a, 10b, thereby requiring the load to be supported under the on-road autonomous vehicles by two narrow extension shafts 11-ext-1, 11-ext-2 (FIG. 8C).

[0156] FIG. 9A illustrates one embodiment of an on-road autonomous vehicle 10 getting into position behind a target vehicle 10-target. 10-mech is a mechanical hook or grabbing mechanism.

[0157] FIG. 9B illustrates one embodiment of the on-road autonomous vehicle 10 now mechanically connected via the mechanical hook or grabbing mechanism 10-mech to the target vehicle 10-target. The target vehicle 10-target pulls the on-road autonomous vehicle thereby allowing the on-road autonomous vehicle to self generate electrical energy.

[0158] FIG. 9C illustrates one embodiment of an on-road autonomous vehicle 10a getting into position behind another on-road autonomous vehicle 10b in order to be pulled thereby.

[0159] FIG. 9D illustrates one embodiment of a method for charging batteries of an on-road autonomous vehicle on the move. In step 1081, identifying, by an on-road autonomous vehicle 10 (FIG. 9A), a target vehicle 10-target (FIG. 9A) operative to mechanically pull another vehicle. In step 1082, self driving (FIG. 2A), by the on-road autonomous vehicle 10, into a position behind the target vehicle identified (FIG. 9A). In step 1083, connecting autonomously (FIG. 9B), by the on-road autonomous vehicle 10, to the target vehicle 10-target, by performing an autonomous maneuver in conjunction with a mechanical hook or grabbing mechanism 10-mech, so as to mechanically connect between the on-road autonomous vehicle 10 and the target vehicle 10-target (FIG. 9B). In step 1084, being pulled, by the target vehicle 10-target, thereby causing at least one wheel 1d (FIG. 9B) of the on-road autonomous vehicle 10 to rotate as the on-road autonomous vehicle 10 is pulled by the target vehicle 10-target. In step 1085, generating electrical energy from at least one dynamo 6d (FIG. 9B) connected to the at least one wheel 1d now rotating, thereby allowing the dynamo 6d to charge a battery 12c of the on-road autonomous vehicle 10.

[0160] In one embodiment, said connecting autonomously is done during forward movement of both the on-road autonomous vehicle 10 and the target vehicle 10-target.

[0161] In one embodiment, said connecting autonomously is done in a stationary state, before forward movement of both the on-road autonomous vehicle 10 and the target vehicle 10-target, in which said forward movement by the target vehicle causes said pulling.

[0162] In one embodiment, said mechanical hook or grabbing mechanism 10-mech is a part of the on-road autonomous vehicle 10. In one embodiment, the hook or grabbing mechanism 10-mech is a moving hook. In one embodiment, the hook or grabbing mechanism 10-mech is a mechanical connector.

[0163] In one embodiment, the target vehicle 10-target is a second on-road autonomous vehicle 10b (FIG. 9C), in which the second on-road autonomous vehicle 10b synchronizes said autonomous connection with the on-road autonomous vehicle 10 or 10a. In one embodiment, said self driving, by the on-road autonomous vehicle 10 or 10a, into a position behind the target vehicle 10-target, is done in conjunction with the second on-road autonomous vehicle 10b, 10-target also self driving into a position in front of the on-road autonomous vehicle 10 or 10a.

[0164] In one embodiment, the dynamo 6d is an electrical engine of the on-road autonomous vehicle 10, in which the electrical engine is operated in a breaking mode, thereby facilitating said generation of electrical energy.

[0165] In one embodiment, said identification is done so as to identify the target vehicle 10-target as a vehicle currently traveling in a direction similar to a direction desirable by the on-road autonomous vehicle 10.

[0166] FIG. 10A illustrates one embodiment of an on-road autonomous vehicle 10 carrying a passenger cabin 11.

[0167] FIG. 10B illustrates one embodiment of the on-road autonomous vehicle 10 providing air gap protection 11-10-gap-1, 11-10-gap-2 to the passenger cabin 11.

[0168] FIG. 10C illustrates one embodiment of the on-road autonomous vehicle 10 providing further air gap protection 11-10-gap-3, 11-10-gap-4, 11-10-gap-5, 11-10-gap-6 to the passenger cabin 11.

[0169] One embodiment is a system operative to provide a hybrid air-gap and mechanical protection for a passenger cabin. The system includes a passenger cabin 11 (FIG. 10A), and an on-road autonomous vehicle 10 (FIG. 10A) operative to straddle over a passenger cabin 11 and then to pick up the passenger cabin 11, such that the passenger cabin 11 is carried underneath the on-road autonomous vehicle 10 (FIG. 10A). The on-road autonomous vehicle 10 is operative to mechanically enclose the passenger cabin 11 from at least 4 (four) directions, such as to provide mechanical protection from impact with foreign objects, in which said mechanical protection is enhanced by maintaining air-gaps 11-10-gap-1, 2, 3, 4, 5, 6 between the on-road autonomous vehicle 10 and the passenger cabin 11 in conjunction with the at least 4 (four) directions.

[0170] In one embodiment, said at least 4 (four) directions are front, rear, left, and right, in which the front direction is associated with one of the air-gaps 10-11-gap-5 (FIG. 10C) located in front of the passenger cabin 11, the rear direction is associated with one of the air-gaps 10-11-gap-3 (FIG. 10C) located behind the passenger cabin 11, the left direction is associated with one of the air-gaps 10-11-gap-1 (FIG. 10B) located to the left of the passenger cabin 11, and the right direction is associated with one of the air-gaps 10-11-gap-2 (FIG. 10B) located to the right of the passenger cabin 11. In one embodiment, said at least 4 (four) directions are at least 5 (five) directions comprising also an up direction associated with one of the air-gaps 10-11-gap-6 (FIG. 10C) located above the passenger cabin 11. In one embodiment, said at least 5 (five) directions are 6 (directions) directions comprising also a down direction associated with one of the air-gaps 10-11-gap-4 (FIG. 10C) located below the passenger cabin 11.

[0171] FIG. 11A illustrates one embodiment of an on-road autonomous vehicle 10 about to be hit by a foreign object 21. The on-road autonomous vehicle 10 comprises a rear section 10-rear, a front section 10-front, and a piston or spring 10-sp.

[0172] FIG. 11B illustrates one embodiment of the on-road autonomous vehicle 10 being hit by the foreign object 21h.

[0173] One embodiment is a system operative to protect an on-road vehicle from impact with foreign objects. The system includes a front section 10-front (FIG. 11A) of an on-road vehicle 10 (FIG. 11A), in which the front section comprises two front wheels 1c, 1d on which the on-road vehicle is supported. The system further includes a rear section 10-rear (FIG. 11A) of the on-road vehicle 10, in which the rear section comprises two rear wheels 1a, 1b on which the on-road vehicle is further supported, and in which the front section 10-front is mechanically connected to the rear section 10-rear so as to allow movement of the entire

front section 10-front relative to the rear sections 10-rear. The system further includes a passenger or cargo cabin 11 of the on-road vehicle 10, in which the passenger or cargo cabin is mechanically connected to the rear section 10-rear without touching the front section 10-front. The system further includes a linear horizontal actuator or horizontal spring 10-sp (FIG. 11A) having two sides, in which the linear horizontal actuator or horizontal spring is mechanically connected to the front section 10-front on one side and to the rear section 10-rear on the other side, and in which the linear horizontal actuator or horizontal spring 10-sp is operative to generate a reaction force pushing the front section 10-front away from the rear section 10-rear when the front section moves toward the rear section. During a collision of the front section 10-front with a foreign object 21h (FIG. 11B), the front section 10-front moves toward the rear section 10-rear (FIG. 11B), thereby causing said reaction force to accelerate the rear section 10-rear away from the foreign object 21h, and thereby avoiding a collision between the passenger or cargo cabin 11 and the foreign object 21h or at least reducing a relative velocity between the passenger or cargo cabin 11 and the foreign object 21h. In one embodiment, the linear horizontal actuator or horizontal spring is a piston.

[0174] One embodiment is an on-road vehicle operative to control a length thereof. The on-road vehicle includes a front section 10-front (FIG. 11A) of the on-road vehicle 10 (FIG. 11A), in which the front section comprises two front wheels 1c, 1d on which the on-road vehicle is supported. The on-road vehicle further includes a rear section 10-rear (FIG. 11A) of the on-road vehicle 10, in which the rear section comprises two rear wheels 1a, 1b on which the on-road vehicle is further supported, and in which the front section 10-front is mechanically connected to the rear section 10-rear so as to allow movement of the entire front section 10-front relative to the rear sections 10-rear. The on-road vehicle further includes a linear horizontal actuator or a piston 10-sp (FIG. 11A) having two sides, in which the linear horizontal actuator or piston is mechanically connected to the front section 10-front on one side and to the rear section 10-rear on the other side, and in which the linear horizontal actuator or piston 10-sp is operative to adjust a distance between the front section 10-front and the rear section 10-rear so as to control a length of the on-road vehicle.

[0175] In one embodiment, said length is adjusted to support loads 11 of different lengths to be carried by the on-road vehicle.

[0176] In one embodiment, said length is adjusted to support compact length during parking of the on-road vehicle 10.

[0177] In one embodiment, said length is adjusted to support extended length during high speed driving of the on-road vehicle 10 in order to protect a load 11 carried by the on-road vehicle by increasing an air-gap 11-10-gap-5 (FIG. 10C) between the load 11 the front section 10-front.

[0178] FIG. 12A illustrates one embodiment of an on-road autonomous vehicle 10 carrying a load 11 having a certain aerodynamic design and extending beyond a length of the on-road autonomous vehicle.

[0179] One embodiment is a combined vehicle-and-load arrangement operative to reduce drag on the vehicle. The combined vehicle-and-load arrangement includes an on-road autonomous vehicle 10 (FIG. 12A) having a first

aerodynamic drag coefficient and a load **11** (FIG. 12A) having a certain aerodynamic design. The on-road autonomous vehicle **10** is configured to straddle autonomously over the load **11**, thereby allowing the on-road autonomous vehicle **10** to grab and lift autonomously the load **11** above ground in a linear upward movement that creates a full clearance of the load above ground. The combined vehicle-and-load arrangement **10+11** of the on-road autonomous vehicle together with the load now lifted is operative to reduce the first aerodynamic drag coefficient of the on-road autonomous vehicle.

[0180] FIG. 12B illustrates one embodiment of a method for adjusting an on-road autonomous vehicle to carry a long load. In step **1091**, straddling autonomously, by an on-road autonomous vehicle **10** having a certain length **10-L** (FIG. 12), over a load **11** having a greater length **11-L** (FIG. 12), thereby allowing the on-road autonomous vehicle **10** to grab and lift autonomously the load **11** above ground in a linear upward movement that creates a full clearance of the load above ground, and resulting in the load **11** extending beyond said certain length **10-L**. In step **1092**, detecting, by sensors **4a**, **4b**, **4c** (FIG. 12) belonging to the on-road autonomous vehicle **10**, said extension of the load **11** beyond said certain length **10-L**. In step **1093**, altering, according to said detection, at least a first parameter in conjunction with an autonomous driving procedure of the on-road autonomous vehicle, thereby adapting the on-road autonomous vehicle **10** to self-drive with the load **11** extending beyond said certain length **10-L**.

[0181] In one embodiment, the first parameter is an effective length of the on-road autonomous vehicle, in which said effective length is increased from the certain length **10-L** to the greater length **11-L**.

[0182] One embodiment is a system operative to autonomously collect and transport a passenger **30** in a cabin **11** (FIG. 5C), comprising: an autonomous on-road vehicle **10** operative to straddle over (FIG. 3B, and the transition from FIG. 3A to FIG. 3B) loads **11** such as cabins operative to contain passengers **30** (FIG. 5C); and a cabin **11** operative to contain passengers **30** (FIG. 5B), in which the cabin **11** is currently located at a certain parking location without any passengers inside (the cabin **11** as shown in FIG. 5B, but without the passenger **30** inside); wherein, the system is configured to: receive a request to collect-and-transport a passenger **30** which is currently located or is soon-to-be-located in a pick-up location; self-drive, as a response to said request, the on-road autonomous vehicle **10**, to said certain parking location; upon arrival of the on-road autonomous vehicle **10** to said certain parking location: straddle autonomously (FIG. 3B, and the transition from FIG. 3A to FIG. 3B), the on-road autonomous vehicle **10**, over the cabin **11**, thereby allowing the on-road autonomous vehicle **10** to grab and lift autonomously (the transition between FIG. 3C and FIG. 3D) the cabin **11**; self-drive the on-road autonomous vehicle **10** to said pick-up location, while the cabin **11** is hanging underneath the on-road autonomous vehicle; and pick-up the passenger **30** (FIG. 5C) at the pick-up location.

[0183] One embodiment is a system operative to transport passengers in a cabin and also transport cargo loads, comprising: an autonomous on-road vehicle **10** operative to straddle over (FIG. 3A, FIG. 3B) loads **11** such as cabins operative to contain passengers and such as cargo loads; a cabin **11** operative to contain passengers **30** (FIG. 5C), in which the cabin is currently located at a certain parking

location; and a cargo load **11** (FIG. 2C); wherein, the system is configured to: autonomously transport the cargo load **11** (FIG. 4B), using the autonomous on-road vehicle **10**, to a certain location; autonomously release the cargo load **11** at the certain location (the transition from FIG. 3D to FIG. 3C to FIG. 3B and to FIG. 3A); self-drive the on-road autonomous vehicle **10** (FIG. 2A), which is now free of the cargo load, to the certain parking location; upon arrival of the on-road autonomous vehicle **10** to said certain parking location: straddle autonomously, the on-road autonomous vehicle, over the cabin, thereby allowing the on-road autonomous vehicle to grab and lift autonomously the cabin; self-drive the on-road autonomous vehicle to a pick-up location, while the cabin is hanging underneath the on-road autonomous vehicle; and pick-up a passenger at the pick-up location (FIG. 5C) into the cabin.

[0184] One embodiment is a system operative to transport both passengers in a towed cabin and towed cargo loads, comprising: an autonomous on-road vehicle operative to tow loads such as towed cabins operative to contain passengers and such as towed cargo loads; a towed cabin operative to contain passengers, in which the towed cabin is currently located at a certain parking location; and a towed cargo load; wherein, the system is configured to: autonomously transport the towed cargo load, using the autonomous on-road vehicle, to a certain location; autonomously release the towed cargo load at the certain location; self-drive the on-road autonomous vehicle, which is now free of the towed cargo load, to the certain parking location; upon arrival of the on-road autonomous vehicle to said certain parking location: connect the towed cabin, thereby allowing the on-road autonomous vehicle to autonomously tow the towed cabin; self-drive the on-road autonomous vehicle to a pick-up location, while the towed cabin is hanging behind the on-road autonomous vehicle; and pick-up a passenger at the pick-up location into the cabin.

[0185] FIG. 13A illustrates one embodiment of an on-road autonomous vehicle **10** currently integrated with a first object **11-obj-1** having drawers **11-drawer** that are presently closed. The drawers **11-drawer** may contain packages to be delivered or other items to be delivered and/or stored. The on-road autonomous vehicle **10** is operative to self-integrate with the first object **11-obj-1**, thereby creating a first resultant vehicle **10+11-obj-1** that is the combination of both the autonomous vehicle **10** and the first object **11-obj-1** and that has a first specific purpose, in which such a first specific purpose may be the delivery of packages. For example, the first resultant vehicle **10+11-obj-1** may self drive to a logistics center, load the drawers **11-drawer** with packages, and then self-drive with the packages onboard to several delivery destinations, in which each of the drawers **11-drawer** gets open as the resultant vehicle **10+11-obj-1** arrives at the designated delivery destination, and the respective package is then picked up, perhaps by a certain person who ordered the respective package. Self integration of on-road autonomous vehicle **10** with the first object **11-obj-1** facilitates the first specific purpose of package delivery by: (i) combining the features of both the on-road autonomous vehicle **10** and the first object **11-obj-1** in creation of a combined set of features that facilitates self package delivery, and (ii) maintaining cooperation/communication between the on-road autonomous vehicle **10** and the first object **11-obj-1** during the different phases of package delivery. For example, self-driving the first resultant vehicle

10+11-obj-1 to the logistics center is based on sensors (e.g., **4a**, **4b**, **4c**, FIG. 1A) and self-driving directives embedded in the on-road autonomous vehicle **10**, but loading packages into the drawers **11-drawer** requires further cooperation and communication with the first object **11-obj-1**, for example when the sensors onboard the on-road autonomous vehicle **10** conclude that the first resultant vehicle **10+11-obj-1** is in the correct loading spot in the logistics center and consequently the on-road autonomous vehicle **10** instructs the first object **11-obj-1** to open all drawers **11-drawer** thereby allowing the loading of packages into the first object **11-obj-1**. When the drawers **11-drawer** are loaded with packages, the first resultant vehicle **10+11-obj-1** again uses sensors and directives embedded in the on-road autonomous vehicle **10** to self drive to several delivery locations. Upon arrival to a certain delivery location, the sensors **4a**, **4b**, **4c** onboard on-road autonomous vehicle **10** can be used again to identify a specific person to whom one of the packages is intended, and the on-road autonomous vehicle **10** can then order a particular one of the drawers **11-drawer** in the first object **11-obj-1** to open up and allow this person access to the respective package. As is evident from the above example, a synergy is created between the on-road autonomous vehicle **10** and the first object **11-obj-1**, in which such a synergy is the result of a deep integration between the on-road autonomous vehicle **10** and the first object **11-obj-1**, which now operate together as a synchronized and managed single resultant vehicle **10+11-obj-1**. It is noted that after said self integration of (i.e., deep integration between) the on-road autonomous vehicle **10** and the first object **11-obj-1**, the resultant vehicle **10+11-obj-1** has a resultant specific outer shape, for example by now including drawers **11-drawer** that are accessible to people standing alongside the resultant vehicle **10+11-obj-1**, in which such a resultant specific outer shape is critical for achieving the first specific purpose of package delivery. In one embodiment, an interface **5-int-1** is included in the on-road autonomous vehicle **10**, in which maintaining the cooperation and communication between the on-road autonomous vehicle **10** and the first object **11-obj-1** is done via such interface. The interface **5-int-1** may support power transfer, such as electricity, from the on-road autonomous vehicle **10** to the first object **11-obj-1** and vice versa, and may further support data communication between the two, in which commands can be relayed through such interface and sensory data can be shared. In one embodiment, the self integration of the on-road autonomous vehicle **10** with the first object **11-obj-1** is achieved via a straddling process, in which the on-road autonomous vehicle **10** straddles over the first object **11-obj-1**, lowers itself or a connector thereof **5-cnct**, grabs the first object **11-obj-1**, and lifts the first object **11-obj-1**, as described by the sequence shown in FIG. 3A, FIG. 3B, FIG. 3C, FIG. 3D, and in which the first object **11-obj-1** is represented by the label **11**.

[0186] FIG. 13B illustrates one embodiment of the on-road autonomous vehicle **10** still integrated with the first object **11-obj-1** and getting one of the drawers **11-drawer** opened, perhaps for loading or unloading a package into the drawer **11-drawer**.

[0187] FIG. 14A illustrates one embodiment of an on-road autonomous vehicle **10** currently integrated with a second object **11-obj-2** having a door **11-door** that is presently closed. The door **11-door** may allow passengers to get in and out of object **11-obj-2**, which may be a passenger cabin. The

on-road autonomous vehicle **10** is operative to self-integrate with the second object **11-obj-2**, thereby creating a second resultant vehicle **10+11-obj-2** that is the combination of both the autonomous vehicle **10** and the second object **11-obj-2** and that has a second specific purpose, in which such a second specific purpose may be transporting passengers in conjunction with a taxi service. For example, the second resultant vehicle **10+11-obj-2** may self drive to a pick-up destination, in which the door **11-door** gets open as the second resultant vehicle **10+11-obj-2** arrives at the pick-up destination, and a passenger can then get into the passenger cabin **11-obj-2**. Self integration of on-road autonomous vehicle **10** with the second object **11-obj-2** facilitates the second specific purpose of transporting passengers by: (i) combining the features of both the on-road autonomous vehicle **10** and the second object **11-obj-2** in creation of a combined set of features that facilitates self transporting of passengers, and (ii) maintaining cooperation/communication between the on-road autonomous vehicle **10** and the second object **11-obj-2** during the different phases of transporting passengers. For example, self-driving the second resultant vehicle **10+11-obj-2** to the pick-up location is based on sensors (e.g., **4a**, **4b**, **4c**, FIG. 1A) and self-driving directives embedded in the on-road autonomous vehicle **10**, but letting a passenger get onboard the passenger cabin **11-obj-2** by opening the door **11-door** requires further cooperation and communication with the second object **11-obj-2**, for example when the sensors onboard the on-road autonomous vehicle **10** conclude that the second resultant vehicle **10+11-obj-2** is in the correct address and further recognizes the passenger to be picked-up, then consequently the on-road autonomous vehicle **10** instructs the second object **11-obj-2** to open the door **11-door** thereby allowing the recognized passenger in. As is evident from the above example, a different synergy is created between the on-road autonomous vehicle **10** and the second object **11-obj-2** (as compared to the synergy created with the first object **11-obj-1**), in which such a different synergy is the result of a deep integration between the on-road autonomous vehicle **10** and the second object **11-obj-2**, which now operate together as a differently synchronized and managed single second resultant vehicle **10+11-obj-2**. It is noted that after said self integration of the on-road autonomous vehicle **10** and the second object **11-obj-2**, the second resultant vehicle **10+11-obj-2** has a second resultant specific outer shape (as compared to the first resultant outer shape in conjunction with the first object **11-obj-1**), for example by now including a door **11-door** that is located at the right height allowing passengers in the resultant second vehicle **10+11-obj-2**, in which such a second resultant specific outer shape is critical for achieving the second specific purpose of transporting passengers. In one embodiment, the self integration of the on-road autonomous vehicle **10** with the second object **11-obj-2** is achieved via a straddling process, in which the on-road autonomous vehicle **10** straddles over the second object **11-obj-2**, lowers itself or a connector thereof **5-cnct**, grabs the second object **11-obj-2**, and lifts the second object **11-obj-2**, as described by the sequence shown in FIG. 3A, FIG. 3B, FIG. 3C, FIG. 3D, and in which the second object **11-obj-2** is represented by the label **11**. In one embodiment, the self integration of the on-road autonomous vehicle **10** with the second object **11-obj-2** is done after de-integrating the on-road autonomous vehicle **10** with the first object **11-obj-1**, which may be achieved autonomously using a

straddling-off process, in which the on-road autonomous vehicle 10 lowers the second object 11-obj-2, and straddles off the second object 11-obj-2, as can be visualized by “playing in reverse” the sequence shown in FIG. 3A, FIG. 3B, FIG. 3C, FIG. 3D, i.e., starting from FIG. 3D and ending with FIG. 3A.

[0188] FIG. 14B illustrates one embodiment of the on-road autonomous vehicle 10 still integrated with the second object 11-obj-2 and getting the door 11-door opened, perhaps for letting passengers get into and out of the passenger cabin 11-obj-2.

[0189] One embodiment is a system operative to autonomously alter functionality of an on-road vehicle. The system includes: an on-road vehicle 10 (FIG. 13A, FIG. 13B, FIG. 14A, FIG. 14B), operative to straddle over objects 11; a first object 11-obj-1 (FIG. 13A, FIG. 13B) operative to facilitate a first function when integrated with the on-road vehicle 10, in which the first object 11-obj-1 is currently integrated with the on-road vehicle 10 (as shown in FIG. 13A, FIG. 13B), thereby currently enabling the on-road vehicle 10 together with the first object 11-obj-1 to perform said first function; and a second object 11-obj-2 (FIG. 14A, FIG. 14B) operative to facilitate a second function when integrated with the on-road vehicle 10, in which the second object 11-obj-2 is currently located at a certain location (e.g., at location 20-p as shown in FIG. 2B, in which object 11-obj-2 is represented by the label 11).

[0190] In one embodiment, as a response to a specific request received in the system, the system is configured to autonomously alter functionality of the on-road vehicle 10 from a first functionality associated with the first function into a different functionality associated with the second function, in which as a part of said autonomous alteration and said response, the on-road vehicle is configured to: release autonomously the first object 11-obj-1; self drive from a current location of the on-road vehicle 10 to said certain location 20-p of the second object 11-obj-2; upon arrival to said certain location 20-p: straddle autonomously over the second object 11-obj-2 (e.g., as shown in FIG. 3A, FIG. 3B, in which object 11-obj-2 is represented by the label 11), thereby allowing the on-road vehicle 10 to grab and lift autonomously the second object 11-obj-2 above ground (e.g., as shown in FIG. 3C, FIG. 3D), thereby integrating autonomously the second object 11-obj-2 with the on-road vehicle 10; and perform said second function in conjunction with the second object 11-obj-2 now integrated with the on-road 10 vehicle (as shown in FIG. 14A, 14B).

[0191] In one embodiment, the first object 11-obj-1 is essentially a first type of container having a surface with a first type of interface 11-drawer (FIG. 13A, FIG. 13B), in which the first type of interface is operative to facilitate a certain first way of interfacing with people; and the second object 11-obj-2 is essentially a second type of container having a surface with a second type of interface 11-door (FIG. 14A, FIG. 14B), in which the second type of interface is operative to facilitate a certain second way of interfacing with people.

[0192] In one embodiment, the first type of container 11-obj-1 is a container operative to contain packages in drawers 11-drawer; the first function is autonomous package delivery; the first type of interface is associated with at least one of the drawers 11-drawer getting opened (FIG. 13A, FIG. 13B); the first way of interfacing with people comprises people accessing the drawers 11-drawer and collect-

ing a package delivered by the on-road vehicle 10; and the first function of autonomous package delivery is facilitated by said integration of the first object 11-obj-1 with the on-road vehicle 10, in which said integration enables the system to both: (i) facilitate said delivery by driving autonomously the on-road vehicle 10 with packages onboard, and (ii) facilitate said collection of the packages by people accessing the drawers 11-drawer.

[0193] In one embodiment, the second type of container 11-obj-2 is a container operative to accommodate passengers; the second function is an autonomous taxi service; the second type of interface is associated with a door 11-door (FIG. 14A, FIG. 14B) operative to allow the passengers getting into and out-of the second type of container 11-obj-2; the second way of interfacing with people comprises people opening and closing the door 11-door; and the second function of autonomous taxi service is facilitated by said integration autonomously of the second object 11-obj-2 with the on-road vehicle 10, in which said integration autonomously enables the system to both: (i) facilitate said taxi service by autonomously transporting the passengers by the on-road vehicle 10 and in conjunction with the second object 11-obj-2, and (ii) further facilitate said taxi service by said allowing the passengers to get into and out-of the second type of container 11-obj-2 using the door 11-door.

[0194] In one embodiment, said lifting autonomously of the second object 11-obj-2 above ground is done so as to position the door 11-door at a certain height above ground that is operative to allow said passengers getting into and out-of the second type of container 11-obj-2, in which said certain height is between 20 (twenty) centimeters and 70 (seventy) centimeters above ground.

[0195] In one embodiment, the second object 11-obj-2 is selected from a group consisting of: (i) a container operative to contain packages in drawers 11-drawer, in which the second function is autonomous package delivery, (ii) a container operative to accommodate passengers 30 (FIG. 5B), in which the second function is a taxi service, (iii) a container operative to contain a load, in which the second function is transporting loads, (iv) a power source such as battery (e.g., 11-func, FIG. 7B), a fuel cell, and a generator, in which the second function is charging the on-road vehicle 10 while on the move, (v) a mobile vending machine, in which the second function is selling goods at different locations, (vi) a tank, in which the second function is transporting substances such as liquids and compressed gas, (vii) transportable electronic communication equipment such as a radio access network (RAN), in which the second function is providing electronic communication services from different locations.

[0196] In one embodiment, the on-road vehicle 10 comprises: an upper horizontal structure 3a, 3b, 3c (FIG. 1A) elevated above ground by vertical structures 2a, 2a', 2b, 2b', 2c, 2c', 2d, 2d' (FIG. 1A) mounted on at least four wheels 1a, 1b, 1c, 1d (FIG. 1A) touching ground 9-ground, so as to create a certain clearance 9-clr above ground for at least a first connector 5-cnct associated with the upper horizontal structure 3b and attached thereunder; a control sub-system 4, 6, 7, 8 comprising a processing unit 8 (FIG. 1D) and a plurality of sensors 4a, 4b, 4c (FIG. 1A) and actuators 6, 7 (FIG. 1D), in which the control sub-system is configured to generate, in real-time, a three-dimensional representation of surrounding environment using data collected by the plurality of sensors; and at least a first linear actuator 2'+2 (2'

moving up and down relative to 2, i.e., 2a' moving relative to 2a, 2b' moving relative to 2b, 2c' moving relative to 2c, and 2d' moving relative to 2d, FIG. 1A) configured to control and set said certain clearance 9-clr of the first connector 5-cnct, by causing the first connector, or the entire upper horizontal structure 3a, 3b, 3c including the first connector, to move up or down relative to ground 9-ground. In one embodiment, the control sub-system 4, 6, 7, 8 is further configured to use said three-dimensional representation, said actuators 6, 7, and said processing unit 8 in conjunction with a set of public-road self-driving directives, to: (i) facilitate said self-driving (FIG. 2A, FIG. 2B) of the on-road vehicle 10, over public roads 20a (FIG. 2A), 20b (FIG. 2B) and alongside regular car traffic 21a, 21b, 21c, 21d, 21e, 21f (FIG. 2A, FIG. 2B, FIG. 2C), to said certain location 20-p, (ii) position the on-road autonomous vehicle 10 in front of the second object 11-obj-2 (11 in FIG. 2C, FIG. 3A), and (iii) facilitate said straddling of the on-road vehicle 10 over the second object 11-obj-2 (11 in FIG. 3B), such that said first connector 5-cnct is brought to a predetermined position 5-pos over the second object 11-obj-2 (11 in FIG. 3B); and the control sub-system 4, 6, 7, 8 is further configured to use the first linear actuator 2'+2 to: (i) facilitate said grabbing by lowering the first connector 5-cnct into mechanical contact with the second object 11-obj-2 (11 in FIG. 3C) thereby allowing the connector to connect to or otherwise grab the second object, and (ii) facilitate said lifting by lifting (FIG. 3D) the second object 11-obj-2 above ground into a position operative to self-transport the second object.

[0197] In one embodiment, said first linear actuator 2'+2 is a distributed linear actuator comprising several sub-actuators 2a, 2a', 2b, 2b', 2c, 2c', 2d, 2d' (FIG. 1A), in which each sub-actuator is associated with one of the wheels 1a, 1b, 1c, 1d (FIG. 1A), such that the entire upper horizontal structure 3a, 3b, 3c is operative to move up and down relative to the wheels 1a, 1b, 1c, 1d and ground 9-ground, in which the linear actuators 2'+2 are also operative to act as springs/mechanical dampers for the wheels 1a, 1b, 1c, 1d relative to the upper horizontal structure 3a, 3b, 3c.

[0198] In one embodiment, said first linear actuator 2'+2 is embedded in the first connector 5-cnct, thereby causing only the connector to move up and down relative to ground 9-ground, and such that the upper horizontal structure 3a, 3b, 3c remains in place.

[0199] In one embodiment, said integration of the first object 11-obj-1 with the on-road vehicle 10 was previously achieved by the on-road vehicle by performing a previous autonomous maneuver as a response to a particular request received in the system, in which as part of the previous autonomous maneuver, the on-road vehicle 10 is configured to: self drive from a previous location of the on-road vehicle to a specific location at which the first object 11-obj-1 is located; upon arrival to said specific location: straddle autonomously over the first object 11-obj-1, thereby allowing the on-road vehicle 10 to grab and lift autonomously the first object above ground, thereby facilitating said integration of the first object 11-obj-1 with the on-road vehicle 10.

[0200] In one embodiment, as a part of said releasing autonomously of the first object 11-obj-1, the on-road vehicle 10 is configured to: (i) lower the first object, (ii) un-grab the first object and (iii) straddle off the first object.

[0201] In one embodiment, said autonomously altering of functionality is facilitated by a combination of different

autonomous functions working in synchronization, in which said combination of different autonomous functions comprises: (i) said self-driving, thereby allowing the on-road vehicle 10 to autonomously access the second object 11-object-2, (ii) said straddling autonomously, thereby allowing the on-road vehicle 10 to self-align with the second object 11-object-2, and (iii) said grabbing and lifting autonomously of the second object 11-object-2, thereby allowing the on-road vehicle 10 to self-integrate with the second object, in which said autonomously altering of functionality is further facilitated by an ability of the on-road vehicle 10 to straddle over the respective objects 11-object-1, 11-object-2, and in which said autonomously altering of functionality comprises at least said alteration of functionality taking place without relying on external support such as support from people for driving, support from people for lifting and displacing/aligning loads, and support from external mechanical devices for lifting and displacing/aligning loads; and said autonomously altering of functionality is further facilitated by the on-road vehicle eclectically interfacing 5-int-1 with the respective objects, in which: said electrically interfacing is done in conjunction with said grabbing of the respective object; said electrically interfacing comprises at least one of: (i) supplying electrical power to the respective object, (ii) supplying communication services to the respective object, and (iii) supplying sensory information to the respective object, thereby allowing the respective object to better interact with people in conjunction with accomplishing the respective functionality; and said electrically interfacing 5-int-1 is a part of said integration.

[0202] In one embodiment, said autonomously altering of functionality comprises switching/changing/modifying autonomy mode by the on-road vehicle, from a first autonomy mode into a second autonomy mode, in which the first autonomy mode is operative to support a first automatic on-road behavior that facilitates the first function, and the second autonomy mode is operative to support a second and different automatic on-road behavior that facilitates the second function, in which said switching/changing of the autonomy mode is a part of said integration. In one embodiment, the first object 11-obj-1 is a container operative to contain packages in drawers 11-drawer; the first function is autonomous package delivery; and the first autonomy mode is a mode that automatically facilitates arriving with the packages to a vicinity of people and enabling the people to access the drawers 11-drawer and collect a package delivered by the on-road vehicle 10. In one embodiment, the second object 11-obj-2 is a container operative to accommodate passengers; the second function is an autonomous taxi service; and the second autonomy mode is a mode that facilitates said taxi service by autonomously transporting the passengers by the on-road vehicle 10, and autonomously allowing the passengers to get into and out-of the second type of container 11-obj-2 using a door 11-door.

[0203] FIG. 15 illustrates one embodiment of a method for autonomously altering functionality of an on-road vehicle. The method includes: In step 1101, performing a first function by an on-road vehicle 10, in which the first function is performed by the on-road vehicle in conjunction with a first object 11-obj-1 that is currently integrated with the on-road-vehicle and that is operative to facilitate said first function. In step 1102, receiving, in conjunction with the on-road vehicle 10, a request associated with altering functionality of the on-road vehicle from a first functionality

associated with the first function into a different functionality associated with a second function. In step 1103, releasing autonomously the first object 11-obj-1 by the on-road vehicle 10 as a response to said request. In step 1104, self driving by the on-road vehicle 10, as a further response to said request, from a current location of the on-road vehicle to a certain location at which a second object 11-obj-2 is located, in which the second object is associated with said different functionality. In step 1105, upon arrival to said certain location: (i) straddling autonomously, by the on-road vehicle 10, over the second object 11-obj-2, (ii) grabbing and lifting, autonomously, the second object 11-obj-2 above ground by the on-road vehicle 10, and thereby autonomously integrating the second object 11-obj-2 with the on-road vehicle 10. In step 1106, performing said second function by the on-road vehicle in conjunction with the second object 11-obj-2 now integrated with the on-road vehicle 10.

[0204] In one embodiment, the method further comprises: receiving, in conjunction with the on-road vehicle 10, and prior to said performing of the first function, a prior request associated with altering functionality of the on-road vehicle into the first functionality associated with the first object 11-obj-1; self driving by the on-road vehicle 10, as a response to said prior request, from a previous location of the on-road vehicle to a particular location at which the first object 11-obj-1 is located; and upon arrival to said particular location: (i) straddling autonomously, by the on-road vehicle 10, over the first object 11-obj-1, (ii) grabbing and lifting, autonomously, the first object 11-obj-1 above ground by the on-road vehicle 10, thereby facilitating said integration of the first object 11-obj-1 with the on-road vehicle 10 and said performing of the first function by the on-road vehicle 10.

[0205] In one embodiment, said releasing autonomously of the first object 11-obj-1 by the on-road vehicle 10 comprises: lowering the first object 11-obj-1 by the on-road vehicle 10; un-grabbing the first object 11-obj-1 by the on-road vehicle 10; and straddling off the first object 11-obj-1 by the on-road vehicle 10.

[0206] One embodiment is a system operative to respond to a dynamic demand for various functions by autonomously altering shape and functionality of on-road vehicles. The system includes: a fleet of on-road vehicles comprising a plurality of on-road vehicles such as vehicle 10, in which each of the on-road vehicles 10 is configured to autonomously-upon-demand pick-up and integrate-with various objects 11-obj-1, 11-obj-2; and a pool of objects 11-obj-1, 11-obj-2 comprising said various objects, in which each of the objects is associated with a respective functionality, thereby collectively supporting a variety of functionalities, and in which there are more objects 11-obj-1, 11-obj-2 in the pool than on-road vehicles 10 in the system.

[0207] In one embodiment, the system is configured to: determine current demands for various functionalities; determine, based on said current demands, a new assignment of functionalities for at least some of the on-road vehicles 10, in which said new assignment is expected to allow the fleet of on-road vehicles to better respond to the demands; and per each of the on-road vehicles 10 for which a new assignment of functionality was determined, the on road vehicle 10 is configured to: (i) release one of the objects 11-obj-1 that is currently integrated therewith, (ii) self drive from a current location of the on-road vehicle 10 to a location of parking of another one of the objects 11-obj-2 that is associated with the

respective functionality newly assigned, and (iii) upon arrival to the location of parking: straddle autonomously over the another object 11-obj-2, thereby allowing the on-road vehicle 10 to grab and lift autonomously said another object 11-obj-2 above ground, and thereby integrating autonomously the another object 11-obj-2 with the on-road vehicle 10, thus embedding in the on-road vehicle 10 the respective functionality newly assigned.

[0208] In one embodiment, per each of the on-road vehicles 10 for which a new assignment of functionality was determined: said integrating autonomously of the respective another object 11-obj-2 with the on-road vehicle 10 results in an alteration of an outer shape of the on-road vehicle 10 from a previous outer shape associated with the respective previously integrated object 11-obj-1 into a new outer shape associated with the respective newly integrated object 11-obj-2.

[0209] In one embodiment, said alteration of the outer shape facilitates said new functionality assigned.

[0210] In one embodiment, said previous outer shape is associated with a first outer interface (e.g., 11-drawer) in the previously integrated object 11-obj-1, in which said first outer interface is associated with a first way of interacting with people; and said new outer shape is associated with a second outer interface (e.g., 11-door) in the newly integrated object 11-obj-2, in which said second outer interface is associated with a second way of interacting with people.

[0211] FIG. 16 illustrates several embodiments of on-road autonomous vehicles having several different sizes respectively. A large on-road autonomous vehicle 10-large is operative to straddle over and pick up large-sized containers/loads 11-large, in which large-sized containers are containers having at least one dimension (e.g., length) that is longer than 2 (two) meters, and in which such containers are operative to be used in conjunction with various functions, such as transporting passengers and transporting/delivering packages. A medium on-road autonomous vehicle 10-medium is operative to straddle over and pick up medium-sized containers/loads 11-medium, in which medium-sized containers are containers having at least one dimension between 1 (one) meter and 2 (two) meters, and in which such containers are operative to be used in conjunction with various functions, such as transporting/delivering groceries and packages. A small on-road autonomous vehicle 10-small is operative to straddle over and pick up small-sized containers/loads 11-small, in which small-sized containers are containers having at least one dimension between 50 (fifty) centimeters and 1 (one) meter, and in which such containers are operative to be used in conjunction with various functions, such as transporting/delivering packages and delivering food. In one embodiment, the small load 11-small is, by itself, a package being delivered by the small on-road autonomous vehicle 10-small. A tiny on-road autonomous vehicle 10-tiny is operative to straddle over and pick up tiny containers/loads 11-tiny, in which tiny containers are containers having at least one dimension between 20 (twenty) centimeters and 50 (fifty) centimeters, and in which such containers are operative to be used in conjunction with various functions, such as transporting/delivering packages, documents, and medicine. In one embodiment, the tiny load 11-tiny is, by itself, a package being delivered by the small on-road autonomous vehicle 10-small.

[02112] FIG. 17A illustrates one embodiment of a wheel robot 10 carrying a disposable package 11-crb using straps 92-strap.

[02113] FIG. 17B illustrates one embodiment of a wheel robot 10 carrying a disposable package 11-crb by inserting grabbing elements 5-cnct into handle holes 92-ph embedded in the package.

[02114] FIG. 17C illustrates one embodiment of a wheel robot 10 carrying a disposable package 11-crb using side arms 5-side-arm operative to move sideways to release the package.

[02115] FIG. 17D illustrates one embodiment of a wheel robot 10 carrying a disposable package 11-bag comprising a bag.

[02116] One embodiment is a system operative to autonomously deliver consumer items in disposable packages. The system includes: a wheel robot 10 (FIG. 17A) operative to straddle over and off objects; at least one type of disposable package 11 (FIG. 17A) comprising a respective type of disposable handle 92 (FIG. 17A); and a grabbing mechanism 5-cnct (FIG. 17A) embedded in the wheel robot 10.

[02117] In one embodiment, the wheel robot 10 is configured to carry a consumer item 93-ci (FIG. 17A) in said disposable package 11 from a pick-up location to a drop-off location, in which the grabbing mechanism 5-cnct is configured to grab the disposable package 11 by the respective disposable handle 92 and under the wheel robot during said carriage. Upon arrival of the wheel robot 10 to the drop-off location, the grabbing mechanism 5-cnct is configured to un-grab the disposable package 92, thereby releasing the disposable package on ground under the wheel robot (e.g., on a sidewalk, on a road, on grass, or on a flat object placed on ground); and the wheel robot 10 is further configured to straddle off the disposable package 92 and the consumer item 93-ci, thereby leaving the consumer item in the disposable package on the ground for a consumer to pick up.

[02118] In one embodiment, the disposable package 11 is of a cardboard packaging box type 11-crb (FIG. 17A, FIG. 17B, FIG. 17C). In one embodiment, the respective disposable handle type 92 comprises at least one of: (i) at least one strap 92-strap (FIG. 17A) attached to the top of the cardboard packaging box 11-crb, (ii) punch-holes 92-ph (FIG. 17B, FIG. 17C) in opposite two sides of the cardboard packaging box 11-crb, (iii) a cardboard handle element at the top of the cardboard packaging box, and (iv) a rope and/or a string and/or a strap around and/or through the cardboard packaging box.

[02119] In one embodiment, the disposable package 11 is of a plastic bag type 11-bag (FIG. 17D). In one embodiment, the respective disposable handle type 92 is an integral part 92-in (FIG. 17D) of the plastic bag 11-bag. In one embodiment, the plastic bag 11-bag is made of a material comprising at least one of: (i) high-density polyethylene (HDPE), (ii) low-density polyethylene (LDPE), and (iii) linear low-density polyethylene (LLDPE). In one embodiment, the plastic bag 11-bag is made of a plastic net, and the respective disposable handle type 92 comprises a part of the plastic net. In one embodiment, the plastic bag 11-bag is made of polyvinyl chloride (PVC) clear semi-rigid plastic structure.

[0220] In one embodiment, the disposable package 11 is of a paper bag type, in which the respective disposable handle type 92 comprises at least one of: (i) punch-holes in the paper bag, and (ii) at least one strap and/or a string attached to the paper bag.

[0221] In one embodiment, the disposable package 11 is of a cloth and/or canvas bag type, in which the respective disposable handle type comprises at least one strap and/or string attached to the top of the cloth and/or canvas bag.

[0222] In one embodiment, the wheel robot 10 is small in dimensions relative to an average on-road vehicle, measuring less than 2.5 (two point five) meters in length, less than 1.5 (one point five) meters in height, and weighing less than 700 (seven hundred) kilograms, thereby allowing the wheel robot access to sidewalks. In one embodiment, the drop-off location is situated on a sidewalk and in proximity to an address associated with the consumer; and the wheel robot 10 is configured to travel on the sidewalk while avoiding pedestrians. In one embodiment, the disposable package 11 comprises a cardboard packaging box 11-crb, and prior to said releasing of the disposable package with the consumer item inside, the wheel robot is configured to detect that the drop-off location is at least dry, thereby assuring survivability of the cardboard packaging box and the consumer item inside. In one embodiment the cardboard packaging box 11-crb comprises a disposable extension attached to a bottom side of the cardboard packaging, thereby assuring survivability of the cardboard packaging when the drop-off location is wet. In one embodiment, said extension comprises cardboard structures such as cardboard legs.

[0223] In one embodiment, the wheel robot 10 is small in dimensions relative to an average human being, measuring less than 1.5 (one point five) meters in length, less than 1 (one) meter in height, and weighing less than 200 (two hundred) kilograms; the disposable package 11 is small in dimensions relative to an average human being, has at least one dimension of less than 1 (one) meter, and is therefore carriable by the average human being; and the disposable package 11 comprises at least one of: (i) a plastic bag 11-bag, (ii) a paper bag, (iii) a cardboard packaging box 11-crb, and (iv) a polystyrene foam structure. In one embodiment, the wheel robot 10 is configured to move at speeds of less than 2 (two) meters per second, thereby increasing safety in conjunction with said sidewalk travel and pedestrian avoidance. In one embodiment, the disposable handle 92 is dual purposed, in which a first purpose is associated with said grabbing mechanism 5-cnct and the second purpose is associated with the consumer picking up the disposable package 11 with the consumer item 93-ci inside.

[0224] In one embodiment, prior to said carriage from the pick-up location to the drop-off location: the wheel robot 10 is further configured to straddle over the disposable package 11 containing the consumer item 93-ci at the pick-up location; and

[0225] the grabbing mechanism 5-cnct is further configured to grab the disposable package 11 by the respective disposable handle 92.

[0226] FIG. 18 illustrates one embodiment of a method for autonomously delivering consumer items in disposable packages. The method includes: In step 1111, receiving an order to deliver a consumer item 93-ci (FIG. 17A, FIG. 17D) to a consumer. In step 1112, grabbing, by a grabbing mechanism 5-cnct (FIG. 17A) in a wheel robot 10 (FIG. 17A), a disposable package 11 (FIG. 17) containing the consumer item 93-ci at a pick-up location. In step 1113, carrying, by a wheel robot 10, the consumer item 93-ci in the disposable package 11 from the pick-up location to a drop-off location, in which the grabbing mechanism 5-cnct is configured to grab the disposable package during said car-

riage and under the wheel robot. In step 1114, upon arrival of the wheel robot 10 to the drop-off location, un-grabbing the disposable package 11, thereby releasing the disposable package on ground under the wheel robot. In step 1115, straddling, by the wheel robot 10, off the disposable package 11 and the consumer item 93-ci, thereby leaving the consumer item in the disposable package on the ground for a consumer to pick up.

[0227] In one embodiment, said grabbing is done so as to place a bottom of the disposable package 11 at a height of no more than 1 (one) meter above ground, and no less than 10 (ten) centimeters above ground during said carriage; and said releasing results in said disposable package 11 with the consumer item inside 93-ci falling a short distance of between 10 centimeters and one meter to the ground, in which a survival of the consumer item when the disposable package hits the ground is facilitated by at least one of: (i) said distance being short, and (ii) said disposable package comprising a soft and/or collapsible cushioning material 94-cm (FIG. 17D) that is an integral part of the disposable package 11 and/or that is attached to the bottom side of the disposable package.

[0228] In one embodiment, the method further comprises prior to said grabbing: straddling, by the wheel robot 10, over the disposable package 11 containing the consumer item 93-ci at the pick-up location; lowering the grabbing mechanism 5-cnct to a position to perform said grabbing; and lifting above ground, by the grabbing mechanism 5-cnct, the disposable package 11 together with the consumer item 93-ci; in which said releasing comprises the grabbing mechanism 5-cnct lowering the disposable package 11 together with the consumer item 93-ci to the ground prior to said un-grabbing.

[0229] In one embodiment, the consumer item 93-ci comprises at least one of: (i) groceries, (ii) a home appliance, (iii) clothing, (iv) a consumer electronic device, (v) books and/or other paper products, (vi) a pharmaceutical product, (vii) a toy, (viii) a takeout item, (ix) a parcel, (x) a kitchenware item, (xi) a furniture item, (xii) a care product, (xiii) a beauty product, (xiv) a toiletry product, and (xv) a tool item.

[0230] FIG. 19A illustrates one embodiment of an on-road autonomous vehicle 10 carrying a first package container 11m-4. The on-road vehicle 10 is operative to straddle-over and grab the package container using a grabbing mechanism 5-cnct-8 in accordance with some embodiments. A second grabbing mechanism 5-cnct-9 is also shown.

[0231] FIG. 19B illustrates one embodiment of the on-road autonomous vehicle 10 carrying the first package container 11m-4 and a second package container 11m-5 at the same time and in tandem. The on-road vehicle 10 is operative to straddle-over and grab the second package container 11m-5 using a grabbing mechanism 5-cnct-9 in accordance with some embodiments. The on-road vehicle 10 may straddle-over and grab the second package container 11m-5 via 5-cnct-9 while the first package container 11m-4 is already attached via 5-cnct-8. The on-road vehicle 10 may straddle-over and grab package container 11m-4 first and then straddle-over and grab package container 11m-5, or grab package container 11m-5 first and then straddle-over and grab package container 11m-4. The on-road vehicle 10 is operative to release and straddle-off the second package container 11m-5 in accordance with some embodiments. The on-road vehicle 10 may release and straddle-off the second package container 11m-5 while the first package container

11m-4 is still attached via 5-cnct-8. The on-road vehicle 10 may release and straddle-off package container 11m-4 first and then release and straddle-off package container 11m-5, or release and straddle-off container 11m-5 first and then release and straddle-off package container 11m-4.

[0232] In one embodiment, the on-road vehicle 10 may transport only one of the package containers in some occasions, as shown in FIG. 19A, and may transport two or more of the package containers in other occasions as shown in FIG. 19B, according to a current need and in conjunction with some embodiments.

[0233] It is noted that on-road vehicle 10 is depicted as carrying two of the packages 11m-4, 11m-5, but on-road vehicle 10 may carry more packages in tandem using additional grabbing mechanisms 5-cnct (not shown), or it may carry more packages side-by-side rather than in tandem (not shown).

[0234] FIG. 19C illustrates one embodiment of the first package container 11m-4 comprising a plurality of lockers 11L-41, 11L-42 containing a respective plurality of packages 11pkg-41, 11pkg-42, in which each of the packages in the lockers 11L-41, 11L-42 is accessible via a respective locker door 11dr-41, 11dr-42. The package container 11m-4 includes a grabbing point 5-grb-4 via which the on-road vehicle 10 is able to grab and lift the container, using 5-cnct-8 for example, into a transport position as illustrated in FIG. 19A.

[0235] FIG. 19D illustrates one embodiment of the second package container 11m-5 comprising a plurality of lockers 11L-51, 11L-52 containing a respective plurality of packages 11pkg-51, 11pkg-52, in which each of the packages in the lockers 11L-51, 11L-52 is accessible via a respective locker door 11dr-51, 11dr-52. The package container 11m-5 includes a grabbing point 5-grb-5 via which the on-road vehicle 10 is able to grab and lift the container, using 5-cnct-9 for example, into a transport position as illustrated in FIG. 19B.

[0236] In one embodiment, each of the lockers 11L-41, 11L-42, 11L-51, 11L-52 may include a locking mechanism, e.g., the locking mechanism 11CLK of locker 11L-52 is shown. The locking mechanisms may be individually commanded externally to allow unlocking of a respective one of the doors 11dr-41, 11dr-42, 11dr-51, 11dr-52 in conjunction with a respective customer approaching the container to pick-up a respective one of the packages 11pkg-41, 11pkg-42, 11pkg-51, 11pkg-52.

[0237] FIG. 19E illustrates one embodiment of the on-road autonomous vehicle 10 and package containers 11m being in communicative contact 10-comm, e.g., wireless contact, with a server 8-server that may be external to vehicle 10, e.g., located in a control center, or embedded in the vehicle, e.g., part of processing unit 8 (FIG. 1D). The package containers 11m may include a locally embedded processor 8-embd operative to receive instruction and send updates from/to server 8-server. Any operation and/or action executed in the computers 8, 8-server, 8-embd may be facilitated and governed by a non-transitory and/or transitory machine readable code stored locally in conjunction with the respective computer using memory components such as flash memory, random-access-memory (RAM), and hard-disk-drives (HDD).

[0238] FIG. 19F illustrates one embodiment of an example of a route 8-leg-1, 8-leg-2, 8-leg-3, 8-leg-4, 8-leg-5, 8-leg-6, 8-leg-7, 8-leg-8 taken by the on-road autonomous vehicle 10

in conjunction with moving and placing the package containers **11m-4**, **11m-5** in different delivery locations **7-loc**, as described in the following paragraphs:

[0239] According to one example, on-road vehicle **10** may first grab both of containers **11m-4**, **11m-5** from a location of origin **7-loc-0** associated, perhaps, with a warehouse and/or a store, in which the containers **11m-4**, **11m-5** are pre-loaded with the packages **11pkg-41**, **11pkg-42**, **11pkg-51**, **11pkg-52** to be delivered en-route **8-leg**. On-road vehicle **10** then transports the containers **11m-4**, **11m-5**, together, to a first location **7-loc-1**, perhaps a parking location near a house in which a first customer resides. On-road vehicle **10** then releases the first package container **11m-4** at location **7-loc-1**, and server **8-server** receives **10-comm**, either from container **11m-4** or vehicle **10**, an indication that package **11pkg-41** is at location **7-loc-1**, following which the server **8-server** may relate this information to the first customer, perhaps via his smartphone. However, vehicle **10** does not wait for the first customer to pick-up package **11pkg-41**, but instead continues on its journey **8-leg-2** toward location **7-loc-3** with the second package container **11m-5** still onboard, while leaving container **11m-4** at location **7-loc-1** together with package **11pkg-41** (and **11pkg-42** as well).

[0240] Upon arrival of vehicle **10** to location **7-loc-3**, it releases the second package container **11m-5**, and server **8-server** receives an indication that package **11pkg-51** is at location **7-loc-3**, following which the server **8-server** may relate this information to another customer associated with location **7-loc-3**.

[0241] Then, upon an indication received in server **8-server**, that the first customer has picked up package **11pkg-41** from locker **11L-41** of container **11m-4** at location **7-loc-1**, the vehicle **10** returns **8-leg-3** to location **7-loc-1**, without any container onboard, to grab container **11m-4** and transport it **8-leg-4** to location **7-loc-2**. Upon arrival of container **11m-4** to location **7-loc-2**, server **8-server** receives an indication that package **11pkg-42** is at location **7-loc-2**, following which the server **8-server** may relate this information to yet another customer associated with location **7-loc-2**.

[0242] Then, upon an indication received in server **8-server**, that package **11pkg-51** has been picked up by the respective customer from locker **11L-51** of container **11m-5** at location **7-loc-3**, the vehicle **10** returns **8-leg-5** to location **7-loc-3**, without any container onboard, to grab container **11m-5** and transport it **8-leg-6** to location **7-loc-4**. Upon arrival of container **11m-5** to location **7-loc-4**, server **8-server** receives an indication that package **11pkg-52** is at location **7-loc-4**, following which the server **8-server** may relate this information to yet another customer associated with location **7-loc-4**.

[0243] Then, upon an indication received in server **8-server**, that packages **11pkg-52** and **11pkg-42** have been picked up by the respective customers from lockers **11L-42** of container **11m-4** at location **7-loc-2** and **11L-52** of container **11m-5** at location **7-loc-4**, the vehicle **10** grabs container **11m-5** at location **7-loc-4**, returns **8-leg-7** to location **7-loc-2**, with container **11m-5** onboard, to grab container **11m-4** as well, and transport both containers **8-leg-8** back to the origin **7-loc-0**. Upon arrival of containers **11m-4**, **11m-5** to the origin **7-loc-0**, the containers may be loaded again with additional packages to be delivered in conjunction with a next delivery round.

[0244] One embodiment is a system operative to autonomously deliver packages using a plurality of package containers, comprising: at least a first **11m-4** (FIG. 19C) and a second **11m-5** (FIG. 19D) package containers, in which each of the package containers comprises a plurality of lockers **11L** (FIG. 19C, FIG. 19D) each comprising a respective door **11dr** (FIG. 19C, FIG. 19D) operative to be locked and unlocked independently of the other doors in the respective package container, and in which each of at least some of the lockers contains a respective package **11pkg** (FIG. 19C, FIG. 19D) to be delivered; and an on-road vehicle **10** (FIG. 19A) operative to straddle over the package containers **11m-4**, **11m-5**.

[0245] In one embodiment, upon reception of an indication pertaining to a delivery status associated with a first package **11pkg-41** in one of the lockers **11L-41** of the first container **11m-4** parked at a first location **7-loc-1** (FIG. 19F), the on-road vehicle **10** is configured to: straddle-over and grab the first container **11m-4** from the first location **7-loc-1**, transport **8-leg-4** (FIG. 19F) the first container **11m-4** to a second location **7-loc-2**, and straddle-off the first container to be left parked at the second location; then upon reception of an indication pertaining to a delivery status associated with a second package **11pkg-51** in one of the lockers **11L-51** of the second container **11m-5** parked at a third location **7-loc-3**, the on-road vehicle **10** is configured to: straddle-over and grab the second container **11m-5** from the third location **7-loc-3**, transport **8-leg-6** the second container **11m-5** to a fourth location **7-loc-4**, and straddle-off the second container to be left parked at the fourth location; in which the system is configured to: in conjunction with arrival of the first container **11m-4** at the second location **7-loc-2**, send an indication that a third package **11pkg-42** is now and/or will be in place to be picked-up from another one of the lockers **11L-42** of the first container **11m-4**; and in conjunction with arrival of the second container **11m-5** at the fourth location **7-loc-4**, send an indication that a fourth package **11pkg-52** is now and/or will be in place to be picked-up from another one of the lockers **11L-52** of the second container **11m-5**.

[0246] In one embodiment, said indication pertaining to the delivery status associated with the first package **11pkg-41** is an indication that the first package was picked-up out of one of the lockers **11L-41** of the first container **11m-4** parked at the first location **7-loc-1**.

[0247] In one embodiment, said indication pertaining to the delivery status associated with the first package **11pkg-41** is an indication that the first package has been failed to be picked-up out of one of the lockers **11L-41** of the first container **11m-4** parked at the first location **7-loc-1**.

[0248] In one embodiment, each of the lockers **11L** is integrated in one of the sides of the respective package container **11m** so as to allow direct access to a person picking-up the respective package **11pkg**.

[0249] In one embodiment, each of said doors **11dr** constitutes a part of said side of the respective package container **11m**.

[0250] In one embodiment, at least one of the doors, e.g., **11dr-52**, is operative to be opened via a respective hinge in one of the sides of the door.

[0251] In one embodiment, at least one of the doors **11dr** is a part of drawer containing the respective package **11pkg**, e.g., similar to **11-drawer** in FIG. 13B.

[0252] In one embodiment, each of the lockers 11L is operative to be unlocked using an electronic lock mechanism 11LCK (FIG. 19D) embedded in the locker.

[0253] In one embodiment, upon reception of a respective unlock indication, the system is configured to electronically unlock the electronic lock mechanism 11LCK of a certain one of the lockers 11L in order to allow a respective authorized person to pick-up the respective package 11pkg.

[0254] In one embodiment, each of the package containers 11m has a weight of at least 50 (fifty) kilograms, so as to make it difficult for a single person to manually carry the package container, thereby preventing theft.

[0255] In one embodiment, each of the package containers 11m has a weight of at least 100 (one hundred) kilograms, so as to make it difficult for a non-straddling on-road vehicle to transport the package container away from the respective parking location, thereby preventing theft.

[0256] In one embodiment, prior to said transports 8-leg-4, 8-leg-6, the on-road vehicle 10 is further configured to: straddle-over and grab the first package container 11m-4 parked at a warehouse location 7-loc-0, as illustrated in FIG. 19A; straddle-over and grab the second package container 11m-5 parked at the same warehouse location 7-loc-0, thereby grabbing both the first and second package containers 11m-4, 11m-5 at the same time and in tandem, as illustrated in FIG. 19B; carry 8-leg-1 (FIG. 19F) the first and second package containers 11m-4, 11m-5 together to the first location 7-loc-1 and straddle-off the first package container 11m-4; and carry 8-leg-2 the remaining second package container 11m-5 to the third location 7-loc-3 and straddle-off the second package container.

[0257] One embodiment is a system operative to autonomously deliver packages using a plurality of package containers, comprising: at least a first 11m-4 (FIG. 19C) and a second 11m-5 (FIG. 19D) package containers, in which the package containers are operative to contain packages 11pkg (FIG. 19C, FIG. 19D) to be delivered; and an on-road vehicle 10 (FIG. 19A, FIG. 19B) operative to straddle over the package containers 11m-4, 11m-5.

[0258] In one embodiment, the on-road vehicle 10 is configured to: straddle-over and grab the first package container 11m-4 parked at a warehouse location 7-loc-0 (FIG. 19F), as illustrated in FIG. 19A; straddle-over and grab the second package container 11m-5 parked at the same warehouse location 7-loc-0, thereby grabbing both the first and second package containers 11m-4, 11m-5 at the same time and in tandem, as illustrated in FIG. 19B; carry 8-leg-1 (FIG. 19F) the first and second package containers together 11m-4, 11m-5 to a first location 7-loc-1 (FIG. 19F) and straddle-off the first package container 11m-4, thereby leaving the first package container parked at the first location so as to allow pick-up of a first package 11pkg-41 (FIG. 19C); and carry 8-leg-2 (FIG. 19F) the remaining second package container 11m-5 to a third location 7-loc-3 (FIG. 19F) and straddle-off the second package container, thereby leaving the second package container parked at the third location so as to allow pick-up of a second package 11pkg-51 (FIG. 19D).

[0259] In one embodiment, the on-road vehicle 10 is further configured to: straddle-over grab and carry the first package container 11m-4 from the first location 7-loc-1 to a second location 7-loc-2; and straddle-off the first package

container 11m-4, thereby leaving the first package container parked at the second location 7-loc-2 so as to allow pick-up of a third package 11pkg-42.

[0260] In one embodiment, the on-road vehicle 10 is further configured to: straddle-over grab and carry the second package container 11m-5 from the third location 7-loc-3 to a fourth location 7-loc-4; and straddle-off the second package container 11m-5, thereby leaving the second package container parked at the fourth location 7-loc-4 so as to allow pick-up of a fourth package 11pkg-52.

[0261] In one embodiment, the on-road vehicle 10 is further configured to: straddle-over grab and carry 8-leg-7 the second package container 11m-5 from a fourth location 7-loc-4 to a second location 7-loc-2; and straddle-over and grab the first package container 11m-4 parked at the second location 7-loc-2, thereby grabbing both the first and second package containers 11m-4, 11m-5 at the same time and in tandem; and carry 8-leg-8 the first and second package containers together 11m-4, 11m-5 to the warehouse location 7-loc-0 and straddle-off the first and second package container, thereby returning the first and second package containers to the warehouse location.

[0262] In one embodiment, at least one of the package containers 11m-4, 11m-5 is a vending machine.

[0263] FIG. 20 illustrates one embodiment of a method for autonomously delivering packages using a plurality of package containers. In conjunction with receiving a plurality of indications associated respectively with a plurality of packages 11pkg, in which each of the indications is an indication pertaining to a delivery status associated with a respective one of the packages, e.g., 11pkg-41, in a respective one of a plurality of lockers, e.g., 11L-41, of a respective one of the package containers, e.g., 11m-4, that is parked in a current respective place, e.g., 7-loc-1, the method includes (per each of the indication received):

[0264] In step 1121: straddling-over and grabbing, by an on-road vehicle 10, the respective package container, e.g., 11m-4, from the current respective place, e.g., location 7-loc-1.

[0265] In step 1122: releasing and straddling-off, by the on-road vehicle 10, said respective package container 11m-4 at a different respective place, e.g., 7-loc-2.

[0266] In step 1123 sending an indication that said respective package container 11m-4 is now and/or will be in the different respective place 7-loc-2 for another of the packages, e.g., 11pkg-42, to be picked-up from a different one of the lockers, e.g., 11L-42, of said respective package container 11m-4.

[0267] FIG. 21A illustrates one embodiment of an autonomous vehicle 10 featuring a hybrid locomotion system configured for enhanced mobility and load handling. The vehicle 10 comprises a main body 3-body, which may house control systems, power sources, and potentially serve as a support structure. Connected to the main body 3-body are a plurality of articulated legs, such as legs 2a-1, 2a-2, 2c-1, 2c-2, 2d-1, and 2d-2. These legs are actuated via respective joints, such as joints 2a-j1, 2a-j2, 2c-j2, and 2d-j2, allowing for complex movements and height adjustments. Wheels 1a, 1b, 1c, and 1d are attached to the distal ends of respective articulated legs, providing efficient locomotion over suitable ground surfaces. The vehicle 10 may further include sensors 4-sense for perceiving the environment, actuators such as actuator 6d associated with the wheels or joints, and a

connector mechanism 5-cnct integrated with or positioned relative to the main body 3-body for interacting with loads. [0268] FIG. 21B illustrates one embodiment of the autonomous vehicle 10 engaging with a load 11. In this depiction, the system actuates the articulated legs (e.g., 2a-1, 2a-2) via their respective joints (e.g., 2a-j1, 2a-j2, 2c-j2) to modify the vehicle's stance relative to the ground (e.g., in a bending-like motion). This adjustment configures the vehicle 10 such that the main body 3-body is lowered and positioned right over the load 11. This demonstrates the capability of the system to straddle over a load 11 and come in contact therewith, as a preliminary step before lifting or securing the load for transport.

[0269] FIG. 21C illustrates one embodiment of the autonomous vehicle 10 subsequent to engaging with and securing the load 11, potentially following the straddling configuration shown in FIG. 21B. After positioning and lowering itself over the load 11, the system is configured to actuate a connector mechanism (such as 5-cnct shown in FIG. 21A, or another mechanism associated with the main body 3-body) to grab the load 11. Following the grabbing action, the system actuates the articulated legs (e.g., 2a-1, 2a-2) and joints (e.g., 2a-j1, 2a-j2, 2c-j2) to lift the load 11 off the ground surface, thereby securing it underneath the main body 3-body. The figure shows the vehicle 10 transporting the now-lifted load 11, maintaining it at a specific elevation range 11-ele relative to the ground surface. This variable ground clearance, provided by the articulated legs and joints, enables safe transport of the under-body load 11. This under-body transport configuration leverages the integrated capabilities of grabbing, lifting via leg actuation, and maintaining variable height afforded by the articulated legged-wheeled structure.

[0270] FIG. 22 illustrates one embodiment of a manipulation capability of the autonomous vehicle 10, demonstrating interaction with objects at elevated heights. To achieve the necessary reach, the system is configured to actuate its rear articulated legs to adopt a stabilized, upright or elevated stance, primarily supported by these rear legs and their respective wheels. While maintaining this elevated stance, the vehicle 10 utilizes its front articulated legs, configured to act as manipulators or arms, to interact with a load 11. In this specific embodiment, the front articulated legs are shown positioning the load 11 onto an elevated external surface, depicted as a shelf structure 110-shelf. This capability highlights the system's versatility, extending beyond ground-level operations and under-body transport to include complex interactions such as retrieving or placing items on shelves or other raised platforms by dynamically altering its posture and using specific legs for manipulation.

[0271] In one embodiment demonstrating advanced manipulation, particularly for handling loads (11) requiring precise orientation, the system (10) utilizes a specialized posture and coordinated wheel control. The system first actuates its rear articulated legs to achieve a stable, elevated stance described previously (FIG. 22). The front articulated legs are then positioned such that the front wheels (1c, 1d) make direct contact with the load (11), acting as primary gripping or contact points. Instead of or in addition to using dedicated grippers, the system relies on the friction of the wheels (1c, 1d) against the load surface, possibly augmented by pressure applied by the front legs. Critically, as the system manipulates the load (11) using its front legs—for example, lifting it from/to an elevated surface (110-shelf),

repositioning it, or preparing to place it underneath the main body (3-body)—the control sub-system (8, FIG. 26) actively rotates the front wheels (1c, 1d). This wheel rotation compensates for the changes in the front leg joint angles (e.g., 2c-j2) during the manipulation, thereby ensuring the load (11) maintains a desired orientation, such as remaining substantially horizontal, throughout the posture change and movement sequence. This coordinated control of front leg joints and front wheel rotation enables delicate handling and stable orientation control of the grasped load.

[0272] FIG. 23 illustrates one embodiment of enhanced navigational capabilities of the autonomous vehicle 10 enabled by its hybrid locomotion system. The vehicle 10, while carrying a load 11 underneath its main body, is depicted successfully navigating a significant obstacle, specifically stairs 111-stairs. The coordinated use of the articulated legs and wheels allows the vehicle 10 to traverse complex terrain that would typically be inaccessible to purely wheeled autonomous systems, demonstrating a key advantage for applications requiring delivery or operation in varied environments.

[0273] One embodiment is a system (10) operative to deliver loads autonomously comprising: a main body (3-body) configured to support a load (11) positioned underneath said main body; a plurality of articulated legs (e.g., 2a-1, 2a-2, 2c-1, 2c-2, 2d-1, 2d-2) comprising joints (e.g., 2a-j1, 2a-j2, 2c-j2, 2d-j2), connected to the main body (3-body); and a plurality of wheels (e.g., 1a, 1b, 1c, 1d), each wheel attached to a respective one of the articulated legs; wherein the articulated legs and joints are configured to control elevation of the main body (3-body); and wherein the system is configured to: actuate the wheels (1a, 1b, 1c, 1d) to achieve forward movement of the load (11) supported underneath the main body (3-body); and further actuate the articulated legs (e.g., 2a-1, 2a-2) via the joints (e.g., 2a-j1, 2a-j2) to adjust said elevation for at least one of maintaining ground clearance 11-ele of the load when navigating obstacles (e.g., stairs 111-stairs in FIG. 23) and manipulating the load (11) relative to a ground surface. This system architecture, visualized generally in FIG. 21A and in operation in FIGS. 21B, 21C, and 23, combines the efficiency of wheeled locomotion for travel over relatively flat surfaces with the adaptability of articulated legs for adjusting height, handling loads, and traversing challenging terrain, all while carrying the payload underneath the main structure.

[0274] In one embodiment, manipulating the load (11) relative to the ground surface comprises actuating the articulated legs (e.g., 2a-1, 2a-2) via the joints (e.g., 2a-j1, 2a-j2, 2c-j2) to lower the main body (3-body) and release the load (11) onto the ground surface. For example, after transporting the load as shown in FIG. 21C, the system would perform the reverse operation of lifting, using the legs and joints to carefully decrease the elevation (11-ele) until the load (11) rests securely on the ground, after which a connector mechanism would release it.

[0275] In one embodiment, manipulating the load (11) relative to the ground surface comprises actuating the articulated legs (e.g., 2a-1, 2a-2) via the joints (e.g., 2a-j1, 2a-j2, 2c-j2) to lift the load (11) from the ground surface underneath the main body (3-body). This action is depicted in the transition from a configuration like FIG. 21B (where the vehicle is positioned over the load) to the transport configuration shown in FIG. 21C (where the load is elevated 11-ele).

The legs extend or reconfigure using the joints to increase the ground clearance, lifting the secured load.

[0276] In one embodiment, prior to said lifting of the load (11), the system is further configured to straddle over the load (11). As illustrated in FIG. 21B, the vehicle (10) utilizes its articulated legs (e.g., 2a-1, 2a-2) and overall structure to position its main body (3-body) directly above the load (11) while the load is resting on the ground, ensuring proper alignment for subsequent grabbing and lifting operations.

[0277] In one embodiment, navigating obstacles comprises actuating the articulated legs (e.g., 2a-1, 2a-2) via the joints (e.g., 2a-j1, 2a-j2) to navigate stairs (111-stairs). FIG. 23 explicitly shows the vehicle (10) using its legs to ascend or descend stairs (111-stairs) while carrying the load (11), demonstrating a key advantage over purely wheeled systems.

[0278] In one embodiment, a control sub-system (like processor 8 in FIG. 1D) is configured to independently actuate each articulated leg (e.g., 2a-1, 2c-1, 2d-1) and its respective joint/s (e.g., 2a-j1, 2c-j2, 2d-j2) to maintain a substantially horizontal orientation of the main body (3-body), and consequently the load (11) supported thereunder, while traversing uneven ground surfaces or obstacles. This active stabilization ensures the load remains level and secure even when the wheels (1a, 1b, 1c, 1d) are on different height levels, as might occur on rough terrain or slopes.

[0279] In one embodiment, maintaining the substantially horizontal orientation occurs while navigating stairs (111-stairs), including situations where front ones of the articulated legs (e.g., 2c, 2d) and rear ones of the articulated legs (2a and the other rear leg not numbered) are simultaneously in contact with different steps of said stairs (111-stairs). As shown conceptually in FIG. 23, the independent control allows the front legs to be on a higher step while the rear legs are on a lower step (or vice versa), with the joints actuating to keep the main body (3-body) and load (11) level during the transition between steps.

[0280] In one embodiment, the load is a package (represented conceptually by 11) to be delivered to a designated location. This specifies one application for the system, where the load (11) contains goods intended for delivery.

[0281] In one embodiment, the system further comprising sensors (4-sense) configured to perceive the ground surface, wherein the system utilizes data from said sensors (4-sense) to coordinate actuation of the wheels (1a, 1b, 1c, 1d) and joints (e.g., 2a-j1, 2a-j2, 2c-j2, 2d-j2). As indicated in FIG. 21A, sensors (4-sense), which may include cameras, LiDAR, inertial measurement units, or other types, provide environmental awareness enabling the control sub-system to make informed decisions for navigation, obstacle avoidance, and stabilization.

[0282] In one embodiment, the system is further configured to actuate a first subset of said articulated legs, designated as rear legs (implied rear legs in FIG. 22), and their respective joints to stabilize the main body (3-body) in an elevated stance, primarily supported by said rear legs and their wheels, thereby increasing a reach capability of a second subset of said articulated legs, designated as front legs (e.g., legs used for manipulation in FIG. 22). FIG. 22 illustrates this concept, where the vehicle (10) adopts an upright posture, supported by its rear limbs, allowing the front limbs greater freedom and height for interaction tasks.

[0283] In one embodiment, the system is further configured to utilize said front legs and their respective joints as

manipulators, akin to arms or hands, to grasp and retrieve the load (11) from an elevated surface, such as a shelf (110-self), while the main body (3-body) is maintained in said elevated stance by the rear legs. FIG. 22 explicitly depicts the front legs of the vehicle (10) acting as manipulators to handle the load (11) relative to the shelf (110-self), demonstrating this advanced capability enabled by the elevated stance.

[0284] In one embodiment, the system is further configured to subsequently actuate said front legs, acting as manipulators, to either position the load (11) on ground or attach the load (11) underneath said main body (3-body) for transport. After retrieving the load from the shelf (110-self) as shown in FIG. 22, the front legs can then maneuver the load either to be placed directly on the ground or secured to the underside of the main body (3-body) for subsequent transport, similar to the configuration in FIG. 21C.

[0285] In one embodiment, the system further comprising at least one connector mechanism (e.g., 5-cnct in FIG. 21A) associated with the main body (3-body), wherein said connector mechanism is configured to secure the load (11) underneath the main body (3-body) during transport. This mechanism ensures the load remains firmly attached to the vehicle while it is moving, particularly when lifted as in FIG. 21C.

[0286] In one embodiment, the connector mechanism is configured to actively grab the load (11) prior to the system starting manipulating the load (11) to lift it from the ground surface. Before the legs actuate to lift the load (as in the transition from FIG. 21B to 21C), the connector mechanism engages and secures the load.

[0287] In one embodiment, the connector mechanism is configured to release the load (11) after the system has finished manipulating the load (11) to lower it onto the ground surface. Once the legs have lowered the load (reverse of lifting from FIG. 21C), the connector mechanism disengages, allowing the vehicle to move away.

[0288] In one embodiment, the connector mechanism comprises at least one of: a. an electromagnetic connector; b. a mechanical gripper or clamp; c. a vacuum suction cup; and d. a hook-and-latch mechanism. These represent various technologies that could be implemented for the connector mechanism (e.g., 5-cnct) to achieve the grabbing and releasing functions depending on the nature of the load and the operational requirements.

[0289] In one embodiment, the load (11) comprises a complementary connector interface configured to engage with the connector mechanism (e.g., 5-cnct) of the autonomous vehicle system (10). For reliable connection, the load (11) itself would possess features (e.g., a ferromagnetic plate for an electromagnet, specific geometry for grippers, a suitable surface for suction) designed to mate with the vehicle's connector mechanism.

[0290] FIG. 24 illustrates one embodiment of a method for delivering a load autonomously using a system comprising a main body (3-body), articulated legs (e.g., 2a-1) with joints (e.g., 2a-j1) connected to the main body, and wheels (e.g., 1a) attached to the articulated legs, the method comprising: securing the load (11) underneath the main body (3-body) of the autonomous system (step 1131); actuating the wheels (e.g., 1a, 1c) to achieve forward movement, thereby transporting the secured load (11) towards a designated delivery location over a ground surface (step 1132); upon arrival at or near the designated delivery location, at least reducing said forward movement achieved by actuating

the wheels (step 1133); and actuating the articulated legs (e.g., 2a-1, 2c-1) via the joints (e.g., 2a-j1, 2c-j2) to manipulate the load (11) relative to a ground surface at the designated delivery location, thereby delivering the load (step 1134). The initial securing (1131) might follow a lift operation. The transport phase (1132) primarily utilizes the efficiency of wheeled motion. Upon reaching the destination, the system slows down or stops (1133) before using the adaptability of the legs and joints (1134) to perform the final delivery manipulation, such as lowering and releasing the load. The entire process may be orchestrated by a control sub-system (8).

[0291] FIG. 25 illustrates one embodiment a method for autonomously transporting a load using a system comprising a main body (3-body) supporting the load (11) underneath, articulated legs (e.g., 2a-1) with joints (e.g., 2a-j1) connected to the main body, and wheels (e.g., 1a) attached to the articulated legs, the method comprising: securing the load (11) underneath the main body (3-body) of the autonomous system (step 1141); actuating the wheels (e.g., 1a, 1c) to achieve forward movement, thereby transporting the secured load (11) towards a target location over a ground surface (step 1142); and dynamically actuating the articulated legs (e.g., 2a-1, 2c-1) via the joints (e.g., 2a-j1, 2c-j2) to maintain a ground clearance (e.g., 11-ele in FIG. 21C) between the load (11) supported underneath the main body (3-body) and the ground surface while traversing variations in the ground surface and/or encountering obstacles (step 1143). This method emphasizes the active ride height control during transport. While the wheels (controlled by actuators 6) provide primary propulsion (1142), the control sub-system (8), using inputs from sensors (4), simultaneously commands the legs/joints (actuators 2) to make continuous adjustments (1143) ensuring the under-body load (11) does not collide with the ground or low obstacles.

[0292] In one embodiment, maintaining said ground clearance includes increasing elevation of the main body (3-body) when an obstacle is detected. As the system transports the load using wheels, sensors (4) may detect a curb, rock, stairs, or other obstacle. The control sub-system (8) processes this information and commands the articulated legs and joints (2) to actuate, maintaining or temporarily increasing the ground clearance (11-ele in FIG. 21C) to allow the vehicle and the under-body load (11) to pass safely over the obstacle without interrupting forward movement significantly.

[0293] In one embodiment, maintaining said ground clearance includes decreasing elevation of the main body (3-body) when traversing a relatively smooth and level portion of the ground surface to improve stability and/or lower the center of gravity. During transport over flat terrain, where maximum ground clearance is not needed, the control sub-system (8) may actuate the legs and joints (2) to slightly lower the main body (3-body) and load (11). This lowers the overall center of gravity, enhancing stability during wheeled motion and potentially reducing energy consumption or wind resistance.

[0294] In one embodiment, traversing variations in the ground surface includes traversing uneven terrain and/or stairs (111-stairs). The dynamic actuation of legs and joints to maintain ground clearance is particularly crucial when navigating challenging surfaces such as rough, uneven ground or ascending/descending stairs (111-stairs), as depicted in FIG. 23. The system continuously adjusts leg

positions and joint angles to keep the load (11) clear of the steps or ground variations while the wheels (e.g., 1a, 1c) provide traction or support as needed.

[0295] In one embodiment, maintaining of ground clearance within a predetermined range involves specific numerical targets adjusted dynamically based on terrain and obstacles, as perceived by sensors (4) and processed by the control sub-system (8). For instance, during transport over a smooth, level ground surface, the system might maintain a nominal ground clearance (11-ele in FIG. 21C) between the lowest point of the load (11) and the ground within a relatively low range, perhaps 5 to 10 centimeters, to optimize stability by keeping the center of gravity low. Upon detecting an upcoming obstacle, such as a standard curb of 15 centimeters, the control sub-system (8) would proactively command the articulated legs and joints (actuators 2, 6) to increase the ground clearance to 20 to 25 centimeters just before reaching the obstacle, ensuring the load (11) passes safely over it. When traversing continuously uneven terrain or stairs (111-stairs), the dynamic adjustment of ground clearance becomes even more critical. For stairs specifically (FIG. 23), as the front wheels approach the first step, the front legs may lift to place the wheels onto the step while simultaneously adjusting joint angles to begin raising the main body (3-body). Throughout this maneuver, where front and rear legs are often on different steps, the system continuously adjusts the height of all legs to maintain the load (11) at a safe clearance, perhaps 5-10 centimeters above the edge (nose) of each stair step it is currently passing over, while also potentially maintaining the main body and load in a substantially horizontal orientation. The specific values within these clearance ranges are predetermined based on the vehicle's (10) limits, load (11) characteristics, and the operating environment, and are continuously monitored and adjusted.

[0296] FIG. 26 illustrates one embodiment of a control sub-system 8 configured to manage the autonomous operation of the vehicle 10. The control sub-system 8 receives input data from various sensors 4 (e.g., corresponding to 4-sense in FIG. 21A), which perceive the external environment and internal vehicle state. Within the control sub-system 8, processing resources include memory 9-mem, a central processing unit (CPU) 9-CPU, and potentially a graphics processing unit (GPU) 9-GPU, which is often beneficial for parallel processing tasks common in robotics and machine learning. These resources execute control logic and algorithms, including, in this embodiment, a machine learning (ML) model 9model. The ML model 9model may be trained to perform complex tasks such as object recognition, terrain classification, path planning, gait generation for the articulated legs, or determining optimal coordination between wheels and legs based on the sensor 4 inputs. The outputs generated by the control sub-system 8, derived from processing sensor data potentially using the ML model 9model, are command signals sent to actuators 2 and 6. These actuators correspond to the motors and mechanisms controlling the articulated legs, joints, and wheels of the vehicle 10, thereby translating computational decisions into physical movement and manipulation actions of the autonomous system.

[0297] In one embodiment, the adaptable legged-wheeled locomotion system described herein (e.g., 10 in FIG. 21A) can be scaled to address the diverse load types and operational environments depicted in FIG. 16. A large-scale

version, analogous to vehicle **10**-large, could transport substantial loads (**11**-large) exceeding 2 meters in dimension, such as specialized equipment containers or building materials. Its articulated legs would enable navigation over rough construction sites or uneven loading docks inaccessible to standard wheeled platforms, and allow precise positioning or straddling over ground-level obstacles during handling. A medium-scale implementation, similar to **10**-medium, carrying loads like furniture, appliance boxes, or potentially adapted passenger cabins (**11**-medium, approx. 1-2 meters), could leverage its legs to climb entrance stairs (like **111**-stairs in FIG. 23), delivering bulky items directly inside buildings. A small-scale version, akin to **10**-small, would excel at last-meter delivery, carrying parcels or grocery totes (**11**-small, approx. 0.5-1 meter) from a curbside point, autonomously navigating sidewalks, ascending porch steps, and using leg manipulation to place the package securely. Finally, a tiny-scale variant (**10**-tiny) carrying small, critical items like documents or medical supplies (**11**-tiny, <0.5 meters) could navigate complex indoor environments, overcoming thresholds, small steps, or clutter found in offices or hospitals. Across all scales, the core benefit remains: combining efficient wheeled travel with legged adaptability for obstacle negotiation and load manipulation appropriate to the size and nature of the cargo.

[0298] Exemplary Scenario 1: Curbside to Apartment Door Delivery. In one delivery scenario, the autonomous vehicle (**10**) might be deployed from a larger delivery van parked at the curb. Initially, it would actuate its wheels (e.g., **1a**, **1c**) to efficiently traverse the sidewalk. Upon encountering a curb or uneven section, the control sub-system (**8**), using input from sensors (**4**), would actuate the articulated legs and joints to increase ground clearance (**11**-ele), stepping over the obstacle while maintaining stability and load orientation (claim 1.3b). If the destination involves navigating entryway stairs (**111**-stairs, FIG. 23), the system transitions to primarily using its articulated legs, independently actuating each leg (claim 1.3b) to ascend the steps while keeping the under-body package level and clear of the stair edges. Finally, upon reaching the recipient's door (the designated location), the vehicle stops wheeled motion and actuates the legs and joints to lower the main body (**3**-body) and release the package (**11**) onto the floor or doorstep.

[0299] Exemplary Scenario 2: Multi-Story Building Delivery via Elevator. For deliveries within a multi-story building, the autonomous vehicle (**10**) would first navigate the lobby using its wheels. Upon reaching an elevator bank, sensors (**4**) would identify the call button. The system would then actuate its rear legs to achieve a stable, elevated stance. Utilizing its front legs as manipulators, it would extend one leg to press the elevator call button. Once the elevator arrives and the doors open, the vehicle (**10**) might need to temporarily lower its stance slightly or adjust its legs to enter. To safely ride the elevator while potentially needing its legs for stability or height adjustment, the system could be configured to place the load (**11**) onto the elevator floor. During the ride, it might maintain the elevated stance for stability. After exiting the elevator, it would retrieve the load (**11**) back underneath its main body (**3**-body), resume wheeled locomotion to the recipient's office or apartment door, and finally deliver the load by lowering and releasing it.

[0300] Exemplary Scenario 3: Warehouse Shelf Retrieval and Loading. In a warehouse or fulfillment center setting,

the autonomous vehicle (**10**) can automate the initial loading process. It would navigate aisles using its wheels. Upon reaching a designated shelf (similar to **110**-self in FIG. 22), the system actuates its rear legs to adopt the elevated stance (claim 1.6), increasing the reach of its front legs. The front legs, acting as manipulators would then grasp a specific package or load (**11**) from the shelf. Once the load is securely grasped, the system actuates the front legs again to maneuver the load underneath the main body (**3**-body) and secure it using a connector mechanism. The vehicle (**10**) is then ready to transport the load using its wheels.

[0301] Exemplary Scenario 4: Uneven Terrain Site Delivery. Consider a scenario delivering supplies or tools (load **11**) on a construction site or across uneven outdoor terrain. The autonomous vehicle (**10**) would primarily use its wheels on flat sections. However, when encountering rough patches, debris, or significant slopes, the control sub-system (**8**) would rely heavily on dynamically actuating the articulated legs and joints to maintain a safe ground clearance (**11**-ele) and stability. The system would continuously adjust leg height and joint angles to keep the main body (**3**-body) and load (**11**) level while stepping over rocks or navigating muddy sections, ensuring the load arrives safely at the designated work zone where it can be lowered and released.

[0302] Exemplary Scenario 5: In a common urban or suburban delivery scenario, the autonomous vehicle (**10**), carrying a package secured underneath its main body (**3**-body), might start its journey by navigating public roads alongside regular vehicle traffic. During this phase, the system primarily actuates its wheels for efficient forward movement, relying heavily on its sensors and control sub-system (**8**) for lane keeping, speed control, and collision avoidance relative to other vehicles. Upon approaching the designated delivery address, the system identifies a suitable location to transition off the road, such as a driveway entrance or a clear section of curb. It then carefully actuates its articulated legs and joints to mount the curb, dynamically adjusting ground clearance (**11**-ele) to ensure the load (**11**) clears the elevation change while maintaining stability. Once on the sidewalk or pathway, it may resume primarily wheeled locomotion towards the building entrance. If the final approach involves navigating stairs (**111**-stairs, FIG. 23), the system again transitions its locomotion strategy. It ceases significant forward wheel motion and predominantly uses the articulated legs and joints to ascend the steps, independently actuating each leg to maintain a level orientation for the main body and load and ensuring safe clearance over each stair edge. Having successfully navigated the stairs, the vehicle reaches the doorstep (final designated location), stops, and performs the final manipulation by actuating the legs and joints to lower and release the package (**11**).

[0303] In one embodiment, in preparation for manipulating a load (**11**), such as performing a straddle-and-lift operation or retrieving an item from a shelf, the autonomous system (**10**) employs its sensors (**4**) to first perceive the target load. The control sub-system (**8**), potentially utilizing image processing algorithms or machine learning models (**9**model, FIG. 26), analyzes the sensor data to gauge the size and estimate the dimensions of the load (**11**). Based on these determined dimensions, the system preemptively adjusts the variable ground clearance (**11**-ele, FIG. 21C) by actuating the articulated legs (e.g., **2a-1**) and joints (e.g., **2a-j1**). This proactive adjustment ensures sufficient vertical and lateral

space is created underneath the main body (3-body) before the vehicle attempts to position itself over or grasp the load, preventing potential collisions. Furthermore, the control sub-system (8) may also adjust actuator settings for the legs and joints, such as movement speed, torque limits, or joint stiffness, based on the perceived size or potentially estimated weight of the load (11) to ensure safe and appropriate handling during manipulation (claim 1.0).

[0304] In one embodiment, prior to manipulating a load (11), particularly for operations like straddling over it or lifting it, the autonomous system (10) utilizes its sensors (4) to perceive the load. The control sub-system (8) processes the sensor data to specifically determine the height or vertical dimension of the load (11) relative to the ground surface. Based on this perceived load height, the control sub-system (8) calculates the required minimum elevation for the main body (3-body) to ensure adequate clearance. The system then actuates the articulated legs (e.g., 2a-1) and joints (e.g., 2a-j1) to adjust the main body's elevation (affecting 11-ele, FIG. 21C) to meet or exceed this calculated height. This proactive height adjustment based on perceived load dimensions prevents collision between the main body or connector mechanism and the load itself during the approach and manipulation phases.

[0305] In one embodiment, during forward movement utilizing the wheels, the system continuously monitors the upcoming terrain using its sensors (4). The control sub-system (8) processes this sensor data, potentially using the ML model (9model), to determine the height profile of the ground surface and detect the height of any potential obstacles in the vehicle's path. This real-time height information is crucial for the dynamic actuation of the articulated legs and joints. Based on the perceived height of the upcoming ground or obstacles relative to the current elevation of the load (11) supported underneath the main body (3-body), the control sub-system (8) calculates and commands precise adjustments to the articulated legs and joints. This ensures that the variable ground clearance (11-ele, FIG. 21C) is constantly maintained within the predetermined safe range (claim 3.0), proactively increasing clearance to step over raised obstacles (claim 3.1) or accommodating dips and variations in the terrain, thereby preventing the under-body load (11) from impacting the ground during transport.

[0306] In one embodiment, before initiating or during autonomous transport, the system (10) may utilize its sensors (4) not only to perceive the ground but also to gauge or confirm the height (vertical dimension) of the load (11) itself before it has been secured underneath the main body (3-body). This measured load height is critical input for the control sub-system (8). To ensure safe transport, the control sub-system determines a target minimum ground clearance required between the bottom of the load (11) and the ground surface (e.g., the lower bound of the predetermined safe range). Knowing the load's height, the control sub-system calculates the necessary minimum elevation for the main body (3-body) relative to the ground that will achieve this target load-to-ground clearance. It then actuates the articulated legs (e.g., 2a-1) and joints (e.g., 2a-j1) to position the main body (3-body) at or above this calculated minimum elevation, establishing the desired initial height (representing the clearance under the body, which indirectly sets clearance under the load) before or as wheeled motion

commences. This ensures the load (11) starts its journey with sufficient separation from the ground, based directly on its own measured dimensions.

[0307] In one embodiment, in certain locomotion scenarios where precise foot placement and stability are paramount, such as when navigating obstacles like stairs (111-stairs, FIG. 23), the control sub-system (8) may be configured to operate the system (10) in a predominantly legged walking gait, temporarily deviating from primary reliance on wheeled motion. During this legged gait, the control sub-system (8) may command actuators (2, 6, FIG. 26) associated with the wheels (e.g., 1a, 1b, 1c, 1d) to actively lock or apply brakes to prevent unintended rolling. By locking the wheels when a specific leg is being used as a stationary support point (akin to a foot placement), the system enhances stability and positional accuracy. This allows the articulated legs (e.g., 2a-1, 2c-1) and joints (e.g., 2a-j1, 2c-j2) to execute precise movements for lifting the body, stepping onto the next stair, and maintaining balance, ensuring safe and controlled traversal of complex terrain where wheeled rolling would be ineffective or detrimental. Once back on a suitable surface, the wheel locks can be released to resume efficient wheeled locomotion.

[0308] In one embodiment, the autonomous system (10), particularly in its legged-wheeled configuration (FIG. 21A), is designed with versatility to handle a wide range of loads (11), adapting to diverse delivery and transport needs. As explicitly shown, the system can transport generic boxed loads (11 in FIGS. 21B, 21C, 23, 22). This load (11) may represent a standard package to be delivered. Furthermore, drawing context from related embodiments, the load (11) handled by the legged-wheeled vehicle (10) can encompass more specialized items. These include detachable containers (e.g., analogous to 11m-4, 11m-5 in FIGS. 19C, 19D) which may themselves contain multiple lockers (e.g., 11L-41, 11L-51) for secure, multi-recipient deliveries. The system can also be configured to handle functional loads (e.g., analogous to 11-func in FIG. 7B), such as portable power sources, fuel tanks, or communication equipment. Additionally, adapting concepts from FIGS. 17A-17D, the system, potentially via a specialized connector or gripper, could be configured to handle disposable packages, such as cardboard boxes (11-crb), plastic bags (11-bag), or items secured with straps (92-strap), especially for last-meter delivery scenarios where the legged mobility is advantageous for reaching doorsteps or navigating complex pathways after initial transport by another system. The ability to perceive the load's dimensions allows the system to adjust its handling strategy for these varied load types.

[0309] In one embodiment, subsequent to lifting the load (11) underneath the main body (3-body), the system (10) provides enhanced load securing through active leg actuation. The control sub-system (8, FIG. 26) may be configured to command multiple articulated legs via their respective joints to contract inwards towards the secured load (11). This inward motion applies controlled pressure, effectively clamping or bracing the load (11). This leg-based securing mechanism can complement or form an integral part of a dedicated connector mechanism (5-cnct). Beyond providing stability during transport over uneven terrain or stairs (111-stairs, FIG. 23) and physical protection, this inward leg contraction also serves as an active anti-theft measure. The control sub-system (8), using input from sensors (4, 4-sense) such as force sensors, proximity sensors, or cameras, can be

configured to detect an attempt at unauthorized removal of the load (11). Upon detecting such an attempt, the system can increase the clamping force applied by the articulated legs, making it significantly more difficult to forcibly detach the load (11) from the vehicle (10), thereby deterring theft while the load is in transit or temporarily placed.

[0310] In one embodiment, while the embodiments illustrated herein, particularly in FIGS. 21A-23, depict an autonomous vehicle (10) with four articulated legs, each equipped with a wheel, this configuration is presented by way of example and not limitation. The fundamental principles of the system—combining articulated legs and wheels for hybrid locomotion, providing variable ground clearance for under-body load transport, and utilizing legs for manipulation and obstacle negotiation—can be implemented using alternative leg configurations. Therefore, the scope of the invention encompasses autonomous vehicles utilizing a different plurality of articulated legs, such as three, six, or potentially more legs, depending on specific design requirements for stability, payload capacity, redundancy, or maneuverability, while still incorporating wheels on at least some of the legs and retaining the core functionalities described.

[0311] In various embodiments, at least some of the joints (e.g., 2a-j1, 2a-j2, 2c-j2, 2d-j2 in FIG. 21A) connecting the segments of the articulated legs (e.g., 2a-1, 2a-2) and connecting the legs to the main body (3-body) are configured to allow multiple degrees of freedom (DOF). For example, a proximal joint connecting a leg to the body might allow for rotation akin to a hip (providing forward/backward and side-to-side motion), while a distal joint might function like a knee or ankle (providing primarily pitching motion). Having multiple joints per leg, each potentially offering 2-DOF or 3-DOF (e.g., incorporating pitch, roll, and yaw axes), facilitates a large workspace and complex kinematics. This multi-DOF configuration enables the control sub-system (8) to orchestrate intricate movements, facilitating a full or partial humanoid-like motion span for the articulated legs. Examples include not only lifting the body vertically or stepping forward/backward but also enabling side-stepping, torso twisting relative to the legs, complex leg folding for compact storage, and the versatile manipulation capabilities shown in FIG. 22 where the front legs act akin to arms with shoulder, elbow, and wrist-like articulations. This enhanced range of motion is crucial for navigating highly complex terrain, maintaining stability across varied postures, and performing sophisticated load manipulation tasks.

[0312] In one embodiment, the system's capability for manipulating the load relative to a ground surface extends beyond simple lowering and releasing to achieve centimeter-level accuracy in spatial placement. This precision is enabled by the complex kinematics afforded by the multi-jointed articulated legs (e.g., 2a-1, 2a-2) and the sophisticated algorithms executed by the control sub-system (8, FIG. 26). Utilizing feedback from high-resolution sensors (4, 4-sense), including potentially proprioceptive sensors within the joints themselves, the control sub-system can orchestrate intricate, fine-tuned movements of the individual legs and joints. This allows the main body (3-body), and consequently the load (11) held beneath it or manipulated by front legs, to be positioned with high accuracy in three-dimensional space relative to the target delivery surface. This capability is crucial for scenarios requiring precise placement, such as depositing a package into a narrow designated receptacle, aligning the load with specific mark-

ers on the ground, or carefully setting down fragile items within tight tolerances, ensuring not just delivery but accurate and safe final positioning of the load.

[0313] In various embodiments, actuation of the system's moving components, including providing rotational force to the wheels (e.g., 1a, 1b, 1c, 1d) for locomotion and controlling the movement of the articulated legs (e.g., 2a-1, 2a-2) via their respective joints (e.g., 2a-j1, 2a-j2, 2c-j2, 2d-j2), is accomplished using one or more actuators (represented generally by 2, 6 in FIG. 26, and 6d in FIG. 21A). In various embodiments, these actuators may comprise electric motors, such as servo motors or brushless DC motors, known for their precise control and efficiency, often integrated near or within the joints. Alternatively, particularly for applications demanding high force or torque for lifting heavy loads or robust leg articulation, hydraulic actuation systems utilizing pressurized fluid and cylinders may be employed. Other technologies, including pneumatic systems or advanced materials-based actuators, could also be implemented depending on the specific scale, power, speed, and precision requirements of the autonomous vehicle (10). The selection of the actuation method is a key design choice impacting the overall performance, capabilities, and operational characteristics of the system.

[0314] In one embodiment, the autonomous delivery system (10) offers capabilities beyond simple ground-level load release, enabling precision placement of packages (11) tailored to specific recipient needs or environmental conditions. Upon arriving at the designated delivery location and ceasing forward wheeled motion, the control sub-system (8) utilizes sensor data (4-sense) to identify the precise target placement spot. This might be a slightly elevated porch bench to keep the package off wet ground, a specific designated area within a larger porch, or inside a customer-provided secure delivery receptacle. The system then finely actuates the articulated legs and joints not just to lower the load, but to carefully maneuver the main body (3-body) and potentially utilize the front legs as fine manipulators to position the package (11) at the exact target coordinates. Furthermore, for packages marked as fragile or requiring a specific orientation (e.g., 'This Side Up'), the system can control the final release from the connector mechanism and the leg movements to ensure the package is deposited gently and maintains the required orientation on the surface. This precision enhances package safety, security, and meets specific customer requirements.

[0315] In one embodiment, in a typical operational scenario, the autonomous vehicle (10) is loaded at a depot with multiple distinct packages (conceptually represented by load 11, but understood to be multiple individually addressable items secured underneath or within the main body 3-body), each destined for a different location along a pre-planned route. The control sub-system (8, FIG. 26) optimizes the delivery sequence. The vehicle (10) departs the depot, primarily utilizing its wheels for efficient travel along roads and sidewalks towards the first designated delivery location. During transit, it continuously uses its articulated legs and joints to maintain safe ground clearance (11-ele) over varying terrain and obstacles. Upon arriving near the first address, the vehicle may need to navigate specific obstacles like curbs or a few stairs (111-stairs, FIG. 23) using primarily its legs, potentially locking its wheels for stability and maintaining a level orientation. Reaching the precise drop-off point (e.g., the front porch), the vehicle stops, actuates its

legs and joints to lower the main body, and releases only the first specific package using its connector mechanism. Having completed the first delivery, the system secures any remaining packages, adjusts its stance, and proceeds towards the next delivery location on its route, again primarily using wheeled locomotion, repeating the process of navigation, obstacle traversal (if needed), and precise package release until all packages for that route have been delivered.

[0316] In one embodiment, the powered actuation system as described herein may be embodied, for example, by the linear actuators (2'+2) enabling vertical movement of the wheel structures (2, 2') relative to the main body (3) in the embodiment of FIG. 1A, and/or by the articulated legs (e.g., 2a-1) and joints (e.g., 2a-j1) enabling movement of the wheels (e.g., 1a) relative to the main body (3-body) in the embodiment of FIG. 21A, and/or by any other means.

[0317] One embodiment is an autonomous vehicle (e.g., 10 in FIG. 1A; 10 in FIG. 21A), comprising: a main body structure (e.g., upper horizontal structure 3 in FIG. 1A; main body 3-body in FIG. 21A); a plurality of wheels (e.g., 1, 1a, 1b, 1c, 1d in FIG. 1A; 1a, 1b, 1c, 1d in FIG. 21A); a plurality of powered actuation systems (e.g., linear actuators 2'+2 in FIG. 3C/3D; articulated legs 2a-1/joints 2a-j1 in FIG. 21A), wherein for each wheel (e.g., 1a) of the plurality of wheels, a respective one of the powered actuation systems (e.g., associated linear actuator 2a'/2a; associated leg 2a-1/joints 2a-j1) is configured to actively modify a relative positioning between said wheel (e.g., 1a) and the main body structure (e.g., 3; 3-body); and a grabbing mechanism (e.g., 5-cnct in FIG. 1A/3B; 5-cnct in FIG. 21A) coupled to an underside of the main body structure, the grabbing mechanism configured to hold a load (e.g., 11 in FIG. 3D; 11 in FIG. 21C) beneath the main body structure and above ground; wherein the autonomous vehicle (10) is configured to control the plurality of powered actuation systems such that said active modification of the relative positioning between each of the wheels and the main body structure serves dual purposes of (i) providing active suspension for each said wheel at least when the vehicle is in forward motion and (ii) enabling coordinated modification of said relative positioning so as to manipulate the load (e.g., 11) relative to the ground.

[0318] In one embodiment, the load (e.g., 11) comprises a consumer item (e.g., 93-ci in FIG. 17A) to be delivered to a drop-off location.

[0319] In one embodiment, the consumer item (93-ci) is contained within a disposable package (e.g., 11-crb in FIG. 17A; 11-bag in FIG. 17D), and wherein the grabbing mechanism (e.g., 5-cnct in FIG. 17A/17B; 5-side-arm in FIG. 17C) is configured to directly hold said disposable package.

[0320] In one embodiment, the dimensions of the autonomous vehicle (10) are configured to allow the vehicle to traverse a sidewalk prior to reaching the drop-off location.

[0321] In one embodiment, said dimensions comprise a length of less than 2.5 meters, a height of less than 1.5 meters, and a weight of less than 700 kilograms.

[0322] In one embodiment, the vehicle (10) is configured to traverse roads (e.g., 20a in FIG. 2A; 20c in FIG. 4B) prior to said sidewalk traversal. The vehicle is capable of operating on standard roadways alongside traffic before transitioning to sidewalks for final delivery approach.

[0323] In one embodiment, said dimensions comprise a length of less than 1.5 meters, a height of less than 1 meter, and a weight of less than 200 kilograms. These smaller

dimensions are suitable for very compact vehicles potentially operating primarily in pedestrian areas or indoors.

[0324] In one embodiment, said manipulation of the load (11) relative to the ground comprises the vehicle (10) being configured to: control the plurality of powered actuation systems (e.g., 2'+2; legs/joints) to coordinately modify said relative positioning to lower the load (11) towards the ground surface (e.g., transition from FIG. 3D to 3C; or from FIG. 21C to 21B); and operate the grabbing mechanism (e.g., 5-cnct) to release the load (11) onto the ground surface.

[0325] In one embodiment, the autonomous vehicle is further configured to subsequent to the grabbing mechanism releasing the load (11), straddle off the load.

[0326] In one embodiment, said manipulation of the load (11) relative to the ground comprises the vehicle (10) being configured to: position the main body structure (e.g., 3; 3-body) over the load (11) at a pick-up location (e.g., FIG. 3B; FIG. 21B); operate the grabbing mechanism (e.g., 5-cnct) to hold the load (11); and subsequently control the plurality of powered actuation systems (e.g., 2'+2; legs/joints) to coordinately modify said relative positioning to lift the load (11) from the ground surface (e.g., transition from FIG. 3C to 3D; transition from FIG. 21B to 21C).

[0327] In one embodiment, said coordinated modification of said relative positioning, enabled by the control of the plurality of powered actuation systems (e.g., 2'+2; legs/joints) to manipulate the load (11) relative to the ground, includes maintaining a clearance (e.g., 9-clr in FIG. 1A; 11-ele in FIG. 21C) between a bottom surface of the load (11) and a ground surface during transport of the load by the autonomous vehicle (10).

[0328] In one embodiment, said clearance (e.g., 9-clr; 11-ele) is at least 10 (ten) centimeters.

[0329] In one embodiment, said providing active suspension comprises each respective powered actuation system (e.g., 2'+2; legs/joints) functioning as at least one of a spring and a damper between its associated wheel (e.g., 1a) and the main body structure (e.g., 3; 3-body).

[0330] In one embodiment, said providing active suspension enables vertical movement of each said wheel (e.g., 1a) relative to the main body structure (e.g., 3; 3-body) responsive to variations in the ground surface encountered during said forward motion. This describes the outcome of the active suspension—allowing wheels to move up and down independently to absorb bumps, curbs and stairs while the main body remains relatively stable.

[0331] In one embodiment, the autonomous vehicle further comprises a plurality of sensors (e.g., 4, 4a, 4b, 4c in FIG. 1A; 4-sense in FIG. 21A; 4 in FIG. 26) configured to collect data representing a surrounding environment of the autonomous vehicle (10).

[0332] In one embodiment, the autonomous vehicle (10) is further configured to generate, based on the data collected by the plurality of sensors (e.g., 4; 4-sense), a real-time three-dimensional representation (e.g., via model 9model in FIG. 26) of the surrounding environment, said representation including the load (11) when present within the surrounding environment.

[0333] In one embodiment, said configuration to control the plurality of powered actuation systems (e.g., 2'+2; legs/joints) to manipulate the load (11) relative to the ground is based at least in part on said three-dimensional representation (e.g., 9model) of the surrounding environment. The

vehicle uses its environmental understanding to accurately position itself and manipulate the load.

[0334] FIG. 27 illustrates one embodiment is a method for autonomously delivering a load (e.g., 11), the method comprising: receiving, by a control system (e.g., 8 in FIG. 26) of an autonomous vehicle (10) comprising a main body structure (e.g., 3; 3-body) and a plurality of wheels (e.g., 1; 1a), instructions associated with delivering the load to a drop-off location (step 1151); transporting the load (11), held by a grabbing mechanism (e.g., 5-cnct) coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels (e.g., step 1132 in FIG. 24), wherein during said transporting, active suspension is provided by a plurality of powered actuation systems (e.g., 2'+2; legs/joints), wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is configured to actively modify a relative positioning between said wheel and the main body structure (step 1152); upon arrival at the drop-off location, performing, via control of said plurality of powered actuation systems, a coordinated modification of said relative positioning between each of the wheels and the main body structure to manipulate the load (11) relative to a ground surface (step 1153); and operating the grabbing mechanism to release the load (step 1154). This method claim outlines the delivery process utilizing the dual-purpose actuation systems.

[0335] FIG. 28 illustrates one embodiment of a method for autonomously delivering a load (e.g., 11), the method comprising: receiving, by a control system (e.g., 8) of an autonomous vehicle (10) comprising a main body structure (e.g., 3; 3-body) and a plurality of wheels (e.g., 1; 1a), instructions associated with delivering the load to a drop-off location (step 1161); transporting the load (11), held by a grabbing mechanism (e.g., 5-cnct) coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active suspension is provided by a plurality of powered actuation systems (e.g., 2'+2; legs/joints), wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is configured to actively modify a relative positioning between said wheel and the main body structure (step 1162); and coordinately controlling, via the control system (8) during at least a portion of said transporting, said plurality of powered actuation systems (e.g., 2'+2; legs/joints) to maintain a clearance (e.g., 9-clr; 11-ele) between a bottom surface of the load (11) and a ground surface (step 1163).

[0336] In one embodiment, said clearance (e.g., 9-clr; 11-ele) is at least 10 (ten) centimeters.

[0337] In one embodiment, the autonomous vehicle (10) actively maintains a safe ground clearance (e.g., 9-clr; 11-ele) for the load (11) carried underneath its main body (3, 3-body) using a closed-loop sensing and control process. Primarily, forward-looking sensors (4, 4-sense, potentially LiDAR, cameras, radar) perceive the upcoming terrain and obstacle height profile. This data feeds into the control sub-system (8, FIG. 26), which calculates and commands dynamic adjustments to the powered actuation systems (2'+2 or articulated legs/joints via actuators 2) to proactively modify the wheel-body relative positioning, thus ensuring adequate clearance before encountering variations. Additionally, the system may employ complementary sensing methods for reactive adjustments or verification. For instance, downward-facing visual sensors (part of 4 or

4-sense) could directly monitor the actual clearance between the load's bottom surface and the ground in real-time. Furthermore, inertial measurement units (IMUs) within the sensor suite (4, 4-sense) could detect sudden accelerations or jolts indicative of an unintended contact, while acoustic sensors might detect the sound of such contact. In some embodiments, mechanical contact sensors or force/torque sensors associated with the load (11) or the grabbing mechanism (5-cnct) could provide direct feedback upon an initial, minor collision of the load's bottom with the ground. Upon detecting such contact via these secondary sensing methods, the control sub-system (8) would immediately trigger a corrective action, commanding the powered actuation systems to rapidly increase the ground clearance to prevent further impact or damage.

[0338] In one embodiment, the autonomous vehicle (10), irrespective of whether its powered actuation system comprises linear actuators (e.g., 2'+2 in FIG. 3C/3D) or articulated legs with joints (e.g., 2a-1, 2a-j1 in FIG. 21A), is configured to autonomously recharge its internal power source by directly interfacing with a stationary charging station (interface 29-rec at location 29 shown in FIG. 6D) or a similar interface located on ground or elevated above ground). Upon navigating to the charging location (29), the vehicle (10) utilizes its sensors (4, 4-sense) to precisely locate and orient itself relative to the charging interface (29-rec). The control sub-system (8, FIG. 26) then commands the respective powered actuation system to manipulate the position and/or orientation of the main body structure (e.g., 3 in FIG. 1A; 3-body in FIG. 21A). This manipulation serves to align a designated vehicle-mounted connector with the charging interface (29-rec). While this connector could be the under-body connector (5-cnct as shown in FIG. 1A or 21A) if the charging interface is appropriately positioned (e.g., on the ground), it could alternatively be a dedicated charging port located elsewhere on the main body structure (e.g., side, front, or rear). The actuation—whether primarily vertical adjustment via linear actuators (2'+2) or multi-axis positioning via articulated legs and joints—achieves the fine alignment necessary to establish secure physical and electrical contact between the vehicle's connector and the stationary charging source (29-rec). Once charging is complete, the control sub-system (8) commands the powered actuation system to perform the reverse maneuver, disengaging the vehicle (10) from the charging station.

[0339] In this description, numerous specific details are set forth. However, the embodiments/cases of the invention may be practiced without some of these specific details. In other instances, well-known hardware, materials, structures and techniques have not been shown in detail in order not to obscure the understanding of this description. In this description, references to “one embodiment” and “one case” mean that the feature being referred to may be included in at least one embodiment/case of the invention. Moreover, separate references to “one embodiment”, “some embodiments”, “one case”, or “some cases” in this description do not necessarily refer to the same embodiment/case. Illustrated embodiments/cases are not mutually exclusive, unless so stated and except as will be readily apparent to those of ordinary skill in the art. Thus, the invention may include any variety of combinations and/or integrations of the features of the embodiments/cases described herein. Also herein, flow diagrams illustrate non-limiting embodiment/case examples

of the methods, and block diagrams illustrate non-limiting embodiment/case examples of the devices. Some operations in the flow diagrams may be described with reference to the embodiments/cases illustrated by the block diagrams. However, the methods of the flow diagrams could be performed by embodiments/cases of the invention other than those discussed with reference to the block diagrams, and embodiments/cases discussed with reference to the block diagrams could perform operations different from those discussed with reference to the flow diagrams. Moreover, although the flow diagrams may depict serial operations, certain embodiments/cases could perform certain operations in parallel and/or in different orders from those depicted. Moreover, the use of repeated reference numerals and/or letters in the text and/or drawings is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments/cases and/or configurations discussed. Furthermore, methods and mechanisms of the embodiments/cases will sometimes be described in singular form for clarity. However, some embodiments/cases may include multiple iterations of a method or multiple instantiations of a mechanism unless noted otherwise. For example, when a controller or an interface are disclosed in an embodiment/case, the scope of the embodiment/case is intended to also cover the use of multiple controllers or interfaces.

[0340] Certain features of the embodiments/cases, which may have been, for clarity, described in the context of separate embodiments/cases, may also be provided in various combinations in a single embodiment/case. Conversely, various features of the embodiments/cases, which may have been, for brevity, described in the context of a single embodiment/case, may also be provided separately or in any suitable sub-combination. The embodiments/cases are not limited in their applications to the details of the order or sequence of steps of operation of methods, or to details of implementation of devices, set in the description, drawings, or examples. In addition, individual blocks illustrated in the figures may be functional in nature and do not necessarily correspond to discrete hardware elements. While the methods disclosed herein have been described and shown with reference to particular steps performed in a particular order, it is understood that these steps may be combined, subdivided, or reordered to form an equivalent method without departing from the teachings of the embodiments/cases. Accordingly, unless specifically indicated herein, the order and grouping of the steps is not a limitation of the embodiments/cases. Embodiments/cases described in conjunction with specific examples are presented by way of example, and not limitation. Moreover, it is evident that many alternatives, modifications and variations will be apparent to those skilled in the art. Accordingly, it is intended to embrace all such alternatives, modifications and variations that fall within the spirit and scope of the appended claims and their equivalents.

What is claimed is:

1. An autonomous vehicle, comprising:

a main body structure;

a plurality of wheels;

a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is configured to actively modify a relative positioning between said wheel and the main body structure; and

a grabbing mechanism coupled to an underside of the main body structure, the grabbing mechanism configured to hold a load beneath the main body structure and above ground;

wherein the autonomous vehicle is configured to control the plurality of powered actuation systems such that said active modification of the relative positioning between each of the wheels and the main body structure serves dual purposes of (i) providing active suspension for each said wheel at least when the vehicle is in forward motion and (ii) enabling coordinated modification of said relative positioning so as to manipulate the load relative to the ground.

2. The autonomous vehicle of claim 1, wherein the load comprises a consumer item to be delivered to a drop-off location.

3. The autonomous vehicle of claim 2, wherein the consumer item is contained within a disposable package, and wherein the grabbing mechanism is configured to directly hold said disposable package.

4. The autonomous vehicle of claim 2, wherein dimensions of the autonomous vehicle are configured to allow the vehicle to traverse a sidewalk prior to reaching the drop-off location.

5. The autonomous vehicle of claim 4, wherein said dimensions comprise a length of less than 2.5 meters, a height of less than 1.5 meters, and a weight of less than 700 kilograms.

6. The autonomous vehicle of claim 5, wherein the vehicle is configured to traverse roads prior to said sidewalk traversal.

7. The autonomous vehicle of claim 4, wherein said dimensions comprise a length of less than 1.5 meters, a height of less than 1 meter, and a weight of less than 200 kilograms.

8. The autonomous vehicle of claim 1, wherein said manipulation of the load relative to the ground comprises the vehicle being configured to:

control the plurality of powered actuation systems to coordinately modify said relative positioning to lower the load towards the ground surface; and

operate the grabbing mechanism to release the load onto the ground surface.

9. The autonomous vehicle of claim 8, further configured to subsequent to the grabbing mechanism releasing the load, straddle off the load.

10. The autonomous vehicle of claim 1, wherein said manipulation of the load relative to the ground comprises the vehicle being configured to:

position the main body structure over the load at a pick-up location;

operate the grabbing mechanism to hold the load;

and subsequently control the plurality of powered actuation systems to coordinately modify said relative positioning to lift the load from the ground surface.

11. The autonomous vehicle of claim 1, wherein said coordinated modification of said relative positioning, enabled by the control of the plurality of powered actuation systems to manipulate the load relative to the ground, includes maintaining a clearance between a bottom surface of the load and a ground surface during transport of the load by the autonomous vehicle.

12. The autonomous vehicle of claim 11, in which said clearance at least 10 (ten) centimeters.

13. The autonomous vehicle of claim **1**, wherein said providing active suspension comprises each respective powered actuation system functioning as at least one of a spring and a damper between its associated wheel and the main body structure.

14. The autonomous vehicle of claim **1**, wherein said providing active suspension enables vertical movement of each said wheel relative to the main body structure responsive to variations in the ground surface encountered during said forward motion.

15. The autonomous vehicle of claim **1**, further comprising a plurality of sensors configured to collect data representing a surrounding environment of the autonomous vehicle.

16. The autonomous vehicle of claim **15**, wherein the autonomous vehicle is further configured to generate, based on the data collected by the plurality of sensors, a real-time three-dimensional representation of the surrounding environment, said representation including the load when present within the surrounding environment.

17. The autonomous vehicle of claim **16**, wherein said configuration to control the plurality of powered actuation systems to manipulate the load relative to the ground is based at least in part on said three-dimensional representation of the surrounding environment.

18. A method for autonomously delivering a load, the method comprising:

receiving, by a control system of an autonomous vehicle comprising a main body structure and a plurality of wheels, instructions associated with delivering the load to a drop-off location;

transporting the load, held by a grabbing mechanism coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active

suspension is provided by a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is actively modifying a relative positioning between said wheel and the main body structure;

upon arrival at the drop-off location, performing, via control of said plurality of powered actuation systems, a coordinated modification of said relative positioning between each of the wheels and the main body structure to manipulate the load relative to a ground surface; and operating the grabbing mechanism to release the load.

19. A method for autonomously delivering a load, the method comprising:

receiving, by a control system of an autonomous vehicle comprising a main body structure and a plurality of wheels, instructions associated with delivering the load to a drop-off location;

transporting the load, held by a grabbing mechanism coupled to an underside of the main body structure, towards the drop-off location by operating the plurality of wheels, wherein during said transporting, active suspension is provided by a plurality of powered actuation systems, wherein for each wheel of the plurality of wheels, a respective one of the powered actuation systems is actively modifying a relative positioning between said wheel and the main body structure; and coordinately controlling, via the control system during at least a portion of said transporting, said plurality of powered actuation systems for maintaining a clearance between a bottom surface of the load and a ground surface.

20. The method of claim **19**, in which said clearance at least 10 (ten) centimeters.

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